Institut d’études politiques de Paris

ÉCOLE DOCTORALE DE SCIENCES PO

Programme doctoral d’économie

Département d’économie

Doctorat en Sciences économiques

Essays on Rational Bubbles

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to be defended on September 21, 2026

Jury

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Note to the Reader / Note au lecteur

The three chapters of this dissertation are written as research articles. Chapters 1 and 2 are closely linked and should be read together, while Chapter 3 is self-contained. The term "paper" is used to refer to the chapters. The language of the dissertation is English. A summary in French is provided at the end, and the acknowledgements are written in French with an English translation. To facilitate consultation, each chapter includes its own bibliography and corresponding appendices at the end of the chapter. During the preparation of the three chapters, I acknowledge the use of LLM tools to assist with sentence reformulation and code writing. All AI outputs were used only after critical review.

Les trois chapitres de cette thèse sont rédigés sous la forme d'articles de recherche. Les chapitres 1 et 2 sont étroitement liés et doivent être lus ensemble, tandis que le chapitre 3 est indépendant. Le terme "papier" est utilisé pour faire référence aux chapitres. La langue de la thèse est l'anglais. Un résumé en français est fourni à la fin du manuscrit. Les remerciements sont rédigés en français, accompagnés d'une traduction anglaise. Afin de faciliter la consultation, chaque chapitre comprend sa propre bibliographie ainsi que les annexes correspondantes en fin de chapitre. Lors de la préparation des trois chapitres, des outils d'intelligence artificielle ont été utilisés afin d'aider à la reformulation de certaines phrases et à l'écriture de code. Les résultats produits par ces outils d'IA n'ont été intégrés qu'après avoir fait l'objet d'un examen critique.

Remerciements

Une thèse, on sait quand ça commence, pas forcément quand ça finit ! Il semblerait pourtant qu'au bout de six ans, ce chapitre se referme et il est temps de remercier toutes celles et tous ceux sans qui elle n'aurait pas été possible (ou beaucoup plus fade !).

Je tiens tout d'abord à remercier les trois personnes qui m'ont suivi pendant cette thèse, mon directeur de thèse, Stéphane Guibaud, ainsi que les deux autres membres de mon comité de thèse, Nicolas Coeurdacier et Jeanne Commault.

Être sous la direction de Stéphane a été pour moi la meilleure école pour gagner en rigueur académique. Ses conseils, questions et remarques ont eu, à ce titre, une valeur inestimable. Je le remercie également pour sa grande disponibilité tout au long de ces six années. Je ne compte plus ses relectures de l'abstract de mon JMP et me souviendrai longtemps de nos rendez-vous pouvant facilement durer deux, voire trois heures !

Avec Nicolas, qui m'a suivi depuis mon mémoire de master, ils ont su trouver un équilibre subtil en m'accompagnant de près tout en me laissant une grande liberté académique. Ainsi, ils m'ont encouragé quand j'ai pris la voie de la macroéconomie théorique et de l'étude des préférences pour la richesse, deux choix potentiellement un peu casse-binette. Par ailleurs, à chaque étape de la thèse où j'ai pu être bloqué, je savais qu'aller les voir me permettrait d'avancer et de revenir avec les bonnes questions. À cet égard, je tiens à souligner l'attention particulière portée aux intuitions économiques de Nicolas, qui a été aussi importante que stimulante.

Par la suite, ce fut une grande chance que Jeanne soit venue compléter ce trio. Par la proximité de nos sujets de recherche, son aide et ses remarques ont été très précieuses. Je tiens donc à les remercier tou.te.s les trois le plus chaleureusement possible pour ces années passées à m'accompagner avec une grande bienveillance.[1] Merci d'avoir été là de présentations en relectures et de la recherche d'idées à la recherche d'emploi !

Je tiens ensuite à remercier Alberto Martin et Jaume Ventura pour leur accueil au printemps 2024 au CREI et à l'Université Pompeu Fabra. Je mesure la chance d'avoir eu deux hôtes si impliqués dans la réussite de mon séjour de recherche. Les deux premiers chapitres de cette thèse doivent beaucoup aux discussions que j'ai pu avoir avec eux. Revenant tout juste de Barcelone au moment où j'écris ces lignes, je constate avec joie que leur accueil y est toujours aussi chaleureux et remercie Alberto d'être rapporteur de cette thèse.

[1] On notera le patient combat de Stéphane vis-à-vis de mes fautes d'anglais les plus récalcitrantes (comme les « g » qui se prononcent « j » et vice-versa pour n'en citer qu'une).

Je remercie également Jean-Baptiste Michau d'avoir accepté d'être rapporteur. Au cours des dernières années, ce fut toujours un plaisir de le retrouver aux sessions du Paris Macro Group ou de venir à Palaiseau pour de longues discussions. J'ai grandement bénéficié de son appréciation de problèmes inhérents aux préférences pour la richesse et de son goût communicatif pour les modèles élégants.

Enfin, merci à Almuth Scholl de compléter ce jury en tant qu'examinatrice externe. Je lui suis reconnaissant de sa confiance pendant ma recherche d'emploi et me réjouis de la rejoindre à l'Université de Constance pour les trois prochaines années.

Par ailleurs, je suis reconnaissant à l'ensemble des professeur.e.s du département d'économie de Sciences Po. J'ai été porté par le dynamisme de cette communauté, où tout le monde assiste au séminaire du vendredi, se retrouve à la machine à café et a à cœur la réussite des doctorant.e.s..

Je tiens en particulier à remercier les membres du reading group de macroéconomie (et de feu le macro/finance/trade reading group) pour leurs nombreux et précieux retours tout au long de ces années : Johannes Boehm, Paul Bouscasse, Antoine Camous, Axelle Ferrière, François Geerolf, Kerstin Holzheu, Thierry Mayer, Isabelle Méjean, Jean-Stéphane Messonier, Éric Mengus, Guillaume Plantin, Morgane Richard, Xavier Ragot, Julia Schmidt et Moritz Schularick. Cette thèse doit beaucoup à leurs conseils, leurs questions, aux références qu'ils m'ont partagées et à leurs encouragements ! Merci également à Jean-Marc Robin et Franz Ostrizek pour nos discussions au long cours sur la nature des préférences pour la richesse.

Je voudrais dire aussi un grand merci à Antoine Ferey et Claire Montialoux de nous avoir guidés avec générosité dans notre recherche d'emploi. Leur investissement pour nous aider à monter notre dossier, nous préparer aux divers entretiens et nous conseiller a été déterminant tout au long de cette dernière année. Par ailleurs, je remercie sincèrement Pierre-Philippe Combes de nous avoir accompagnés en tant que directeur des études doctorales avec beaucoup de rigueur et d'enthousiasme.

Toute personne passée par le département sait que sa bonne santé doit beaucoup à sa précieuse équipe administrative. Un grand merci à Stéphanie Berrebi, Lucie Desramaut, Christelle Hotait, Sandrine Le Goff, Melissa Mundell, Maria Ohlund et ~~le lapin de Pâques~~ Guillaume Sarrat de Tramezaigues d'avoir contribué aux très bonnes conditions de recherche dont j'ai pu bénéficier pendant six ans. Au-delà du professionnel, quel plaisir que de trouver la porte du bureau de Stéphanie et Lucie ouverte et d'y être toujours accueilli avec bonne humeur. Je remercie également Hadjila Nezlioui-Serraz du côté de l'École de la Recherche pour, entre autres, l'organisation de la soutenance de cette thèse.

En dehors du département, je tiens à remercier Vladimir Asryan, Alexandre Gaillard et Nicolas Werquin pour leurs commentaires, leurs conseils et leurs encouragements particulièrement importants au cours de cette thèse.

Mais ces six années ne furent pas uniquement consacrées à la recherche. Ce fut un plaisir d'être chargé de TD ou assistant dans toute une variété de cours. En plus des professeur.e.s déjà mentionné.e.s, je souhaite remercier Pauline Wibaux pour sa confiance et sa sympathie au semestre d'automne 2024. Sur le campus de Reims, où j'ai le plus enseigné, je tiens à souligner le professionnalisme de Célia Guiborat et de Lucas Sageot-Chomel. Ce dernier savait chaque année me convaincre de reprendre le TGV l'année suivante.

Cette thèse fut aussi le moment pour moi de m'engager dans la vie associative de Sciences Po via son AMAP : Sciences Potirons. Merci à Myrtille, Adèle et Amélie pour ces trois semestres à organiser les distributions et aux membres de Pavés rencontrés au chalet du 28, en AG[2] ou en manif : Matthieu, Amélie, Manon, Justine, Théo et tout.e.s les autres. Vous étiez vraiment des rayons de soleil et c'est toujours un plaisir de vous recroiser (désormais le plus souvent un certain week-end autour de mi-septembre). Évidemment, je souhaite dire un immense merci à Stéphane, Christophe et Rémi pour leurs délicieux pain, fruits et légumes qui ont fait mon bonheur tout au long de cette thèse !

Je remercie l'ensemble des membres de la communauté de Sciences Po sans qui toutes ses activités de recherche, d'enseignement et de vie de campus ne seraient pas possibles et notamment les agentes et agents d'entretien, les appariteurs et appariteurs, les régisseuses et régisseurs, et les membres des équipes de sécurité. Je tiens à adresser un salut amical à Steeve. Son sourire rayonnait à l'entrée du 28 et je me souviendrai longtemps de ses expressions du type « dans le film de la bêtise humaine, tu peux être figurant ou acteur principal ». Merci également à Layla Mabrouk pour son rôle décisif dans le mouvement de grève exemplaire des agents d'entretien en mars 2025. De même, je tiens à remercier le Collectif Doctorant pour son travail.

Cette thèse s'inscrivait dans la continuité de mes deux années de master à Sciences Po. Merci à Danell et Tancrède (le trio de français !) et à tou.te.s les autres d'avoir partagé ces deux belles années. J'ai un souvenir ému de la période où comprendre le cours de macro de deux heures de la veille pouvait nous prendre toute la journée. Je tiens à remercier Emma pour tous les bons souvenirs de cette époque, qui m'a vu grandir sur tant d'aspects. Pour n'en mentionner qu'un, sache que cette thèse aurait été bien plus dure à écrire sans avoir été au contact de ton approche anglaise de l'argumentation.

Si ces six années de thèse ont été si riches, cela doit beaucoup à la camaraderie de mes co-doctorant.e.s, que je souhaite remercier. Écrire ces remerciements me replonge

[2] Soit dit en passant, quelle découverte que celle d'AG où tout le monde s'écoute !

dans un joyeux trombinoscope : Felipe et ses anecdotes folles avec des prix Nobel ou des stars de téléréalité ; Victor A, Daniel B. et Samuel, qui ont mis les échecs au cœur des combles ; Nathaniel, dont l'accent britannique a parfois validé ma prononciation de lettres grecques contre l'avis du reste du département ; Constantin et ses délicieux gâteaux aux pommes ; Agathe, qui met des souris dans nos vies ; Viktor, son rire inimitable et ses explications sur la manière romanesque de vivre à la russe ; Juliette, notre squatteuse de combles préférée ; Ségal et ses soirées technos plus survoltées que son « pull de prof de maths » ; Venance, jamais contre une balade macroéconomique le long de la Seine ; Mathias, dont l'appétit pour les avancées de l'IA n'a d'égal que celui pour les buffets ; Mattis et sa force tranquille, dont les plats nous obligeaient —par peur de la comparaison— à ramener mieux que des pâtes pesto dans notre tupp' ; Sophia, dont la Freundlichkeit m'a accompagné de la préparation d'examens pour nos étudiants en conférences ; Théodore et ses plans rando ; Nicolas, notre délégué revendicatif (il ne faut pas dire râleur !) toujours prêt à enflammer le dancefloor ; Dániel G. et son calme inébranlable ; Pauline M. et son esprit « Drômeloc » même en dehors de la « Drômeloc » ; Gustave dont l'intérêt sincère pour tout ce qui touche à de l'éco vient avec une grande générosité (et des flux RSS) ; Alaïs, Alexandre, Camille, Claudius, Clémence, Davide, Diego, Dima, Edgard, Eulalie, Fatimetou, Gustavo, Jakob, Jeteesha, Jim, Mélitine, Nelly, Nikiforos, Norbert, Pauline C., Pierre C., Pierre V., Qinci, Riddhi, Swann, Zydney et bien d'autres. D'anciens assistants de recherche ont, bien sûr, leur place dans cette liste et je pense notamment à Pierre C., Mélitine et Jim. Au-delà des bons moments passés ensemble, je garde des combles (et des « nouveaux » bureaux) le souvenir d'un lieu de grande émulation intellectuelle et j'ai beaucoup appris de nos discussions. Quel plaisir d'avoir été ouvert à leurs sujets de recherche, qui allaient tout de même des mouvements féministes du XIXe siècle avec Olivia aux usines de ciment de Niklas.

J'adresse un grand merci aux doctorant.e.s qui ont eu une place particulière dans ce chemin. Il y a tout d'abord celles et ceux avec qui j'ai commencé la thèse (et qui ont heureusement fini avant moi) : Léonard, dont la générosité se lit sur son sourire ; Juan et sa bienveillance de grand frère (mais que je ne remercie pas pour ses choix de poster) ; Moritz, qui nous a initiés plusieurs fois à l'art du Feuerzangenbowle ; Mylène, ma « voisine de diagonale » par-delà les bureaux, qui savait me rappeler que j'étais un salarié plus qu'un étudiant quand je me réjouissais un peu trop de recevoir du matériel informatique par le département ; Clemens, Nourhan et Kevin.

Merci également à Constance, la voisine de bureau qui donne autant la pêche que de très bonnes recommandations de livres ; Alexis, avec qui une discussion donne envie d'écouter de la techno, de regarder du cinéma italien et même parfois de lancer un club de lecture des textes fondateurs de l'économie ; Victor SJ, bien sympathique, même si « on est juste collègues » ; Aurélien, le partenaire idéal de recherche d'emploi avec qui l'on a partagé les premières versions de « Spiel », les craquages de novembre, le stress de début janvier et, au final, les bonnes nouvelles ; Johanna, que j'ai eu la chance de retrouver de Mannheim à Paris pour former notre wunderschön (und fast regelmäßig) tandem linguistique ; Gabrielle que j'ai appréciée dès le premier jour, alors que notre première interaction ressemblait plus à un questionnaire INSEE qu'à autre chose ; Jérôme, avec qui je mets bien volontiers nos différends sur les conventions dans les restaurants libanais de côté pour discuter légèrement de sujets profonds et profondément de sujets légers ; Ariane, la meilleure voisine pour m'accueillir dans le doctorat (quoi de mieux que de mettre deux pipelettes à côté ?) ; Leila, avec qui on a partagé les départs de manifs il y a trois ans, les verres en terrasse ensuite, puis le statut de collègue et les « petites pauses » d'aprèm rarement courtes qui vont avec ; Paloma, avec qui on s'entraide avec sérieux, mais sans jamais se prendre au sérieux et que l'on croise toujours avec plaisir « par hasard » au Bar à Mines— dont elle augmente les prix à ses heures. Merci d'avoir été notre petite sœur de la macro team !

À propos de macro team, il est temps de remercier Naomi. Elle n'est pas seulement l'une des plus grandes interprètes de « Confessions nocturnes » de sa génération, elle est aussi le pilier de ma thèse et de ma recherche d'emploi. Merci pour tes conseils, pour m'avoir poussé à écrire certains méls, pour les dimanches aprèm passés à préparer la galette en écoutant Chante France, pour les photos où chacun.e fait une moitié de cœur avec sa main, pour les soirées au Divan du monde (tellement notre playlist idéale) et pour toutes les choses que tu as déjà mises dans tes remerciements. Ça va me manquer de ne plus t'entendre chanter dans le bureau ou de ne plus reconnaitre ton rire (au loin) dans les couloirs, mais sache que, pendant comme après le doctorat, « on est ensemble » !

Au passage, je redouble de remerciements pour celles et ceux cité.e.s plus haut qui se sont engagé.e.s dans la vie du département en tant que délégué.e.s ou organisateur.rice.s du CLS, du petit-déjeuner des doctorant.e.s, de la conférence des alumnis ou de la newsletter.

Je me sens également très chanceux d'avoir rencontré d'autres doctorant.e.s sur ma route. Merci à Elliot, Hubert et Maren d'avoir fait de mon échange à Barcelone quatre mois de convivialité et de partage. D'ailleurs, on ne remerciera jamais assez Hubert de nous avoir emmené.e.s à la Bodega d'un certain Armando. Merci à Marguerite pour nos pauses pré-TD sur le campus de Reims. Merci à Jean-Baptiste, que j'ai retrouvé par bien des chemins, de la salsa à l'École de la Recherche en passant par les amitiés communes. Merci aux doctorant.e.s du CERI et notamment Elisabeth, Louis, Michaël, Fatoumata, Laurie et Louise. Vous avez été de chouettes voisin.e.s de labo. Je garde de très bons souvenirs de toutes nos discussions que ce soit pour échanger sur l'intérêt de visiter Lübeck, comprendre comment toucher l'allocation chômage à l'aide d'une auto-entreprise de DJ ou pour débattre de la différence de degré ou de nature entre chirurgie esthétique et tatouage.

Je veux remercier celles et ceux qui remplissent ma vie de joie en dehors du doctorat. Il y a d'abord les ami.e.s. Que vous soyez dans ma vie depuis l'impasse des renards (depuis plus longtemps c'est difficile), le collège, le lycée, la licence, l'Erasmus ou même après, j'ai mille souvenirs de moments passés avec vous ces six dernières années : de Nouvel An, de soirées Fleurus ou Koh-Lanta, de festivals, d'anniversaires, de tournées des bars, de petits verres en terrasse, de concerts, de fous rires, de jeux de société, de cinés, de sushis, de vacances en Bretagne, dans les Vosges, à Marseille, Barcelone ou (presque) Angoulême, de faire-part reçus (coucou Charlie, Jila et Augustin !), d'un mariage (en attendant les autres) et j'en passe. Bref, je préfère ne pas entrer davantage dans le détail, sinon ces remerciements doubleraient de volume. Merci d'être là et de m'avoir à la fois accompagné dans cette thèse et bien changé les idées quand j'y pensais trop.

Je tiens également à remercier l'ensemble de ma famille pour son amour et son soutien tout au long de ces années. Merci à mes parents, Denis et Emmanuelle, de m'avoir donné tout ce qu'il faut pour transformer un bébé de 4 kg en thésard. L'histoire ne dit pas si j'aurais pu faire cette thèse sans un bac S, mais merci papa d'avoir continué à m'épauler après que j'ai arrêté d'écouter tes conseils d'orientation. Et merci maman de m'avoir donné, dès le départ, des conseils d'orientation moins directifs. Merci à mon frère, Romain, pour ces chouettes années où nous étions étudiants tous les deux. De ma première visite à Compiègne jusqu'au concert de Damso en passant par Barcelone et la Bretagne, on ne manquera pas d'anecdotes sur ces années-là. J'ai une pensée pour Mamie Monique, avec qui j'aurais aimé partager l'aboutissement de ce travail.

Et il y a toi, Angèle. Le plus beau jour de cette thèse restera assurément celui où Cupidon, aidé d'une flèche houblonnée (voire carrément imbibée de teq' paf), nous a mis sur le même chemin. Merci pour tout ce que tu m'as apporté depuis. Merci d'être à la fois la personne qui me fait le plus rire et mon meilleur public. Merci d'avoir constamment été fière de moi, d'avoir cherché à comprendre le concept de taux d'actualisation, et de m'avoir soutenu dans les moments les plus stressants de ce doctorat. Merci pour ta patience quand j'ai reporté tant de fois la date de fin de cette thèse (normalement la date du 21 septembre, ça ne bouge plus !). Merci de me suivre à Constance, où j'ai déjà hâte d'écrire un nouveau chapitre avec toi. En deux mots : merci gros !

Enfin, je tiens à dédier cette thèse à l'ensemble des professeur.e.s du lycée Jeanne d'Arc qui m'ont mis sur la voie des sciences sociales : Agnès Beck, Françoise Buzzi, Matthieu Delatte, Bernard Jeanningros et François Neis. Je me souviens de cette époque où un livre ou un concept, que vous partagiez, pouvait chambouler mes idées reçues. Je me souviens de vos efforts de pédagogie pour éveiller notre intérêt, que cela passe par l'application des lois normales aux bureaux de vote russes, des traductions de bandes dessinées, des ateliers de cartographie ou des paris de barres de Crunch. Je me souviens également de votre investissement pour nous emmener au théâtre et en voyage, nous donner des retours détaillés sur nos travaux, nous préparer à entrer dans le supérieur, nous accompagner à partir en Allemagne et j'en passe. Ces trois années ont planté les graines de tant de questions qui m'ont conduit vers la recherche. Je vous en suis infiniment reconnaissant.

Paris, le 25 Juin 2026

Acknowledgements - English Translation

A thesis is something you know when you begin, but not necessarily when you will finish! And yet, after six years, this chapter finally seems to be closing, and it is time to thank all those without whom it would not have been possible (or would have been much duller!).

I would first like to thank the three people who supported me throughout this thesis: my supervisor, Stéphane Guibaud, as well as the two other members of my thesis committee, Nicolas Coeurdacier and Jeanne Commault.

Being supervised by Stéphane was, for me, the best possible training in academic rigor. His advice, questions, and comments were, in that respect, invaluable. I am also grateful to him for his remarkable availability throughout these six years. I have lost count of the number of times he reread the abstract of my JMP, and I will long remember our meetings, which could easily last two, or even three, hours!

Together with Nicolas, who had been advising me since my master's thesis, they managed to strike a subtle balance: guiding me closely while leaving me a great deal of academic freedom. They encouraged me when I chose to go down the path of theoretical macroeconomics and the study of preferences for wealth, two potentially risky choices. Moreover, whenever I found myself stuck at any point during the thesis, I knew that going to see them would help me move forward and come back with the right questions. In this regard, I want to emphasize Nicolas's particular attention to economic intuitions, which was as important as it was stimulating.

It was then a great stroke of luck that Jeanne joined this trio. Given the proximity of our research topics, her help and comments were extremely valuable. I therefore want to thank all three of them for guiding me through these years with such care and generosity.[3]. Thank you for being there, from presentations to rereadings, and from the search for ideas to the job search!

I would also like to thank Alberto Martin and Jaume Ventura for welcoming me in spring 2024 at CREI and Pompeu Fabra University. I know how lucky I was to have two hosts so invested in making my visiting a success. The first two chapters of this thesis owe a great deal to the discussions I had with them. Having just returned from Barcelona as I write these lines, I am happy to see that their welcome is still as warm as ever, and I thank Alberto for agreeing to be one of the referees for this thesis.

[3] One should note Stéphane's patient struggle with my most stubborn English mistakes (such as "g" pronounced as "j" and vice versa, to name just one).

I also thank Jean-Baptiste Michau to serve as a referee. Over the past few years, it has always been a pleasure to see him at Paris Macro Group sessions or to come to Palaiseau for long discussions. I benefited greatly from his appreciation of the issues inherent to preferences for wealth and from his contagious taste for elegant models.

Finally, thank you to Almuth Scholl for completing this jury as an external examiner. I am grateful to her for her trust during my job search, and I am delighted to be joining her at the University of Konstanz for the next three years.

In addition, I am grateful to all the professors in the Department of Economics at Sciences Po. I was carried by the energy of this community, where everyone attends the Friday seminar, gathers around the coffee machine, and cares about the success of PhD students.

I would especially like to thank the members of the macroeconomics reading group (and of the former macro/finance/trade reading group) for their many valuable comments throughout these years: Johannes Boehm, Paul Bouscasse, Antoine Camous, Axelle Ferrière, François Geerolf, Kerstin Holzheu, Thierry Mayer, Isabelle Méjean, Jean-Stéphane Messonier, Éric Mengus, Guillaume Plantin, Morgane Richard, Xavier Ragot, Julia Schmidt, and Moritz Schularick. This thesis owes much to their advice, their questions, the references they shared with me, and their encouragement! Thanks also to Jean-Marc Robin and Franz Ostrizek for our long-running discussions on the nature of preferences for wealth.

I would also like to say a big thank you to Antoine Ferey and Claire Montialoux for generously guiding us through our job search. Their commitment to helping us put together our applications, prepare for the various interviews, and think through our choices was decisive throughout this final year. I also sincerely thank Pierre-Philippe Combes for supporting us as director of doctoral studies with great rigor and enthusiasm.

Anyone who has spent time in the department knows that its good health owes a great deal to its invaluable administrative team. A big thank you to Stéphanie Berrebi, Lucie Desramaut, Christelle Hotait, Sandrine Le Goff, Melissa Mundell, Maria Ohlund, and ~~the Easter Bunny~~ Guillaume Sarrat de Tramezaigues for contributing to the very good research conditions I was able to enjoy for six years. Beyond the professional side, what a pleasure it was to find Stéphanie and Lucie's office door open and to always be welcomed there in good spirits. I also thank Hadjila Nezlioui-Serraz on the École de la Recherche side for, among other things, organizing the defense of this thesis.

Outside the department, I would like to thank Vladimir Asryan, Alexandre Gaillard, and Nicolas Werquin for their comments, advice, and encouragement, which were particularly important during this thesis.

But these six years were not devoted only to research. It was a pleasure to teach tutorials or to assist in a wide variety of courses. In addition to the professors already mentioned, I would like to thank Pauline Wibaux for her trust and kindness during the fall 2024 semester. On the Reims campus, where I taught the most, I would like to acknowledge the professionalism of Célia Guiborat and Lucas Sageot-Chomel. The latter always knew how to convince me, year after year, to take the TGV again the following year.

This thesis was also the moment when I became involved in Sciences Po's student associations through Sciences Potirons, its local organic vegetable distribution scheme. Thank you to Myrtille, Adèle, and Amélie for those three semesters spent organizing the distributions, and to the members of Pavés whom I met at the chalet of 28, in general assembly[4], or at demonstrations: Matthieu, Amélie, Manon, Justine, Théo, and everyone else. You truly were rays of sunshine, and it is always a pleasure to run into you again (now, often, on a certain weekend around mid-September). Of course, I also want to say an enormous thank you to Stéphane, Christophe, and Rémi for the delicious bread, fruit, and vegetables that made me so happy throughout this thesis!

I thank all the members of the Sciences Po community without whom its research, teaching, and campus life activities would not be possible, and in particular the cleaning staff, ushers, technical staff, and security teams. I want to send a friendly greeting to Steeve. His smile lit up the entrance to 28, and I will long remember his expressions, such as: "in the film of human stupidity, you can be an extra or the leading actor." Thanks also to Layla Mabrouk for her decisive role in the exemplary strike movement of the cleaning staff in March 2025. Likewise, I want to thank the Collectif Doctorant for its work.

This thesis followed on from my two years in the master's program at Sciences Po. Thank you to Danell and Tancrède (the trio of French students!) and to everyone else for sharing those two wonderful years. I have fond memories of the time when understanding the previous day's two-hour macro lecture could take us the entire day. I want to thank Emma for all the good memories from that period, which helped me grow in so many ways. To mention just one, know that this thesis would have been much harder to write without having been exposed to your English approach to argumentation.

If these six years of PhD life were so rich, it is largely thanks to the camaraderie of my fellow doctoral students, whom I would like to thank. Writing these acknowledgements sends me back into a joyful gallery of faces: Felipe and his wild stories involving Nobel Prize winners or reality TV stars; Victor A., Daniel B., and Samuel, who put chess at the heart of the combles; Nathaniel, whose British accent sometimes validated

[4] By the way, what a discovery it was to attend a general assembly where everyone actually listens to one another!

my pronunciation of Greek letters against the opinion of the rest of the department; Constantin and his delicious apple cakes; Agathe, who brings mice into our lives; Viktor, with his unmistakable laugh and his explanations of how to live grandly, the Russian way; Juliette, our favorite squatter of the combles; Ségal and his techno parties even wilder than his "math teacher sweater"; Venance, always up for a macroeconomic walk along the Seine; Mathias, whose appetite for advances in AI is matched only by his appetite for buffets; Mattis and his quiet strength, whose dishes forced us - for fear of comparison - to bring something better than pesto pasta in our lunchboxes; Sophia, whose Freundlichkeit accompanied me from preparing exams for our students to conference trips; Théodore and his hiking plans; Nicolas, our combative delegate (one must not say grumbler!) always ready to set the dancefloor on fire; Dániel G. and his unshakable calm; Pauline M. and her "Drômeloc" spirit even outside the "Drômeloc"; Gustave, whose sincere interest in everything related to economics comes with great generosity (and RSS feeds); Alaïs, Alexandre, Camille, Claudius, Clémence, Davide, Diego, Dima, Edgard, Eulalie, Fatimetou, Gustavo, Jakob, Jeteesha, Jim, Mélitine, Nelly, Nikiforos, Norbert, Pauline C., Pierre C., Pierre V., Qinci, Riddhi, Swann, Zydney, and many others. Former research assistants, of course, also belong on this list, and I am thinking in particular of Pierre C., Mélitine, and Jim. Beyond the good times spent together, I will remember the combles (and the "new" offices) as places of great intellectual stimulation, and I learned a lot from our discussions. What a pleasure it was to be exposed to their research topics, which ranged, after all, from nineteenth-century feminist movements with Olivia to Niklas's cement plant.

I send a big thank you to the doctoral students who played a special role along this path. First, there are those with whom I started the PhD (and who fortunately finished before me): Léonard, whose generosity can be read in his smile; Juan and his big-brotherly kindness (though I do not thank him for his choice of posters); Moritz, who introduced us several times to the art of Feuerzangenbowle; Mylène, my "diagonal neighbor" across the offices, who knew how to remind me that I was more an employee than a student whenever I got a little too excited about receiving computer equipment from the department; Clemens, Nourhan, and Kevin.

Thanks also to Constance, the office neighbor who brings as much energy as she does excellent book recommendations; Alexis, with whom a conversation makes you want to listen to techno, watch Italian cinema, and sometimes even start a reading group on the foundational texts of economics; Victor SJ, very nice indeed, even if "we are just colleagues"; Aurélien, the ideal job-market partner, with whom I shared the first versions of the "Spiel," the November breakdowns, the early-January stress, and, in the end, the good news; Johanna, whom I was lucky enough to meet again from Mannheim to Paris to form our wundershön (und fast regelmäßig) language tandem; Gabrielle, whom I liked from the very first day, even though our first interaction was more like an

INSEE questionnaire than anything else; Jérôme, with whom I am very happy to put aside our disagreements over conventions in Lebanese restaurants to discuss serious subjects lightly and light subjects seriously; Ariane, the best neighbor to welcome me into the PhD: what could be better than putting two chatterboxes next to each other?; Leila, with whom I shared the departures for demonstrations three years ago, drinks on terraces afterward, then the status of colleague and the "little afternoon breaks" (rarely short) that come with it; Paloma, with whom we help each other seriously without ever taking ourselves too seriously, and whom one is always happy to run into "by chance" at the Bar à Mines - whose prices she raises in her spare time. Thank you for being our little sister of the macro team!

Speaking of the macro team, it is time to thank Naomi. She is not only one of the greatest interpreters of "Confessions nocturnes" of her generation; she is also the pillar of my thesis and of my job search. Thank you for your advice, for pushing me to write certain emails, for the Sunday afternoons spent preparing the galette while listening to Chante France, for the photos where each of us makes one half of a heart with a hand, for the evenings at the Divan du monde (so perfectly our ideal playlist), and for all the things you have already put in your own acknowledgements. I will miss no longer hearing you sing in the office, or no longer recognizing your laugh (from afar) in the corridors, but know that, during the PhD as after it, "on est ensemble"!

In passing, I want to repeat my thanks to those mentioned above who got involved in the life of the department as delegates or organizers of the CLS, the doctoral students' breakfast, the alumni conference, or the newsletter!

I also feel very lucky to have met other doctoral students along the way. Thank you to Elliot, Hubert, and Maren for making my exchange in Barcelona four months of conviviality and sharing. Besides, we will never thank Hubert enough for taking us to the bodega of a certain Armando. Thank you to Marguerite for our pre-tutorial breaks on the Reims campus. Thank you to Jean-Baptiste, whom I encountered again through many paths, from salsa to the École de la Recherche by way of mutual friendships. Thank you to the doctoral students of CERI, and especially Elisabeth, Louis, Michaël, Fatoumata, Laurie, and Louise. You were wonderful lab neighbors. I have very good memories of all our conversations, whether they were about the value of visiting Lübeck, understanding how to receive unemployment benefits with the help of a DJ auto-entreprise, or debating whether the difference between cosmetic surgery and tattooing is one of degree or of kind.

I want to thank those who fill my life with joy outside the PhD. First, there are my friends. Whether you have been in my life since impasse des renards (it is hard to go back much further), middle school, high school, undergraduate studies, Erasmus, or even after, I have a thousand memories of moments spent with you over these past six years: New Year's Eves, evenings at Fleurus or watching Koh-Lanta, festivals, birthdays, bar crawls, little drinks on terraces, concerts, fits of laughter, board games, movies, sushi, holidays in Brittany, in the Vosges, in Marseille, Barcelona, or (almost) Angoulême, birth announcements received (hello Charlie, Jila, and Augustin!), a wedding (while waiting for the others), and much more. I would rather not go into further detail, otherwise these acknowledgements would double in length. Thank you for being there, and for both accompanying me through this thesis and helping me take my mind off it when I was thinking about it too much.

I would also like to thank my entire family for their love and support throughout these years. Thank you to my parents, Denis and Emmanuelle, for giving me everything needed to turn a 4-kilo baby into a PhD student. History does not say whether I could have written this thesis without a scientific baccalauréat, but thank you, Dad, for continuing to support me after I stopped listening to your career-orientation advice. And thank you, Mom, for giving me, from the very beginning, career-orientation advice that was less directive. Thank you to my brother, Romain, for those wonderful years when we were both students. From my first visit to Compiègne to the Damso concert, by way of Barcelona and Brittany, we will not be short of stories from those years. I am thinking of Mamie Monique, with whom I would have loved to share the completion of this work.

And then there is you, Angèle. The most beautiful day of this thesis will definitely remain the day when Cupid, helped by a hop-soaked arrow (or even one completely drenched in teq' paf), put us on the same path. Thank you for everything you have brought me since. Thank you for being both the person who makes me laugh the most and my best audience. Thank you for constantly being proud of me, for trying to understand the concept of a discount rate, and for supporting me through the most stressful moments of this PhD. Thank you for your patience when I postponed the end date of this thesis so many times (normally, the date of 21 September will not move again!). Thank you for following me to Konstanz, where I already cannot wait to write a new chapter with you. In two words: merci gros!

Finally, I would like to dedicate this thesis to all the teachers at Lycée Jeanne d'Arc who set me on the path toward the social sciences: Agnès Beck, Françoise Buzzi, Matthieu Delatte, Bernard Jeanningros, and François Neis. I remember that time when a book or a concept you shared with us could shake up my preconceptions. I remember your pedagogical efforts to spark our interest, whether through applying normal distributions to Russian polling stations, translating comic strips, cartography workshops, or bets over Crunch bars. I also remember your commitment to taking us to the theater and on trips, giving us detailed feedback on our work, preparing us to enter higher education, helping us go to Germany, and so much more. Those three years planted

the seeds of so many questions that led me toward research. For that, I am infinitely grateful.

Paris, 25 June 2026

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Introduction

Rational Bubbles and Extrapolations

What determines the value of an asset? Standard asset-pricing theory values assets as claims to future cash flows. Their price should reflect the discounted present value of these cash flows, commonly referred to as their fundamental value. Yet several observations suggest that this view may be incomplete. A striking example is public debt. Brunnermeier et al. (2023) ask how the high value of Japanese government debt, which amounted to about 249 percent of GDP in 2025, can be reconciled with a purely fundamental valuation, given Japan's persistent primary deficits over recent decades. A similar puzzle arises for U.S. public debt, as documented by Jiang et al. (2024). Beyond public debt, many asset prices fluctuate substantially over the business cycle, in ways that may be difficult to reconcile with movements in fundamentals alone (Carvalho et al., 2012).

There are two broad reasons why an asset price may diverge from its fundamental value. The first is that agents form incorrect expectations. They may overestimate future cash flows or underestimate the discount rates applied to them. The second possibility is that agents correctly understand that the asset price exceeds the present value of its future cash flows, yet remain willing to hold it because they expect it to be sustainably traded above its fundamental value. The resulting deviation constitutes a rational bubble. Its current value derives solely from its future resale price, which in turn reflects future demand for the bubble. This thesis focuses on such bubbles and examines two distinct sources of asset demand that can sustain them.

The first two chapters show how insatiable preferences for wealth can lead to the existence of rational bubbles that grow faster than the economy. These preferences are introduced for the first time in a heterogeneous-agent setting. Together, these elements yield a model in which an increase in labor income inequality can, through a bubble, raise aggregate wealth without generating a corresponding increase in investment. This mechanism can help explain the rise in wealth-to-output ratios alongside stagnant capital-to-output ratios observed in advanced economies over the past four decades.

The third chapter considers an overlapping-generations framework in which a rational bubble can exist because working generations are willing to save substantially for retirement. Building on this classic mechanism (Samuelson, 1958; Diamond, 1965), the chapter analyzes how changes in productivity growth affect the size of the bubble. It relates this question to the ecological transition, which may constrain the production process and thereby reduce productivity growth. The announcement of such a transition can constitute a critical moment for the revaluation of bubbles.

A key interest of studying rational bubbles is that they raise a broader question about extrapolation in economics. Distinguishing between fundamental value and a rational bubble requires taking a stance on future cash flows, discount rates, and asset demand. Yet the future is, by definition, unobserved, and so are, to a large extent, agents' expectations about it. It follows that the existence of a bubble cannot be definitively tested. However, assuming that an asset is priced at its fundamental value or that it contains a rational bubble amounts to deciding which patterns observed in the data should—or should not—be extrapolated into the future. To return to the initial example, if one extrapolates the primary deficits observed in Japan over recent decades into the future, Japanese public debt must be interpreted as containing a bubble. But its current valuation can be fully rationalized by fundamentals if one expects sufficiently large future primary surpluses.

The sources of asset demand that sustain rational bubbles and the question of extrapolation provide the two organizing themes of this introduction. It first briefly revisits the non-testability of rational bubbles and the general issue of extrapolation. It then presents the three chapters of the thesis. For each chapter, it examines the determinants of asset demand, the extrapolations these models entail, and the main results that follow. The introduction concludes by discussing how insights from rational bubble theory can inform policy.

1 Non-Testability of Rational Bubbles

The literature on rational bubbles has a frustrating feature: its central object cannot be directly identified in the data. A bubble is defined as the gap between an asset's observed market price and its unobserved fundamental value. Constructing the latter requires assumptions about future cash flows and discount rates, so any empirical test jointly evaluates the presence of a bubble and the valuation model used to estimate the fundamental. This is the joint-hypothesis problem familiar from tests of market efficiency, which likewise assess whether market prices deviate from fundamentals (Fama, 1970). It further implies that, even if the fundamental benchmark were correctly specified, an observed deviation would not reveal whether it reflected a rational bubble or departures from rational expectations.

For a time, however, an empirical debate reduced the plausibility of rational bubbles. Unable to test bubbles directly, Abel et al. (1989) examined a necessary condition for their existence in canonical models of the time. In these frameworks, such as Tirole (1985), the return on the bubble equals that on capital. Bubbles can arise only under dynamic inefficiency, that is, when the return on capital is below the economy's growth rate. Abel et al. (1989) concluded that major advanced economies were dynamically efficient, thereby ruling out rational bubbles. This conclusion was called into question roughly two decades later. Geerolf (2018) challenged the empirical evidence for dynamic efficiency, while new theories showed that it was not necessary for bubbles to arise. The relevant condition is instead that the return on the bubble lies below the economy's growth rate. This return may remain below that on capital for several reasons, such as liquidity services (Farhi and Tirole, 2012) or financial frictions (Martin and Ventura, 2012).

The resulting difficulty of confirming or rejecting rational bubbles explains why the literature has focused primarily on theory and, more recently, on quantitative analysis. This thesis follows the same pattern: all three chapters make theoretical contributions, while Chapter 2 also includes a quantitative exercise. Tellingly, one of the main recent advances on the bubble existence question is itself theoretical. Hirano and Toda (2025a) show that rational bubbles are *necessary* when long-run output growth exceeds dividend growth of some asset, which in turn exceeds the interest rate in the absence of bubble. Their result relies on an unbalanced-growth condition that requires output and dividends to grow at permanently different rates.

This illustrates a broader feature of rational-bubble models: they become relevant when the extrapolation of a stylized fact conflicts with fundamental valuation. Using the theorem of Hirano and Toda (2025a) to interpret a concrete macroeconomic context requires two elements. First, an asset must have historically exhibited a declining dividend-to-output ratio—as they argue is the case for land (Hirano and Toda, 2025b)—and, second, this decline must be assumed to persist. As they show, this second assumption is ultimately inconsistent with fundamental pricing. Similarly, when Japan's primary deficits are expected to be permanent, its public debt becomes viewed as bubbly.

How should one decide whether to interpret an asset price as containing a rational bubble or as being entirely fundamental, given that the two explanations are often observationally equivalent? A natural criterion is simplicity. A naive application of the parsimony principle might then lead one to systematically prefer fundamental valuation, simply because it is the more standard hypothesis. Yet, the mapping between the extrapolation of certain stylized facts and the presence of a bubble suggests a more nuanced interpretation. It shifts attention to two questions: which stylized facts are being extrapolated under each interpretation, and what are the implications of these alternative extrapolations? Once these questions are examined carefully, the simpler explanation may turn out to be the one that includes a rational bubble.

These two extrapolation-related questions guide the remainder of this introduction, which now turns to the presentation of the chapters of the dissertation.

2 Chapters 1 and 2: A Scrooge McDuck Theory of Wealth Dynamics

The first two chapters are presented jointly, as they share a common motivation and closely related conceptual frameworks. They study how the rise in top income inequality since the 1980s has shaped wealth accumulation. Since saving rates tend to increase with income (Dynan et al., 2004), a key extrapolation question is whether households at the top of the distribution persistently consume less than their income. These chapters depart from standard models, in which this gap is typically transitory, by introducing insatiable preferences for wealth. This specification allows for persistently positive net saving at the top, leading to qualitatively different implications, including the existence of a rational bubble.

Because the mechanism relies on preferences for wealth as a crucial source of asset demand, the following section first clarifies how such preferences should be interpreted. It then discusses the insatiable specification and finally presents its main implications.

How to Interpret Preferences for Wealth. Assuming that economic agents derive intrinsic utility from holding assets is useful for addressing a wide range of questions. In addition to matching high saving rates of the wealthiest households (Gaillard et al., 2023), they can rationalize several macroeconomic stylized facts at the aggregate level. They have been used to study the rise in indebtedness and the associated financial risks (Kumhof et al., 2015; Mian et al., 2021), the effects of monetary policy at the zero lower bound (Michaillat and Saez, 2021), and the increase in the wealth-to-GDP ratio (Elina and Huleux, 2023). The assumption is also plausible given the variety of motives that may lead individuals to derive direct utility from wealth—including status, power, and an intrinsic desire to accumulate—and for which anecdotal evidence can be found (Carroll, 2000).

An interpretive difficulty may nevertheless arise when it is unclear where utility from consumption ends and utility from wealth itself begins. Consider, for example, an agent who owns a valuable painting and displays it at home. Does the resulting utility come from ownership itself or from consuming the aesthetic services provided by viewing the painting? Under the latter interpretation, where does aesthetic enjoyment end and the social status associated with being an art collector begin? More fundamentally, why should these dimensions be treated as distinct sources of utility rather than as services incorporated into a broader measure of consumption?

Although the analysis developed in the main body of the thesis remains agnostic on these questions, a simple conceptual distinction can be proposed. Consider an asset that provides its owner with a good or service at a given point in time. If this good or service can be transferred to another agent without transferring ownership of the asset itself, the utility it generates can be classified as consumption utility accruing to its user. Conversely, if it cannot be transferred without also transferring ownership of the underlying asset, the resulting utility is viewed as utility from wealth. Here, the term "service" is understood broadly and encompasses any right associated with ownership.

This definition can be readily applied to the example of a house. At any point in time, a house gives its owner the possibility of living in it. This housing service can be transferred to another agent, either free of charge or in exchange for rent, without transferring ownership of the property. It therefore enters utility through preferences for consumption. By contrast, the ability to decide on renovations without seeking permission from another private party is inseparable from homeownership. To the extent that this ability raises an agent's utility, it does so through preferences for wealth. The same applies to the symbolic service associated with the status conferred by homeownership.

Consider next the case of a valuable painting, which illustrates potential complementarities between the two sources of utility. Suppose that an agent enjoys displaying it in their living room, especially when they can keep it there for as long as they wish. The aesthetic service derived from displaying the painting can be transferred without transferring ownership and enters utility through consumption. By contrast, the freedom to display it wherever and for as long as one chooses derives from ownership and enters utility through wealth. These two services may be complementary. In that case, the additional utility generated by their combination is fully attributed to utility from wealth, since it can be obtained only through ownership. For collectors, the resulting utility may be substantial.

The first advantage of this classification is that it makes explicit that wealth constitutes a bundle. Holding an asset combines two distinct elements: a current flow of utility from wealth and a claim on future cash flows. This feature is closely aligned with the way preferences for wealth are introduced in models, where asset holdings simultaneously provide direct utility and serve as a vehicle for transferring resources across time. It is precisely this bundle aspect that gives preferences for wealth dynamic implications that differ from those generated by consumption utility alone.

The second advantage is that this definition encompasses not only the standard motives commonly invoked to justify preferences for wealth—such as status or power—but also the broader set of benefits attached to ownership itself, as illustrated by the examples of housing and art. By attributing complementarities between consumption services and ownership rights to utility from wealth, it allows this source of utility to become particularly important at the top of the distribution. Preferences for wealth may therefore account for a substantial share of saving among the richest households.

Standard versus Insatiable Preferences for Wealth. Chapters 1 and 2 depart from conventional specifications of preferences for wealth by positing insatiable preferences. Under the standard formulation (e.g., Kumhof et al., 2015; Mian et al., 2021), the marginal utility of wealth declines more slowly than that of consumption. Yet, both marginal utilities converge to zero as their respective arguments diverge. Insatiable preferences strengthen this asymmetry by assuming that the marginal utility of holding wealth remains bounded away from zero, even at arbitrarily high levels of wealth. Utility from wealth is then asymptotically linear.

This departure from the standard specification is sufficiently parsimonious that it cannot be directly ruled out empirically. Standard and insatiable preferences differ only in their asymptotic implications, which are not themselves observable. In principle, one could test whether the marginal utility of wealth remains above a given positive threshold. In practice, however, such a test would require consumption data for households sufficiently far into the upper tail of the wealth distribution, precisely where such information is most limited. Yet the relevance of the assumption lies less in whether it can be tested directly than in the extrapolation it permits.

Its main direct implication is that households at the top of the wealth distribution may permanently consume less than their income. This possibility provides a different way of extrapolating the persistent accumulation observed among wealthy households. In the data, top-wealth households, taken as a group, have consumed less than they earned over recent decades. Standard models typically interpret this pattern as a transitional phenomenon: wealthy households save today in order to finance higher consumption in the future. The wealth distribution is then assumed to converge toward a stationary steady state—possibly only very slowly (see Gabaix et al., 2016).

Most models with standard preferences for wealth preserve this interpretation. An exception is Michau et al. (2025), in which the wealth of some agents at the top of the distribution may diverge. This divergence is accompanied by unbounded consumption. In general equilibrium, such arbitrarily large consumption can be sustained only by an arbitrarily small fraction of households. The top of the wealth distribution consists asymptotically of a vanishing fraction of households with diverging wealth and other households that decumulate. Taken as a group, they consume an amount equal to their income.

Insatiable preferences support a different extrapolation. They implicitly impose an upper bound on optimal consumption. When this bound binds asymptotically for some households, they optimally consume less than their income indefinitely: their wealth diverges while their consumption converges to a finite level. They are referred to as Scrooge McDuck agents. Under insatiable preferences for wealth, persistent net saving at the top can be interpreted as a permanent feature of the equilibrium rather than a long transition toward higher future consumption.

Insatiability provides a parsimonious way to extrapolate the accumulation of wealthy households observed in the data. As will be shown, this extrapolation helps account for the wealth dynamics observed in advanced economies.

Main Results. The main result of Chapters 1 and 2 is that persistent net saving at the top can sustain a diverging rational bubble. Scrooge McDuck agents hold a strictly positive surplus wealth, defined as the part of their wealth that is not used to finance any future consumption. This surplus wealth generates asset demand that pushes asset prices above their fundamental value. The associated bubble grows at the equilibrium rate of return, which exceeds the growth rate of the economy. It diverges relative to output and crowds out capital. Wealth and capital dynamics become disconnected: aggregate wealth diverges, while the capital-to-output ratio converges.

This mechanism is consistent with two key stylized facts: in advanced economies, wealth-to-output ratios have increased substantially since the 1980s, while capital-to-output ratios have remained comparatively stable. A quantitative exercise calibrated to the United States further suggests that the model can reproduce the joint dynamics of aggregate wealth and wealth inequality. Moreover, given the interpretation of preferences for wealth developed above, the extrapolation of saving behavior at the top implied by insatiable preferences is plausible. Conditional on accepting this extrapolation, the principle of parsimony favors the Scrooge McDuck theory, which accounts jointly for rising wealth, stable capital, and increasing wealth inequality, over a purely fundamental explanation.

In this environment, the distinction between bubbly and non-bubbly dynamics has important implications, notably for taxation. First, capital income and wealth taxes are not equivalent, because the bubble enters the wealth tax base but does not generate taxable capital income. Second, taxation can change the equilibrium itself. By lowering after-tax top incomes, capital income and wealth taxes may shift the economy from a bubbly to a non-bubbly equilibrium. This removes the crowding-out effect of the bubble and can raise capital accumulation. As a result, the standard trade-off between redistribution and efficiency is no longer automatic: some taxes can be both redistributive and growth-enhancing. Further implications for capital gains, the rate of return on wealth, and the dynamics of wealth inequality are discussed in the introductions and main analyses of Chapters 1 and 2.

3 Chapter 3: Rational Bubbles in the Ecological Transition

Chapter 3 studies a rational bubble sustained by retirement saving in an overlapping-generations economy. This framework is canonical in the theory of rational bubbles, following Samuelson (1958) and Diamond (1965). Agents receive labor income when young and save to finance consumption later in life. This retirement motive can generate a positive demand for assets even when their rate of return lies below the growth rate of the economy. In that case, a rational bubble can arise. It is bought by younger cohorts and sold by older cohorts, thereby transferring resources across generations. The bubble partly alleviates the incompleteness of OLG economies, where agents cannot trade directly with future, unborn cohorts.

The source of asset demand in Chapter 3 is less contentious than in Chapters 1 and 2. Assuming that working-age households will continue to save for retirement is a relatively safe extrapolation, but two extrapolation questions remain. One concerns the evolution of demographics and productivity growth, which shape saving demand and affect the size, or even the existence, of the bubble. The other concerns coordination. Conditional on the economy being dynamically inefficient in the absence of a bubble, standard OLG models remain silent about whether households coordinate on a bubble, and, if so, on which one. Chapter 3 abstracts from this indeterminacy by focusing on the unique equilibrium path that converges to a bubbly balanced-growth path.

The chapter uses this environment to ask how a productivity-growth slowdown affects the size of the rational bubble. The effect is ambiguous. On a bubbly balanced-growth path, the rate of return equals the growth rate of the economy. A permanent decline in productivity growth therefore lowers the equilibrium rate of return and raises normalized capital. This affects the bubble in two ways. First, a larger share of aggregate saving must be allocated to capital rather than to the bubble. Second, aggregate saving itself changes. A larger capital stock raises wages, which may increase saving. By contrast, the lower rate of return has an ambiguous effect on saving, depending in particular on the elasticity of intertemporal substitution. When this elasticity is high, the fall in the rate of return tends to reduce saving, making a decline in the bubble more likely.

This framework can be applied to the implications of the ecological transition for asset pricing. Environmental policy may constrain some production processes and reduce output growth relative to an unconstrained benchmark. In this sense, the announcement of such a transition can be understood as an unexpected negative productivity-growth shock. Since rational bubbles are forward-looking, this announcement immediately affects their valuation. This approach complements the stranded-asset literature, which studies how asset values fall when environmental policy reduces the profitability of brown activities. The ecological transition may strand not only productive assets but also part of the bubbly component of asset prices.

A similar policy trade-off arises for stranded bubbles and for standard stranded assets. When the ecological transition lowers the value of the bubble, delaying implementation reduces the initial valuation loss. It gives the economy more time to adjust before the growth slowdown materializes. However, delaying implementation also postpones the ecological transition itself. Policymakers face a trade-off between limiting financial losses at the announcement date and implementing the transition rapidly. A similar trade-off has been documented for assets with brown fundamentals (e.g. Van der Ploeg and Rezai, 2020), and Chapter 3 shows that it extends to rational bubbles.

4 Conclusion

This introduction highlights three reasons to study rational bubbles despite their lack of direct testability. First, they allow one to extrapolate stylized facts that fundamental pricing cannot accommodate, such as persistent net saving at the top in the Scrooge McDuck framework. Second, they show whether competing extrapolations have different implications and whether parsimony favors one over the other. Third, they reveal how sensitive policy conclusions are to the presence or absence of a bubble. Chapter 3 illustrates a case in which the trade-off between limiting asset-price losses and implementing the ecological transition rapidly is similar for stranded bubbles and standard stranded assets. By contrast, in Chapters 1 and 2, the policy implications of taxation differ sharply depending on whether the economy is bubbly or non-bubbly.

This raises a broader question: how should researchers communicate their extrapolation choices in public debate and policy discussions? Stating every implicit extrapolation is impossible and would make public communication unnecessarily technical. When a rational bubble emerges from a standard framework, such as an OLG environment or an Aiyagari model (Domeij and Ellingsen, 2018), it is likely unnecessary to emphasize why agents save in a way that can sustain a bubble. The key point may simply be that the existence of the bubble depends on coordination. Chapter 3 provides such a case: in the real world, an ambitious ecological transition could also affect how agents coordinate on a bubble. This possibility is not taken into account by the model, but it should be acknowledged in public discussions and potentially studied in complementary work.

However, when the extrapolation that sustains the bubble is less standard, it should be stated as transparently as possible. Communicating the policy implications of Chapters 1 and 2 requires making clear that the theory assumes that some agents at the top of the distribution seek to accumulate diverging wealth without ever consuming their entire income. Once this extrapolation is explicit, one can explain why it changes the implications of taxation. Without this step, the policy message would be difficult to understand: taxation would appear to face a standard trade-off between redistribution and growth in one (non-bubbly) economy, but not necessarily in another (bubbly). Stating the underlying extrapolation is also less technical than framing the issue in terms of rational bubbles, which requires defining fundamental value.

Bibliography

Andrew B Abel, N Gregory Mankiw, Lawrence H Summers, and Richard J Zeckhauser. Assessing dynamic efficiency: Theory and evidence. *The Review of Economic Studies*, 56(1):1–19, 1989.

Markus K Brunnermeier, Sebastian A Merkel, and Yuliy Sannikov. The fiscal theory of price level with a bubble. Working paper, National Bureau of Economic Research, 2023.

Christopher D Carroll. *Why Do the Rich Save So Much?* ed. by J. Slemrod, Harvard University Press, 2000.

Vasco M Carvalho, Alberto Martin, and Jaume Ventura. Understanding bubbly episodes. *American Economic Review*, 102(3):95–100, 2012.

Peter A Diamond. National debt in a neoclassical growth model. *The American Economic Review*, 55(5):1126–1150, 1965.

David Domeij and Tore Ellingsen. Rational bubbles and public debt policy: A quantitative analysis. *Journal of Monetary Economics*, 96:109–123, 2018.

Karen E Dynan, Jonathan Skinner, and Stephen P Zeldes. Do the rich save more? *Journal of Political Economy*, 112(2):397–444, 2004.

Eustache Elina and Raphaël Huleux. From labor income to wealth inequality in the u.s. Working paper, 2023.

Eugene F Fama. Efficient capital markets: A review of theory and empirical work. *The journal of Finance*, 25(2):383–417, 1970.

Emmanuel Farhi and Jean Tirole. Bubbly liquidity. *The Review of Economic Studies*, 79(2): 678–706, 2012.

Xavier Gabaix, Jean-Michel Lasry, Pierre-Louis Lions, and Benjamin Moll. The dynamics of inequality. *Econometrica*, 84(6):2071–2111, 2016.

Alexandre Gaillard, Christian Hellwig, Philipp Wangner, and Nicolas Werquin. Consumption, wealth, and income inequality: A tale of tails. Working paper, 2023.

François Geerolf. Reassessing dynamic efficiency. Technical report, 2018.

Tomohiro Hirano and Alexis Akira Toda. Bubble necessity theorem. *Journal of Political Economy*, 133(1):111–145, 2025a.

Tomohiro Hirano and Alexis Akira Toda. Unbalanced growth and land overvaluation. *Proceedings of the National Academy of Sciences*, 122(14):e2423295122, 2025b.

Zhengyang Jiang, Hanno Lustig, Stijn Van Nieuwerburgh, and Mindy Z Xiaolan. The us public debt valuation puzzle. *Econometrica*, 92(4):1309–1347, 2024.

Michael Kumhof, Romain Rancière, and Pablo Winant. Inequality, leverage, and crises. *American Economic Review*, 105(3):1217–1245, 2015.

Alberto Martin and Jaume Ventura. Economic growth with bubbles. *American Economic Review*, 102(6):3033–3058, 2012.

Atif Mian, Ludwig Straub, and Amir Sufi. Indebted demand. *The Quarterly Journal of Economics*, 136(4):2243–2307, 2021.

Pascal Michaillat and Emmanuel Saez. Resolving new keynesian anomalies with wealth in the utility function. *Review of Economics and Statistics*, 103(2):197–215, 2021.

Jean-Baptiste Michau, Yoshiyasu Ono, and Matthias Schlegl. The preference for wealth and inequality: Towards a piketty theory of wealth inequality. Working paper, 2025.

Paul A Samuelson. An exact consumption-loan model of interest with or without the social contrivance of money. *Journal of Political Economy*, 66(6):467–482, 1958.

Jean Tirole. Asset bubbles and overlapping generations. *Econometrica*, pages 1499–1528, 1985.

Frederick Van der Ploeg and Armon Rezai. Stranded assets in the transition to a carbon-free economy. *Annual review of resource economics*, 12(1):281–298, 2020.

Chapter 1

A Scrooge McDuck Theory of Wealth Dynamics: Motivation and Endowment Economy

Can the rise in the wealth-to-output ratio observed in advanced economies over recent decades be explained by an increase in top income inequality? Recent contributions that advance this interpretation struggle to account for the fact that the increase in wealth was driven by asset prices, rather than capital accumulation. Chapter 1 and 2 show that these stylized facts can be reconciled in the presence of *insatiable* preference for wealth. Chapter 1 shows in an endowment economy how such preferences induce a cap on optimal consumption. When it is binding, top-income agents accumulate diverging wealth over time while keeping their consumption bounded. This consumption-saving behavior supports the existence of a uniquely determined rational bubble, which grows at a rate that *exceeds* that of the economy.

I am particularly grateful to Stéphane Guibaud, Nicolas Cœurdacier and Jeanne Commault for their support throughout this project. I thank Alberto Martin and Jaume Ventura for the opportunity to visit Pompeu Fabra University in Spring 2024, which benefited this project. I also thank Mark Aguiar, Adrien Auclert, Vladimir Asriyan, Paul Bouscasse, Antoine Camous, Ali Choudhary, Naomi Cohen, Antoine Ferey, Axelle Ferriere, Lea Fricke, Aurélien Eyquem, Xavier Gabaix, Alexandre Gaillard, Jordi Galí, François Geerolf, Christian Hellwig, Priit Jeenas, Clara Martínez-Toledano, Isabelle Méjean, Eric Mengus, Jean-Baptiste Michau, Andrej Mijakovic, Guillaume Plantin, Stefan Pollinger, Giacomo Ponzetto, Victoria Vanasco, Xavier Ragot, Venance Riblier, Jean-Marc Robin, Matthias Schlegl, Moritz Schularick, David Thesmar, Dawid Trzeciakiewicz, Philipp Wangner, Nicolas Werquin and all participants at the Adres Job Market Conference 2026, CEPR Symposium 2025, Lausanne PhD Macro Conference 2025, Konstanz Doctoral Workshop on Quantitative Dynamic Economics 2025, MMF Annual Conference 2025, EEA Congress 2025, AFSE Annual Congress 2025, MMF PhD Conference 2025, Doctorissimes Conference 2025, BSE PhD Jamboree 2024, RGS Doctoral Conference 2024 as well as seminar attendees at CREI, CREST, the EconophysiX Lab, the Paris Macro Group, the Toulouse School of Economics, the University of Mannheim and Sciences Po for helpful comments and suggestions. All remaining errors are mine.

1 Introduction

Over recent decades, most advanced economies have seen a marked increase in both aggregate wealth and top income inequality. In G7 countries, the average wealth-to-output ratio rose from about three to five between 1980 and today (Figure 1.1a). Meanwhile, the pre-tax income share of the top 1% has been rising: in the United States, it almost doubled over the period from 11% to more than 20% (Figure 1.1b). Could this change in the income distribution explain the rise in aggregate wealth? Empirical evidence indicates that saving rates increase with permanent income (Dynan et al., 2004; Straub, 2019), hence greater income inequality has effectively shifted income from low- to high-saving households. This reallocation may have raised aggregate saving and aggregate wealth.

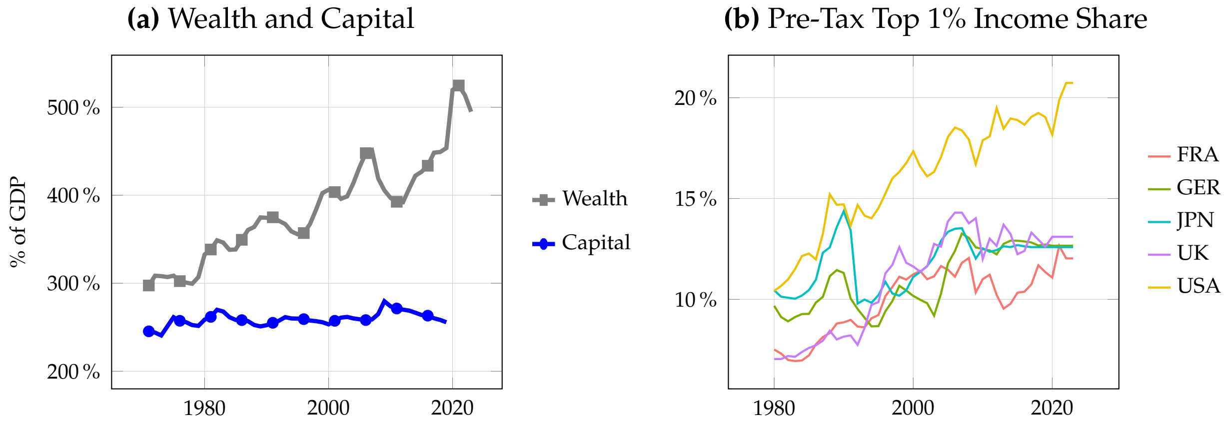
Yet, models that incorporate this mechanism (e.g., Hubmer et al., 2021; Elina and Huleux, 2023) fail to capture the nature of the observed rise in the wealth-to-output ratio. These models often consider capital, whose price is fixed, as the only available asset. By construction, any increase in wealth mechanically corresponds to an increase in capital. But more fundamentally, in such frameworks, an increase in aggregate savings lowers the interest rate, resulting in further capital investment. This stands in contrast to the data, which shows a stable capital-to-output ratio (Figure 1.1a). The observed rise in the wealth-to-output ratio was instead driven by higher asset prices, that is, by capital gains.

This chapter and the next posit insatiable preferences for wealth to generate a capital-gains-driven increase in the wealth-to-output ratio following a rise in income inequality. Preferences for wealth have proven effective in capturing the high saving rates observed at the top of the income distribution (Gaillard et al., 2023). In their standard specification (e.g., Kumhof et al., 2015, Mian et al., 2021), the marginal utility of wealth is assumed to decline more slowly than that of consumption. However, for both wealth and consumption, marginal utility tends to zero as the corresponding argument diverges. Under insatiable preferences for wealth, the asymmetry between utility derived from wealth and consumption is strengthened: the marginal utility of holding wealth remains bounded away from zero, even as wealth diverges.[1] This deviation is sufficiently parsimonious that it cannot be directly ruled out empirically.

The macroeconomic implications of these preferences are analysed within both an endowment (Chapter 1) and a production economy (Chapter 2). The endowment economy helps clarify analytically how asset pricing differs under standard and insatiable preferences for wealth. The production economy, in turn, connects the framework to

[1] Referring to these preferences for wealth as "insatiable" is perhaps slightly misleading, as it could suggest that standard ones admit a bliss point, which they do not. A more precise, albeit more cumbersome, label would be "asymptotically linear" preferences for wealth.

Figure 1.1: Wealth dynamics and income inequality for G7 countries



Notes: Panel (a) reports, as wealth, the aggregate national wealth of G7 countries, measured as the sum of public and private net wealth, relative to total G7 GDP, using data from the World Inequality Database. Similarly, it reports, as capital, the aggregate public and private capital stock of G7 countries as a share of their total GDP, obtained from the IMF Investment and Capital Stock Dataset. Panel (b) presents the pre-tax income share of the top 1 percent in the five main G7 economies, obtained from the World Inequality Database.

the motivating questions that require the presence of capital. Both setups include an infinitely lived, rent-generating asset in fixed supply. Both setups feature an infinitely lived, rent-generating asset in fixed supply, whose price is endogenously determined. Its presence ensures that the rate of return asymptotically exceeds the economy's growth rate.

In contrast to standard preferences for wealth, under insatiable ones, some agents may accumulate unbounded wealth while keeping consumption bounded relative to output. This occurs because insatiable preferences for wealth induce a lower bound on the marginal utility of saving. As the marginal utility of consumption must not fall below that of saving, this implicitly defines an upper bound on consumption. When income inequality is sufficiently high, some agents at the top of the income distribution asymptotically receive an income that exceeds this bound. As a result, they accumulate unbounded wealth relative to output over time while keeping their consumption bounded. They are referred to as Scrooge McDuck agents. By comparison, under standard preferences for wealth, there is no upper bound on optimal consumption and no Scrooge McDuck agents. An agent with diverging wealth would necessarily also have a diverging consumption, albeit typically at a slower rate.

The consumption-saving behavior of Scrooge McDuck agents supports the existence of a diverging rational bubble. The present value of Scrooge McDuck agents' future consumption financed from savings remains below the current value of their wealth. They therefore hold part of their wealth, referred to as surplus wealth, solely for the utility derived from holding it. The associated dividends are fully reinvested. This pushes asset prices above their fundamental value, defined as the present value of future dividends discounted at the equilibrium rate of return. The difference between the price of the rent-generating asset and its fundamental value constitutes a rational bubble, which coincides with surplus wealth. The return on the bubble arises entirely from the appreciation of its price. Being non-dominated, the bubble grows at the equilibrium rate of return, causing its value to diverge.

As Chapter 2 will later show, a rational bubble that grows faster than the economy generate a wedge between wealth and capital dynamics. It is thus the central asset-pricing mechanism through which the model can account for the motivating stylized facts.

Related Literature. Chapter 1 and 2 build on multiple empirical studies documenting, across advanced economies, rising income and wealth inequality (Katz and Murphy, 1992, Piketty and Saez, 2003, Saez and Zucman, 2016; Batty et al., 2019; Chancel et al., 2022; Smith et al., 2023, Blanchet and Martínez-Toledano, 2023), an increase in aggregate wealth driven by capital gains (Piketty and Zucman, 2014, Baselgia and Martínez, 2025), and a declining trend in investment (Gutiérrez and Philippon, 2017). It theoretically contributes to three strands of the literature.

First, motivated by empirical evidence that saving rates increase with income and wealth (Carroll, 2000; Dynan et al., 2004; Straub, 2019; Fagereng et al., 2021) and that aggregate savings are primarily driven by the top of the distribution (Mian et al., 2020; Bauluz et al., 2022), a growing theoretical and quantitative literature highlights the importance of non-homothetic preferences for wealth in explaining wealth inequality and the wealth-to-output ratio (Carroll, 2000; De Nardi, 2004; Kumhof et al., 2015; De Nardi and Yang, 2016; Benhabib et al., 2019; Mian et al., 2021; Elina and Huleux, 2023; Gaillard et al., 2023; Michau et al., 2025). The present framework departs from this literature by positing insatiable preferences for wealth and shows how they alter the predictions for both aggregate and distributional wealth dynamics relative to standard preferences.

Secondly, this work contributes to the rational bubble literature. Previous research has established that rational bubbles can exist in dynamically inefficient economies (Samuelson, 1958; Diamond, 1965; Tirole, 1985; Michau et al., 2023) or under financial frictions (Farhi and Tirole, 2012; Martin and Ventura, 2012; Reis, 2021). In both cases, the rate of return on bubbles is below the economy's growth rate, leading to an asymptotically stationary bubble-to-output ratio. Few studies have explored frameworks with a diverging bubble-to-output ratio, with notable exceptions including Ono (1994) and Kamihigashi (2008). The present analysis departs from these frameworks, which also feature insatiable preferences for liquidity or wealth, by introducing heterogeneous agents that render the equilibrium uniquely determined.

Section 2 describes the model environment. Section 3 introduces some definitions useful for the analysis. Section 4 describes the two possible types of equilibrium bubbly or non-bubbly. Finally, Section 5 characterize under which parameters is the equilibrium bubbly or not.

2 Model Environment

Insatiable preferences for wealth are embedded in a simple two-agents asset pricing model à la Lucas (1978). The economy is deterministic, with discrete time running from $t = 0$ to ∞. The endowment comes from the single unit of Lucas tree, which delivers one unit of the consumption good each period and is priced at q_t at time t after dividend payment. The rate of return of the Lucas tree is defined as:

$$R_{t+1} \equiv \frac{1 + q_{t+1}}{q_t} \tag{1.1}$$

Households. There is a unit mass of infinitely-lived households, divided into two types $i \in \{1, 2\}$, each representing a strictly positive share λ^i of the total population. Agents of type i hold ℓ_t^i units of the Lucas tree at the beginning of period t and consume c_t^i at time t. Agents differ only in their initial endowment of Lucas tree units, with type 1 agents holding the higher initial endowment: $\ell_0^1 \geq \ell_0^2 > 0$. The wealth of a household of type i at the end of period t is defined as $a_{t+1}^i \equiv q_t \ell_{t+1}^i$, and its budget constraint can be written as:

$$c_t^i + a_{t+1}^i = R_t a_t^i. \tag{1.2}$$

Both types of agents share identical preferences. They derive utility each period from consumption and from holding wealth. Agents discount the future at rate $\beta \in [0, 1)$ and maximize intertemporal utility:

$$\sum_{t=0}^{\infty} \beta^t U(c_t^i, a_{t+1}^i) \qquad \text{with} \qquad U(c, a) = u(c) + v(a). \tag{1.3}$$

Utility from consumption is represented by the function $u(c)$, which satisfies the Inada conditions. Preference for wealth is captured by $v(a)$, assumed to be twice continuously differentiable, increasing, and concave. The model remains agnostic about the micro-foundations of the utility derived from wealth, which may arise from bequest motives, the pursuit of power or social status, or an intrinsic desire to accumulate wealth, as discussed, for instance, by Zou (1994) and Carroll (2000).

The solution to the optimization problem for type i households is characterized by the following Euler equation and the transversality condition:[2]

$$u'(c_t^i) = \beta R_{t+1} u'(c_{t+1}^i) + v'(a_{t+1}^i), \tag{1.4}$$

$$\lim_{t \to \infty} \beta^t [u'(c_t^i) - v'(a_{t+1}^i)] a_t^i = 0. \tag{1.5}$$

Wealth-Dependent Consumption-Saving Behavior. In order to match the empirical evidence that saving rates increase with permanent income (Dynan et al., 2004; Straub, 2019), this paper follows the literature on preferences for wealth by assuming that the utility-from-wealth component, $v(a)$, exhibits lower curvature than the utility-from-consumption component, $u(c)$. Formalizing this assumption requires first defining the coefficients of risk aversion for consumption and wealth, denoted respectively by $\theta(c)$ and $\eta(a)$.

$$\theta(c) \equiv -\frac{c\, u''(c)}{u'(c)}, \qquad \eta(a) \equiv -\frac{a\, v''(a)}{v'(a)}. \tag{1.6}$$

The following assumption induces non-homothetic (i.e., wealth-dependent) consumption-saving behavior, whereby wealthier agents save a larger share of their income.

Assumption A. *For any $\{c, a\} \in \mathbb{R}_+^2$, the utility from wealth exhibits lower curvature than the utility from consumption. That is,*

$$\eta(a) < \theta(c). \tag{1.7}$$

Under Assumption A, wealth is a luxury. Intuitively, as the marginal utility of wealth declines more slowly than that of consumption, high-wealth agents value additional wealth relative to additional consumption more than low-wealth agents do. In equilibrium, this leads type 1 agents to save more than type 2 agents, resulting in type 1 agents purchasing Lucas tree units from type 2 agents in each period.

Standard vs Insatiable Preferences for Wealth. The insatiable preferences for wealth studied in this paper depart from standard preferences by imposing a strictly positive lower bound on the marginal utility of holding wealth. As it is concave and strictly increasing, the utility-from-wealth component $v(a)$ admits a well-defined limit for its marginal utility as wealth tends to infinity. Let κ denote this limit:

$$\lim_{a \to \infty} v'(a) = \kappa. \tag{1.8}$$

[2]A proof of the necessity of the transversality condition is provided in Appendix A.1

Preferences for wealth are said to be *insatiable* whenever $\kappa > 0$. In contrast, the case $\kappa = 0$ is referred to as *standard* preferences for wealth, as it corresponds to the usual specification in the wealth-in-utility literature. The parameter κ is taken to be small, so that insatiable preferences for wealth represent a parsimonious deviation from the standard specification.[3] In line with this interpretation, and to rule out situations in which all agents would strictly prefer saving over consumption, the following technical assumption is imposed: $\kappa < \frac{1-\beta}{\beta} u'(1)$.

The insatiability assumption cannot be ruled out empirically. Standard and insatiable preferences for wealth differ only in their asymptotic implications, which are, by definition, not observable. It is therefore impossible to distinguish between $\kappa = 0$ and a strictly positive but arbitrarily small value of κ. In principle, one could test whether κ exceeds a given strictly positive constant by exploiting saving rates across wealth levels at the top of the distribution. In practice, however, such a test is ultimately infeasible due to the lack of consumption data for the relevant high-wealth households. The testability of insatiable preferences for wealth is discussed further in Appendix B. Since insatiable and standard preferences for wealth cannot be distinguished using micro-data, this paper proposes to contrast their implications, particularly for asset pricing, in order to assess which specification is more consistent with observed macroeconomic stylized facts.

Market Clearing Conditions. In each period, there is one unit of the Lucas tree and one unit of the non-storable consumption good. The market clearing condition for assets is expressed as:

$$\lambda^1 \ell_t^1 + \lambda^2 \ell_t^2 = 1, \tag{1.9}$$

and the goods market clearing condition is given by:

$$\lambda^1 c_t^1 + \lambda^2 c_t^2 = 1. \tag{1.10}$$

Equilibrium. Given the initial endowment ℓ_0^1, an equilibrium $\{c_t^1, \ell_t^1, q_t\}_{t=0}^{\infty}$ is characterized by the solutions to the household optimization problems (1.4, 1.5) and the asset market clearing condition (1.9). Goods market clearing (1.10) holds by Walras's Law.

[3] Throughout the analysis, all agents are assumed to share the same preferences, so that the results are not driven by preference heterogeneity. Importantly, the key mechanisms would still hold under heterogeneous preferences, as long as *some* agents—technically, even a single agent—at the top of the wealth distribution exhibit insatiable preferences for wealth.

3 Definitions of Key Concepts

Four key concepts are now introduced to structure the theoretical analysis: (i) the upper bound on consumption implied by insatiable preferences for wealth; (ii) the behavior of *Scrooge McDuck* agents, who accumulate unbounded wealth while keeping consumption bounded; (iii) the notion of *surplus wealth*, defined as the portion of agents' wealth that does not finance any future consumption; and (iv) the rational bubble that emerges in the price of the Lucas tree as a direct consequence of a positive surplus wealth.

Upper Bound on Consumption. Unlike standard preferences for wealth, insatiable ones implicitly define an upper bound on optimal consumption, $\bar{c}_t$.

Definition 1. *Under insatiable preferences for wealth, the upper bound on optimal consumption, $\bar{c}_t$, is defined as:*

$$\bar{c}_t \equiv u'^{-1}\left(\sum_{s=0}^{\infty} \beta^s \left[\prod_{j=1}^{s} R_{t+j}\right] \kappa\right). \tag{1.11}$$

Regardless of agent i's wealth, optimal consumption is always such that $c_t^i < \bar{c}_t$. The lower bound on the marginal utility of holding wealth, κ, induces a corresponding lower bound on the marginal utility of saving, as defined by the inner bracket in Equation 1.11. This lower bound captures the utility derived from saving an additional unit of wealth and reinvesting all associated returns indefinitely, assuming the marginal utility of wealth is fixed to κ. The upper bound $\bar{c}_t$ corresponds to the level of consumption at which the marginal utility of consumption equals the lower bound on the marginal utility of holding wealth. It follows that it is never optimal for any agent i to consume $c_t^i \geq \bar{c}_t$.

As will become clear in the analysis, the fundamental difference between insatiable and standard preferences for wealth stems from the existence of this upper bound on consumption. From Equation 1.11, it can be seen that $\bar{c}_t$ diverges in the limit as κ tends to zero. There is therefore no upper bound on consumption under standard preferences for wealth. This paper focuses on insatiable preferences for wealth as the source of the upper bound on consumption, but similar results could arise from alternative preference specifications, as long as they imply an upper bound on optimal consumption. For instance, one could directly impose an upper bound in the utility-from-consumption component $u(c)$, or alternatively, assume preferences over net saving flows.

Scrooge McDuck Agents. Given the upper bound on consumption, some agents may optimally choose to accumulate unbounded wealth over time while keeping consumption bounded; these agents are referred to as *Scrooge McDuck* agents.

Definition 2. *An agent i is labeled as a Scrooge McDuck agent, if and only if:*

$$\lim_{t\to\infty} a_t^i = \infty, \quad \textit{and} \quad \exists\, M \in \mathbb{R}_+ \textit{ such that } c_t^i < M\ \forall t. \tag{1.12}$$

The existence of Scrooge McDuck agents is closely tied to the presence of an upper bound on consumption, and their consumption asymptotically approaches $\bar{c}_t$. As their wealth diverges, the marginal utility of holding wealth converges to κ, and their marginal utility of saving approaches $\sum_{s=0}^{\infty} \beta^s \left[\prod_{j=1}^{s} R_{t+j}\right] \kappa$. By the definition of $\bar{c}_t$ in Equation 1.11, their asymptotic consumption coincides with the upper bound on optimal consumption. In contrast, under standard preferences for wealth, it is never optimal for an agent to behave as a Scrooge McDuck: diverging wealth necessarily implies diverging consumption, even if consumption may diverge at a slower rate (see, for instance, Michau et al., 2025).

Surplus Wealth. Scrooge McDuck agents hold a portion of their wealth purely for its own sake, referred to as surplus wealth.

Definition 3. *The surplus wealth of an agent of type i at the end of period t, denoted s_{t+1}^i, is defined as the portion of wealth that is not used to finance any future consumption:*

$$s_{t+1}^i \equiv a_{t+1}^i - \sum_{j=1}^{\infty} \frac{c_{t+j}^i}{\prod_{s=1}^{j} R_{t+s}}. \tag{1.13}$$

In this endowment economy, all consumption is financed out of the Lucas tree. As a result, surplus wealth corresponds to total wealth minus the present value of future consumption. All dividends associated with surplus wealth are reinvested rather than consumed, so that surplus wealth grows at rate R_{t+1} from period t to $t+1$.

Rational Bubble. The presence of Scrooge McDuck agents pushes the price of the Lucas tree above its fundamental value and gives rise to a rational bubble, whose size coincides with that of the surplus wealth.

Definition 4. *The asset price q_t is decomposed into a fundamental value f_t, defined as the present value of future dividends discounted at the rate of return, and a bubble component b_t.*

$$f_t \equiv \sum_{j=1}^{\infty} \frac{1}{\prod_{s=1}^{j} R_{t+s}}, \qquad b_t = \lim_{T\to\infty} \frac{q_T}{\prod_{s=1}^{T} R_{t+s}} \qquad \textit{with} \quad q_t = f_t + b_t. \tag{1.14}$$

The fundamental value derives from both the stream of dividends and the utility of holding it. The latter is reflected in the rate of return R_{t+s}, which is pushed down by the preference for wealth. In contrast, the value of the rational bubble arises solely from the utility of holding it. Since agents derive direct utility from it, one might interpret the bubble as a particular form of fundamental. Nevertheless, consistently with the literature (e.g., Michau et al., 2025), it is referred to as a rational bubble for two reasons. First, its current value corresponds to its discounted asymptotic value, reflecting the fact that agents value the bubble only because they anticipate future demand from other agents. Second, if multiple infinitely-lived assets were introduced, the model would leave indeterminate which asset the bubble is attached to.

The following proposition establishes that the rational bubble is the asset pricing counterpart of surplus wealth.

Proposition 1. *The rational bubble coincides with the aggregate surplus wealth:*

$$b_t = \sum_i \lambda^i s^i_{t+1}. \tag{1.15}$$

Proposition 1 follows directly from previous definitions of the surplus wealth (1.13) and the bubble (1.14). Intuitively, in the considered frictionless model, a rational bubble that does not finance consumption can exist only if some wealth is held without a consumption motive. Like surplus wealth, the rational bubble grows at rate R_{t+1} from period t to $t+1$, in order to deliver the same rate of return as fundamental wealth.

4 Bubbly and Non-bubbly Equilibria

The two possible equilibrium types are now described: those featuring a strictly positive bubble and those without a bubble. Discussion of the parameter conditions under which each equilibrium type arises is postponed until after the description of their asymptotic properties. Since the economy is deterministic, all variables either converge or diverge. For notational convenience, asymptotic values are denoted without time subscripts.

Asymptotic Lucas Tree Holdings. The non-homotheticity of preferences implies that type 1 agents asymptotically hold the entire Lucas tree in every equilibrium.

Proposition 2. *Under Assumption A, type 1 agents hold the entire Lucas tree asymptotically:*

$$\lim_{t\to\infty} \ell^1_t = \frac{1}{\lambda^1}, \qquad \lim_{t\to\infty} \ell^2_t = 0. \tag{1.16}$$

Assumption A implies that saving rates increase strictly with wealth. Since type 1 agents begin with greater initial wealth, it follows by induction that they exhibit higher saving rates and greater holdings of the Lucas tree than type 2 agents in every period.

Their relatively stronger marginal preference for wealth over consumption makes them effectively more patient than type 2 agents. Formally, Proposition 2 can be proved by contradiction. From the combination of the two Euler equations (1.4), no equilibrium exists in which both type 1 and type 2 agents asymptotically consume exactly their endowment when $\ell^1 > \ell^2 > 0$.

Non-bubbly Equilibrium. A non-bubbly equilibrium is asymptotically characterized by an upper bound on optimal consumption that exceeds the dividend flow of type 1 agents, inducing them to fully consume their endowment.

Proposition 3. *Asymptotically, if an equilibrium features an upper bound on optimal consumption that is above the endowment received by type* 1 *agents,*

$$\lim_{t\to\infty} \bar{c}_t \geq \frac{1}{\lambda^1}, \tag{1.17}$$

then it is a non-bubbly equilibrium with $b_t = 0\ \forall t$. The asymptotic behavior of the economy is characterized as follows.

(i) *Type 1 agents consume their entire endowment and hold constant wealth:*

$$\lim_{t\to\infty} c_t^1 = \frac{1}{\lambda^1}, \qquad \lim_{t\to\infty} a_t^1 = \frac{1/\lambda^1}{R-1}. \tag{1.18}$$

(ii) *Type 2 agents' consumption and wealth converge to zero:*

$$\lim_{t\to\infty} c_t^2 = \lim_{t\to\infty} a_t^2 = 0. \tag{1.19}$$

(iii) *The price of the Lucas tree equals its fundamental value:*

$$\lim_{t\to\infty} q_t = \lim_{t\to\infty} f_t = \frac{1}{R-1} \tag{1.20}$$

In a non-bubbly equilibrium, all variables converge, and the economy approaches a steady state. Asymptotically, type 1 agents hold the entire stock of the Lucas tree, leaving no additional units to purchase from type 2 agents. As a result, they consume their full endowment, and their wealth converges to the level required to sustain this consumption. They hold no surplus wealth, and no bubble arises in equilibrium, consistent with Proposition 1. Conversely, type 2 agents, whose consumption declines strictly during the transition to the steady state, converge asymptotically to zero consumption and zero wealth.

Bubbly Equilibrium. A bubbly equilibrium is asymptotically characterized by an upper bound on optimal consumption that is below the dividend flow of type 1 agents, leading them to hold a diverging surplus wealth that supports a bubble.

Proposition 4. *Asymptotically, if an equilibrium features an upper bound on optimal consumption that is strictly below the endowment received by type* 1 *agents,*

$$\lim_{t\to\infty} \bar{c}_t < \frac{1}{\lambda^1}, \tag{1.21}$$

then it is a bubbly equilibrium with $b_t > 0\ \forall t$. *The asymptotic behavior of the economy is characterized as follows.*

(i) *Type 1 agents consume at the upper bound on optimal consumption and hold diverging wealth:*

$$\lim_{t\to\infty} c_t^1 = \bar{c}, \qquad \lim_{t\to\infty} a_t^1 = \infty. \tag{1.22}$$

(ii) *Type 2 agents' consumption and wealth converge to positive values:*

$$\lim_{t\to\infty} c_t^2 = c^2 \quad \textit{with} \quad c^2 \equiv \frac{1-\lambda^1\bar{c}}{\lambda^2}, \qquad \lim_{t\to\infty} a_t^2 = \frac{c^2}{R-1}. \tag{1.23}$$

(iii) *The Lucas tree price diverges over time, reflecting the divergence of its bubbly component:*

$$\lim_{t\to\infty} f_t = \frac{1}{R-1}, \qquad \lim_{t\to\infty} b_t = \infty. \tag{1.24}$$

Asymptotically, type 1 agents receive the entire endowment but choose not to consume all of it, as doing so would exceed the upper bound on consumption. They are Scrooge McDuck agents, and hold a strictly positive surplus wealth. They are willing to exchange the difference between their endowment, $1/\lambda^1$, and their consumption, $\bar{c}$, for Lucas tree units regardless of the price, q_t. This non-vanishing demand for new Lucas tree units by type-1 agents, despite their vanishing supply from type-2 agents, drives the Lucas tree price to diverge. The divergence is fueled by a bubbly component in its valuation which, in line with Proposition 1, coincides with the aggregate surplus wealth held by type 1 agents.

The presence of type 2 agents, who are not Scrooge McDuck, ensures that at most one bubbly equilibrium can exist. Type 2 agents' asymptotic consumption level $\lambda^2 c^2$ corresponds to the endowment not consumed by type 1. They hold a vanishing share of the Lucas tree, whose price diverges. The equilibrium price of the Lucas tree, and thus the size of the bubble, is pinned down by the requirement that type 2 agents retain just enough wealth to sustain their asymptotic consumption c^2. If q_t were above its equilibrium value, type 2 agents would demand more consumption, violating the goods market clearing condition (1.10); if it were below, some goods would remain unconsumed, which is inconsistent with strictly positive marginal utility of consumption for both agent types. This contrasts with representative-agent models featuring insatiable preferences for wealth, where a marginal propensity to save of one may lead to equilibrium indeterminacy: additional bubble units can be fully saved without affecting consumption (e.g., Kamihigashi, 2008).

Rate of Return. The following proposition compares the asymptotic equilibrium rate of return between the two types of equilibrium.

Proposition 5. *The asymptotic equilibrium rate of return in any bubbly equilibrium is strictly higher than the asymptotic rate of return in any non-bubbly equilibrium.*

Proposition 5 follows directly from the rewritten Euler equation (1.4) for type 1 agents:

$$\lim_{t\to\infty} \frac{v'(a_t^1)}{u'(c_t^1)} = 1 - \beta R. \tag{1.25}$$

In any bubbly equilibrium, type 1 agents consume less than $1/\lambda^1$, which is their consumption level in any non-bubbly equilibrium. Moreover, they hold asymptotically more wealth in the bubbly case, as their wealth diverges. As a result, the left-hand side of Equation 1.38 is necessarily lower in the presence of a bubble than in its absence, implying that the rate of return must be higher under a bubbly equilibrium than under a non-bubbly one.

The relationship between the equilibrium rate of return and the existence of such bubbles differs fundamentally from that in standard rational bubble models. Rational bubbles that arise under standard preferences for wealth (Michau et al., 2025), as well as those emerging in overlapping generations models (e.g., Samuelson, 1958, Tirole, 1985) or in the presence of financial frictions (e.g., Martin and Ventura, 2012), require the bubble's rate of return to be *sufficiently low* to lie below the growth rate of the economy. This condition ensures that the bubble-to-output ratio remains bounded, in settings where agents do not optimally choose to accumulate unbounded wealth. In contrast, in the present model, the rate of return exceeds the (zero) growth rate, with $R \in (1, 1/\beta)$ in every equilibrium.[4] Still, in a bubbly equilibrium, the rational bubble exists and diverges over time because holding a diverging wealth does not violate the household optimization problem of Scrooge McDuck agents. Furthermore, the bubble exists only because the rate of return is *sufficiently high* to lower the upper bound on optimal consumption (1.11) enough to allow for the existence of Scrooge McDuck agents.

5 Equilibria Characterization

The existence conditions for bubbly and non-bubbly equilibria are now characterized. If the insatiable component of preferences for wealth is sufficiently strong, the

[4]This follows directly from the definition of R, since the price of the Lucas tree, q_t either converges to a positive constant or diverges.

equilibrium is unique and bubbly. Otherwise, whenever κ is strictly positive, both a non-bubbly and a bubbly equilibrium coexist.

Proposition 6. *The existence and uniqueness of equilibrium depend on the strength of the insatiable component in preferences for wealth, as captured by κ.*

- ***(i)*** *If $\kappa = 0$, the equilibrium is unique and non-bubbly.*
- ***(ii)*** *If $0 < \frac{\kappa}{1-\beta} \leq u'\left(\frac{1}{\lambda^1}\right)$, two equilibria exist, one bubbly and one non-bubbly, and equilibrium selection is indeterminate.*
- ***(iii)*** *If $u'\left(\frac{1}{\lambda^1}\right) < \frac{\kappa}{1-\beta}$, the equilibrium is unique and bubbly.*

Under standard preferences for wealth (i), there is no upper bound on optimal consumption, and type 1 agents asymptotically choose to consume their entire endowment, resulting in a unique non-bubbly equilibrium. Understanding cases (ii) and (iii) requires considering the lower bound on the marginal utility of savings, $\sum_{s=0}^{\infty} \beta^s \left(\prod_{j=1}^{s} R_{t+j}\right) \kappa$. Since $R \in (1, 1/\beta)$, this expression attains its lowest asymptotic value when R is arbitrarily close to 1, in which case it equals $\frac{\kappa}{1-\beta}$. In case (iii), the marginal utility associated with consuming the full asymptotic endowment of a type 1 agent, $u'(1/\lambda^1)$, lies below this lower bound. The upper bound on optimal consumption must therefore be, asymptotically, strictly below the endowment of type 1 agents. It follows from Proposition 4 that the resulting equilibrium is necessarily bubbly.

In case (ii), an additional non-bubbly equilibrium exists, characterized by a lower rate of return and a higher upper bound on consumption than the bubbly equilibrium. For this class of parametrizations, in which the two equilibria coexist, equilibrium selection is self-fulfilling. If agents anticipate the high rate of return of the bubbly equilibrium, the upper bound on optimal consumption is low, leading type 1 agents to be Scrooge McDuck. Their surplus wealth results in a rational bubble, which in turn raises the rate of return through its continuous appreciation, thereby validating the initial expectation. Conversely, if agents anticipate the low rate of return associated with the non-bubbly equilibrium, the upper bound on optimal consumption is high, type 1 agents do not behave as Scrooge McDuck agents, and no rational bubble arises.

The existence of multiple equilibria is a strong theoretical result, but its real-world relevance and robustness remain open to discussion. The multiplicity arises from the fact that, under insatiable preferences for wealth, the asymptotic consumption of type 1 agents is particularly responsive to the rate of return. This is evident from the Definition 1 of the upper bound on optimal consumption, which coincides with c^1 in the bubbly case. Empirical evidence is insufficient to assess the plausibility of such a high elasticity of consumption with respect to the rate of return at the top of the income and wealth distribution. The prediction of multiple equilibria may simply reflect that insatiable preferences for wealth constitute an overly stylized preference specification.

The high elasticity and the associated multiplicity of equilibria is not found in alternative preference specifications that impose an upper bound on optimal consumption, such as a satiation point in $u(c)$ or preferences defined over new net savings. Consequently, the multiplicity of equilibria is not central to the analysis.

6 Conclusion

This simple endowment economy shows that insatiable preferences for wealth differ from standard ones by defining an upper bound on optimal consumption. Whenever this upper bound is asymptotically binding for high-wealth agents, these agents behave as Scrooge McDuck: they accumulate an unbounded surplus wealth, supporting the existence of a rational bubble.

Bibliography

Enea Baselgia and Isabel Z Martínez. Wealth-income ratios in free market capitalism: Switzerland, 1900–2020. *Review of Economics and Statistics*, pages 1–21, 2025.

Michael Batty, Jesse Bricker, Joseph Briggs, Elizabeth Holmquist, Susan Hume McIntosh, Kevin B Moore, Eric Reed Nielsen, Sarah Reber, Molly Shatto, Kamila Sommer, et al. Introducing the distributional financial accounts of the united states. Finance and economics discussion series, 2019.

Luis Bauluz, Filip Novokmet, and Moritz Schularick. The anatomy of the global saving glut. Cepr discussion paper no. dp17215, 2022.

Jess Benhabib, Alberto Bisin, and Mi Luo. Wealth distribution and social mobility in the us: A quantitative approach. *American Economic Review*, 109(5):1623–1647, 2019.

Thomas Blanchet and Clara Martínez-Toledano. Wealth inequality dynamics in europe and the united states: Understanding the determinants. *Journal of Monetary Economics*, 133:25–43, 2023.

Christopher D Carroll. *Why Do the Rich Save So Much?* ed. by J. Slemrod, Harvard University Press, 2000.

Lucas Chancel, Thomas Piketty, Emmanuel Saez, and Gabriel Zucman. *World inequality report 2022*. Harvard University Press, 2022.

Mariacristina De Nardi. Wealth inequality and intergenerational links. *The Review of Economic Studies*, 71(3):743–768, 2004.

Mariacristina De Nardi and Fang Yang. Wealth inequality, family background, and estate taxation. *Journal of Monetary Economics*, 77:130–145, 2016.

Peter A Diamond. National debt in a neoclassical growth model. *The American Economic Review*, 55(5):1126–1150, 1965.

Karen E Dynan, Jonathan Skinner, and Stephen P Zeldes. Do the rich save more? *Journal of Political Economy*, 112(2):397–444, 2004.

Eustache Elina and Raphaël Huleux. From labor income to wealth inequality in the u.s. Working paper, 2023.

Andreas Fagereng, Martin Blomhoff Holm, Benjamin Moll, and Gisle Natvik. Saving behavior across the wealth distribution: The importance of capital gains. Working paper, 2021.

Emmanuel Farhi and Jean Tirole. Bubbly liquidity. *The Review of Economic Studies*, 79(2): 678–706, 2012.

Alexandre Gaillard, Christian Hellwig, Philipp Wangner, and Nicolas Werquin. Consumption, wealth, and income inequality: A tale of tails. Working paper, 2023.

Germán Gutiérrez and Thomas Philippon. Investmentless growth: An empirical investigation. *Brookings Papers on Economic Activity*, 2017(2):89–190, 2017.

Joachim Hubmer, Per Krusell, and Anthony A Smith Jr. Sources of us wealth inequality: Past, present, and future. *NBER Macroeconomics Annual*, 35(1):391–455, 2021.

Takashi Kamihigashi. A simple proof of the necessity of the transversality condition. *Economic Theory*, 20:427–433, 2002.

Takashi Kamihigashi. The spirit of capitalism, stock market bubbles and output fluctuations. *International Journal of Economic Theory*, 4(1):3–28, 2008.

Lawrence F Katz and Kevin M Murphy. Changes in relative wages, 1963–1987: supply and demand factors. *The Quarterly Journal of Economics*, 107(1):35–78, 1992.

Michael Kumhof, Romain Rancière, and Pablo Winant. Inequality, leverage, and crises. *American Economic Review*, 105(3):1217–1245, 2015.

Robert E Lucas. Asset prices in an exchange economy. *Econometrica*, pages 1429–1445, 1978.

Alberto Martin and Jaume Ventura. Economic growth with bubbles. *American Economic Review*, 102(6):3033–3058, 2012.

Atif Mian, Ludwig Straub, and Amir Sufi. The saving glut of the rich. Working paper, National Bureau of Economic Research, 2020.

Atif Mian, Ludwig Straub, and Amir Sufi. Indebted demand. *The Quarterly Journal of Economics*, 136(4):2243–2307, 2021.

Jean-Baptiste Michau, Yoshiyasu Ono, and Matthias Schlegl. Wealth preference and rational bubbles. *European Economic Review*, page 104496, 2023.

Jean-Baptiste Michau, Yoshiyasu Ono, and Matthias Schlegl. The preference for wealth and inequality: Towards a piketty theory of wealth inequality. Working paper, 2025.

Yoshiyasu Ono. *Money, interest, and stagnation: Dynamic theory and Keynes's economics*, chapter 11. Oxford University Press, USA, 1994.

Thomas Piketty and Emmanuel Saez. Income inequality in the united states, 1913–1998. *The Quarterly Journal of Economics*, 118(1):1–41, 2003.

Thomas Piketty and Gabriel Zucman. Capital is back: Wealth-income ratios in rich countries 1700–2010. *The Quarterly Journal of Economics*, 129(3):1255–1310, 2014.

Ricardo Reis. The constraint on public debt when r< g but g< m. CEPR Discussion Papers 15950, 2021.

Emmanuel Saez and Gabriel Zucman. Wealth inequality in the united states since 1913: Evidence from capitalized income tax data. *The Quarterly Journal of Economics*, 131(2): 519–578, 2016.

Paul A Samuelson. An exact consumption-loan model of interest with or without the social contrivance of money. *Journal of Political Economy*, 66(6):467–482, 1958.

Matthew Smith, Owen Zidar, and Eric Zwick. Top wealth in america: New estimates under heterogeneous returns. *The Quarterly Journal of Economics*, 138(1):515–573, 2023.

Ludwig Straub. Consumption, savings, and the distribution of permanent income. *Unpublished manuscript, Harvard University*, 2019.

Jean Tirole. Asset bubbles and overlapping generations. *Econometrica*, pages 1499–1528, 1985.

Heng-fu Zou. 'the spirit of capitalism'and long-run growth. *European Journal of Political Economy*, 10(2):279–293, 1994.

Chapter 1 – Appendix

A Proofs

A.1 Proof of the necessity of the transversality condition

The necessity of the transversality condition in the endowment economy (Equation 1.5) is first established. The maximization problem of household i is rewritten as:

$$\begin{cases} \max\limits_{\{a_t^i\}_{t=1}^{\infty}} & \sum\limits_{t=0}^{\infty} g_t^i(a_t^i, a_{t+1}^i) \\ \text{s.t.} & \forall t \in \mathbb{Z}_+, (a_t^i, a_{t+1}^i) \in X_t, \qquad \text{with } a_0^i \text{ given,} \end{cases} \tag{1.26}$$

with $g_t(a_t^i, a_{t+1}^i) \equiv \beta^t \left[u\left(a_t^i R_t - a_{t+1}^i\right) + v(a_{t+1}^i)\right]$, and X_t corresponds to the set of pairs (a_t^i, a_{t+1}^i) satisfying the budget constraint, that is such that: $a_{t+1}^i \leq R_{t+1} a_t$.

The proof follows Kamihigashi (2002), which identifies five conditions under which the transversality condition is a necessary condition. Solely interior solutions of this problem are considered, as it can be easily shown that solutions with $a_t^i = 0$ for some t are dominated by interior solutions. The five conditions identified by Kamihigashi (2002) to prove the necessity of the transversality condition are then satisfied.

Condition 1. $\exists n$, such that $a_0^i \in \mathbb{R}_+^n$ and $\forall t \in \mathbb{Z}_+$, $X_t \subset \mathbb{R}_+^n \times \mathbb{R}_+^n$

This condition is fulfilled for $n = 1$.

Condition 2. $\forall t \in \mathbb{Z}$, X_t is convex and $(0, 0) \in X_t$

It can be shown that if (y, z), $(y', z') \in X_t$, then, for all $\gamma \in [0; 1]$, $(\gamma y + (1-\gamma)y', \gamma z + (1-\gamma)z') \in X_t$

Condition 3. $\forall t \in \mathbb{Z}$, $g_t : X_t \to \mathbb{R}$ is C^1 on $\overset{\circ}{X}_t$ and concave.

As $u(c)$ and the function $a_{t+1}^i :\to v(a_{t+1}^i)$ are C^1, g_t is C^1. Moreover, the consumption level implied by $(a_t^i, a_{t+1}^i) = (y, z)$ is defined as $c(y, z) = yR_t - z$. As $u(c)$ is concave, for all $(y, z) \in \overset{\circ}{X}_t$, it can be shown that $\forall t \in \mathbb{Z}$ and $\forall \gamma \in [0; 1]$

$$\gamma u\Big(c(y,z)\Big) + (1-\gamma) u\Big(c(y',z')\Big) \leq u\Big(\gamma c(y,z) + (1-\gamma)c(y',z')\Big) \tag{1.27}$$

$$= u\bigg(c\Big(\gamma y + (1-\gamma)y', \gamma z + (1-\gamma)\Big)\bigg) \tag{1.28}$$

Given the concavity of both preference terms for consumption and wealth, it follows that:

$$\gamma g_t(y,z) + (1-\gamma)g_t(y',z') \le g_t\Big(\gamma y + (1-\gamma)y', \gamma z + (1-\gamma)z'\Big), \tag{1.29}$$

and hence that g_t is concave.

Condition 4. $\forall t \in \mathbb{Z}, \forall (y,z) \in \overset{\circ}{X}_t, g_{t,1}(y,z) \ge 0.$

This condition follows from $u'(c) \ge 0 \; \forall c$.

Condition 5. For any feasible path a_t^i,

$$\sum_{t=0}^{\infty} g_t(a_t^i, a_{t+1}^i) \equiv \lim_{T\to\infty} \sum_{t=0}^{T} g_t(a_t^i, a_{t+1}^i), \tag{1.30}$$

exists in $(-\infty, \infty)$.

The wealth of an agent cannot grow at a rate above R_t. Given that $R_t < 1/\beta$ and does not converge to $1/\beta$, $\lim\limits_{T\to\infty} \sum_{t=0}^{T} g_t(a_t^i, a_{t+1}^i)$ is not diverging and the condition is satisfied.[5]

The five conditions of Kamihigashi (2002) being satisfied, the transversality condition is a necessary condition of the household maximization problem in the endowment economy. ■

A.2 Proof of Propositions 1-6

Proof of Proposition 1 Proposition 1 follows directly from the definitions of the surplus wealth (1.13) and the bubble (1.14). ■

Proof of Proposition 2 It follows from Blackwell's contraction mapping theorem that the sequence ℓ_t^1 converges. Any trajectory in which ℓ_t^1 does not converge to $1/\lambda^1$ can then be ruled out. First, monotonicity of preferences implies that $\ell_t^1 > \ell_t^2$ for all t. Consequently, the economy cannot converge to a steady state with $\ell^1 < \ell^2$. Second, one can show by contradiction that the egalitarian steady state characterized by $\ell^1 = \ell^2 = 1$ is unstable under Assumption A. If the economy were to converge toward the egalitarian steady state, the growth rate of consumption of type-1 agents would have to be increasing according to the combined Euler equations of both agent

[5] The fact that the equilibrium R_t, taken as given in the household's problem, lies below $1/\beta$ and does not converge to it follows directly from the Euler equation (1.4) under insatiable preferences for wealth.

types (1.4), but decreasing according to the law of motion of their Lucas tree holdings,

$$q_t \times (\ell^1_{t+1} - \ell^1_t) = \ell^1 - c^1_t, \tag{1.31}$$

obtained from their budget constraint (1.2). Finally, the economy cannot converge to a steady state such that $\ell^1 > \ell^2 > 0$, since this would require, from the combined Euler equations (1.4), that

$$\frac{v'(\ell^1 q)}{u'(\ell^1)} = \frac{v'(\ell^2 q)}{u'(\ell^2)}, \tag{1.32}$$

which is violated under Assumption A. Indeed, the left-hand side is necessarily larger than the right-hand side under this assumption. Proposition 2 is therefore satisfied. ■

Proof of Proposition 3 If $\bar{c} > 1/\lambda^1$, then the asymptotic value of q_t must be bounded above. Otherwise, type 1 agents would accumulate unbounded wealth and asymptotically demand consumption exceeding their endowment, $1/\lambda^1$. Given that q_t is bounded, the wealth of type 2 agents converges to zero and so does their consumption given $\ell^2 = 0$:

$$\ell^2 = a^2 = c^2 = 0 \tag{1.33}$$

By the goods market clearing condition (1.10),

$$c^1 = \frac{1}{\lambda^1}. \tag{1.34}$$

Substituting into the budget constraint of type 1 agents (1.2) yields

$$a^1 = \frac{1/\lambda^1}{R-1}. \quad \blacksquare \tag{1.35}$$

Proof of Proposition 4 Given $\bar{c} < 1/\lambda^1$, it cannot be that type 1 agents consume the whole endowment and it follows that $c^2 > 0$. Since $\ell^2 = 0$, this positive asymptotic level of consumption of agents 2 is only possible if:

$$\lim_{t\to\infty} q_t = \infty, \tag{1.36}$$

from the budget constraint (1.2) of type 2 agents. It follows under $R > 1$ that

$$\lim_{t\to\infty} a^1 = s^1 = \infty, \tag{1.37}$$

and consequently,

$$c^1 = \bar{c}.$$

From the goods market clearing condition (1.10), the following expression is obtained

$$c^2 = \frac{1 - \lambda^1 \bar{c}}{\lambda^2},$$

and from the budget constraint (1.2) of type 2 agents

$$a^2 = \frac{c^2}{R-1}. \quad \blacksquare$$

Proof of Proposition 5 As discussed in the main text, Proposition 5 follows directly from the rewritten Euler equation (1.4) for type 1 agents:

$$\lim_{t\to\infty} \frac{v'(a_t^1)}{u'(c_t^1)} = 1 - \beta R. \quad \blacksquare \tag{1.38}$$

Proof of Proposition 6

The existence of a bubbly or non-bubbly equilibrium is established by showing that there exists an asymptotic rate of return consistent with the corresponding asymptotic equilibrium conditions characterized in Propositions 3 and 4, respectively. Indeterminacy remains to be proven and is only a conjecture at this stage.

Existence of a Non-Bubbly Steady State. Let $\overline{R}$ denote the maximal asymptotic rate of return consistent with a non-bubbly steady state, defined implicitly by:

$$u'\left(\frac{1}{\lambda^1}\right) = \frac{\kappa}{1 - \beta\overline{R}}. \tag{1.39}$$

For any $R > \overline{R}$, the marginal utility of consumption at the non-bubbly level $c^1 = 1/\lambda^1$ is lower than the marginal utility from saving. It follows that no non-bubbly steady state exists when $R > \overline{R}$. By Equation 1.39, no non-bubbly steady state exists if $u'\left(\frac{1}{\lambda^1}\right) < \frac{\kappa}{1-\beta}$ as $\overline{R}$ is lower than 1 and the return R necessarily exceeds 1 due to positive payoff from the Lucas tree each period.

In a non-bubbly steady state, the asymptotic rate of return R^{nb} is defined by the asymptotic Euler equation (1.4) of type 1 agents:

$$R^{nb} = \frac{1}{\beta} - \frac{v'(a^1)}{\beta u'(1/\lambda^1)}. \tag{1.40}$$

R^{nb} is uniquely defined from Equation 1.40.

The existence of a non-bubbly steady state is now established by contradiction under $u'\left(\frac{1}{\lambda^1}\right) \geq \frac{\kappa}{1-\beta}$. Specifically, it is shown that $R^{nb} \leq \overline{R}$ in this case. Assume, for the sake of contradiction, that $R^{nb} > \overline{R}$. By rearranging Equations 1.39 and 1.40 yields:

$$\frac{\kappa}{1-\beta\overline{R}} > \frac{v'(a^1)}{1-\beta R^{nb}}, \tag{1.41}$$

which can be equivalently rewritten as

$$\frac{\kappa}{v'(a^1)} > \frac{1-\beta\overline{R}}{1-\beta R^{nb}}. \tag{1.42}$$

This is a contradiction. The left-hand side of Equation 1.42 is strictly below one, whereas, under $R^{nb} < \overline{R}$, the right-hand side is strictly above one. It follows that $R^{nb} \leq \overline{R}$ and a unique non-bubbly steady state exists whenever $u'\left(\frac{1}{\lambda^1}\right) \geq \frac{\kappa}{1-\beta}$.

Existence of a Bubbly Asymptotic Regime. Whenever $\kappa = 0$, $\bar{c}$ is not defined and the condition for the existence of a bubbly asymptotic regime cannot be fulfilled (1.21). Hence, attention is restricted to the case $\kappa > 0$. Asymptotically, in a bubbly equilibrium, the consumption of type-1 agents, $c^{1,b}$, is constrained by the upper bound implied by their optimality condition:

$$c^{1,b} = u'^{-1}\left[\frac{\kappa}{1-\beta R}\right], \tag{1.43}$$

whereas the consumption of type-2 agents, $c^{2,b}$, is determined by their asymptotic Euler equation (1.4):

$$1 - \beta R - \frac{v'\left(\frac{c^{2,b}}{R-1}\right)}{u'(c^{2,b})} = 0. \tag{1.44}$$

The aggregate consumption in a bubbly asymptotic regime for a given R is denoted $g(R)$:

$$g(R) = \lambda^1 c^{1,b}(R) + \lambda^2 c^{2,b}(R) \tag{1.45}$$

Combining Equations 1.43 and 1.44 with the technical assumption $\kappa < \frac{1-\beta}{\beta}u'(1)$, gives

$$\lim_{R \to 1/\beta} g(R) = 0, \quad \text{and} \quad \lim_{R \to 1} g(R) > 1. \tag{1.46}$$

A bubbly asymptotic regime exists for each value of $R \in (1, 1/\beta)$ such that

$$g(R) = 1. \tag{1.47}$$

Given Equation 1.46, a unique bubbly asymptotic regime exists if and only if $g(R)$ is strictly decreasing over the interval $(1, 1/\beta)$. Since the consumption of type-1 agents is trivially decreasing in R, this condition holds provided that $c^{2,b}(R)$ is also strictly decreasing over the interval $(1, 1/\beta)$, which is equivalent to:

$$\frac{\partial h(c^{2,b})}{\partial c^{2,b}} < 0, \quad \text{with} \quad h(c^{2,b}) \equiv \frac{v'\left(\frac{c^{2,b}}{R-1}\right)}{u'(c^{2,b})}. \tag{1.48}$$

This holds true under Assumption A and there exists a unique bubbly asymptotic regime whenever $\kappa > 0$. ■

B Empirical Testability of Insatiable Preferences for Wealth

This appendix discusses whether standard and insatiable preferences for wealth can be empirically distinguished. Two approaches are considered: a non-structural test based on the existence of an upper bound on consumption, and structural tests based on the shape of the consumption-wealth relationship. Neither approach appears fully implementable in practice, and neither can fully rule out insatiable preferences for wealth.

Non-structural Test. A first, relatively model-free test is the following. Under the standard assumption that utility from consumption satisfies the Inada condition, insatiable preferences for wealth imply the existence of an upper bound on consumption, whereas standard preferences for wealth do not. Evidence that consumption reaches a plateau at sufficiently high levels of wealth is thus sufficient to establish that preferences for wealth are insatiable.

The main difficulties in implementing such a test are empirical. Survey data contain very few observations at the very top of the wealth distribution, precisely where the test would need to be implemented. Moreover, consumption is often poorly measured for the richest households. Some forms of spending are missing from standard surveys, and some consumption services are not included in standard measures of consumption. For example, the aesthetic service flow derived a painting is not recorded as conventional consumption. These measurement issues make it difficult to infer from observed consumption whether rich households have truly reached an upper bound.

Structural Tests. A second approach is structural. If no upper bound on consumption is observed, empirical identification must instead rely on other implications of insatiable preferences for wealth at lower wealth levels. Doing so requires additional structural assumptions. For example, in a simple environment with CRRA utility from consumption, CRRA standard preferences for wealth, no precautionary saving, and homogeneous returns, the model implies that $\log(c/a)$ is a linear function of $\log(a)$, where c is consumption and a is wealth. With insatiable preferences for wealth, the same relationship becomes concave in $\log(a)$. A possible structural test is to estimate the shape of the relationship between the log consumption-wealth ratio and log wealth, and test linearity against concavity.

Such tests may detect insatiable preferences at lower levels of wealth, but they are less robust. They are joint tests of insatiable preferences and of the maintained structural assumptions. The example above is only one possible implementation: other models may generate different testable restrictions. In practice, many saving motives—including precautionary saving, bequests, heterogeneous returns, borrowing constraints, entrepreneurial risk, and taxation—can be modeled in different ways and may rationalize a wide range of consumption-wealth relationships observed in the data. This is especially problematic away from the very top of the distribution, where these motives are likely to be quantitatively much more important than the potentially non-vanishing component of preferences for wealth.

Testing for Asymptotic Properties. More fundamentally, even ideal empirical tests cannot fully rule out insatiable preferences for wealth. Standard and insatiable preferences for wealth differ only in their asymptotic implications—that is, when wealth diverges—which are, by definition, not directly observable in finite data. The smaller the parameter κ capturing the linear component of preferences for wealth, the higher the level of wealth at which this component has sizable empirical implications. The implications of insatiable preferences for consumption-saving behavior may become visible only once some agents are sufficiently far along the wealth accumulation path.

As a result, the absence of a visible consumption upper bound, or the failure to detect curvature in the consumption-wealth relationship, should not be interpreted as definitive evidence against κ being strictly positive. Even if the degree of insatiability in preferences for wealth is too small to significantly affect current saving decisions, it may still have substantial effects on future saving behavior and, through agents' anticipation of these future effects, on present asset price dynamics.

Chapter 2

A Scrooge McDuck Theory of Wealth Dynamics: Production Economy and Taxation

Chapter 2 introduces insatiable preferences for wealth into a production economy with fully reproducible capital and an infinitely lived asset in fixed supply. When labor income inequality is sufficiently high, the equilibrium asset price contains a bubbly component. The bubble crowds out investment, while its diverging price sustains a permanent rise in wealth inequality. In this environment, wealth and capital income taxes are not equivalent, but both can shift the equilibrium from bubbly to non-bubbly—thus being potentially redistributive and growth-*enhancing*. A quantitative exercise suggests that a 1.5 percent wealth tax in the U.S. would raise capital by 5 percent over 30 years.

The acknowledgements expressed in Chapter 1 apply equally to this chapter.

While Chapter 1 introduced insatiable preferences for wealth in an endowment economy, the present chapter brings them into a production economy. Chapter 1 clarified analytically how asset pricing differs under standard and insatiable wealth preferences and, in particular, why a diverging rational bubble can arise under the latter. This chapter shows that the result is robust to an environment with reproducible capital and an infinitely lived, rent-generating asset in fixed supply: under sufficiently high income inequality, a diverging rational bubble can also exist in a production economy. This links the framework to the motivating stylized facts of rising income inequality, an increasing wealth-to-output ratio, and a stagnant capital-to-output ratio.

The existence of a rational bubble that expands faster than the economy's growth rate constitutes the central asset-pricing feature generating a wedge between wealth and capital dynamics. The bubble crowds out capital and bubbly equilibria reproduce the key stylized facts motivating this study. They feature a bubble-driven increase in the wealth-to-output ratio without an accompanying rise in the capital-to-output ratio. The recurrent increases in bubble valuation help reconcile the empirical evidence of rising price-to-earnings ratios for various non-reproducible assets with the stability of the return on capital (Gomme et al., 2011; Reis, 2022). Furthermore, the theory provides a unified framework for understanding the rise in the wealth-to-output ratio as a single phenomenon, despite its manifestation on different asset classes across countries. The bubble is formally tied to the rent-generating factor of production in the model, but can be mapped to a broad range of real-world assets such as land, equities, and even public debt.

Whether the equilibrium is bubbly or non-bubbly depends on whether income at the top asymptotically exceeds the upper bound on optimal consumption. If this holds true, agents at the top are necessarily Scrooge McDuck, and the equilibrium is bubbly. In a comparative statics analysis, increasing the insatiable component of preferences for wealth or raising labor income inequality broadens the set of parameters consistent with a bubble. The former operates by lowering the upper bound on optimal consumption, while the latter raises top incomes. In line with the motivating evidence, an increase in top income inequality may shift the equilibrium from non-bubbly to bubbly. A novel numerical method is developed to compute the non-balanced growth path when the economy features a diverging rational bubble.

This study contributes to the ongoing debate on the welfare implications of capital gains. Existing work has primarily focused on fundamentals-driven increases in the price-to-earnings ratio, resulting from a once-and-for-all decline in the discount rate (Fagereng et al., 2021). These capital gains have often been described as "paper gains" (Cochrane, 2020; Krugman), as they do not alter dividend flows and cannot sustainably finance higher consumption. In contrast, this analysis focuses on bubble-driven capital gains, which are recurring. This continuous appreciation allows non-Scrooge McDuck agents to realize capital gains in each period to finance consumption. This is particularly relevant as evidence shows that, in the United States, capital gains among the bottom 90 percent have been used largely to finance consumption through increased collateralized borrowing (Mian et al., 2020).

This model is also the first microfounded general equilibrium that jointly delivers diverging wealth inequality and an asymptotic rate of return, r, above the growth rate, g.[1] In a bubbly equilibrium, Scrooge McDuck agents optimally accumulate unbounded wealth, while the wealth of non-Scrooge McDuck agents converges to non-zero but bounded levels. This aligns with Piketty (2014)'s conjecture that wealth inequality diverges when $r > g$, a result that is not immediate in general equilibrium. In particular, under standard preferences for wealth, Michau et al. (2025) find that diverging inequality prevents r from remaining above g.[2] In their model, as more and more wealth accrues to high-saving agents at the top, aggregate savings, and hence capital, increase. Asymptotically, the capital stock converges to the exact level where $r = g$. Under insatiable wealth preferences, diverging inequality breaks the one-to-one link between savings and capital on which this argument rests, as agents invest part of their wealth in the rational bubble.

The distinction between an economy without a diverging bubble and one with insatiable wealth preferences that sustain a bubble also matters for tax policy. Both capital income and wealth taxes are considered. Progressive taxation lowers top incomes, thereby expanding the parameter space in which the upper bound on optimal consumption does not bind. As a result, introducing a progressive tax can shift the equilibrium from bubbly to non-bubbly. This equilibrium-shift effect of taxation has a positive impact on capital accumulation, as it prevents the crowding out associated with the bubble. At the same time, capital income and wealth taxation lower saving incentives, which has a negative effect on the capital stock. The net effect of taxation is ambiguous: the positive equilibrium-shift effect must be weighed against the disincentive to save. The standard efficiency-redistribution trade-off is no longer automatic: certain taxes can simultaneously increase capital accumulation and output while achieving redistribution.

In the present framework, the equivalence between capital income and wealth taxation breaks down. Although all assets yield a homogeneous return, they differ in their taxable income-to-taxable value ratios. Consequently, they are taxed differently under capital income and wealth taxation. The most striking case is that of the rational

[1] Diverging wealth inequality is understood here as an increasing absolute gap between agents at the top and bottom of the wealth distribution. A diverging ratio of wealth between these groups arises in many more frameworks, notably when wealth at the bottom converges to zero.

[2] As discussed above, under standard preferences for wealth, agents whose wealth diverges relative to output also have diverging consumption. Their presence does not violate the goods market clearing condition in Michau et al. (2025), only because they account for a vanishingly small share of agents.

bubble component of the rent-generating asset. It enters the wealth tax base but not that of the capital income tax, as it pays no dividends. Furthermore, any non-vanishing wealth tax precludes the existence of a bubbly equilibrium. A diverging bubble would yield unbounded tax revenues, the redistribution of which would violate the goods market clearing condition. In a bubbly equilibrium, any wealth tax therefore induces an equilibrium-shift effect. Its disincentive effect, by contrast, increases proportionally with the tax rate. It follows that there exists a range of sufficiently small tax rates for which the equilibrium-shift effect dominates, leading to higher capital accumulation.

Finally, the empirical relevance of the mechanism is assessed through a simple quantitative exercise for the United States. The model parameters are calibrated to match 1989 data under the assumption that the economy is initially in a non-bubbly steady state. An increase in labor income inequality, estimated from SCF data, is then introduced, shifting the economy toward a bubbly equilibrium. Despite its simplicity, the model replicates the evolution of wealth inequality and the capital- and wealth-to-output ratios reasonably well. This outcome is associated with a bubble amounting to 150 percent of GDP in 2024. The effects of introducing a counterfactual 1.5 percent wealth tax in the 2022 economy are then evaluated. The tax reduces the wealth-to-output ratio by eliminating the bubble, while remaining sufficiently small to raise the aggregate capital stock by 5 percent over 30 years. The resulting gains in consumption utility, driven by redistribution and a higher capital stock, more than offset the losses associated with lower utility derived from wealth.

Related Literature. Beyond the strands of literature discussed in Chapter 1, this work also contributes to the literature on capital income and wealth taxation (Judd, 1985; Chamley, 1986; Straub and Werning, 2020; Gaillard and Wangner, 2023). It highlights the equilibrium-shift effect that such taxes may have when they move the economy from a bubbly to a non-bubbly equilibrium. This shift raises both capital and consumption, an effect that is absent under standard non-homothetic preferences (Morrison, 2024). Moreover, this paper sheds new light on the equivalence between capital income and wealth taxation (Allais, 1977; Guvenen et al., 2023; Piketty et al., 2023). It shows that this equivalence may break down even under homogeneous returns in the presence of capital gains. Although the argument can be applied more generally, it is developed here for the first time.

The remainder of the chapter is organized as follows. Section 1 extends the analysis of Chapter 1 to a production economy, linking the model to the motivating stylized facts. Section 2 investigates the positive implications of the model for wealth and capital income taxation. Section 3 presents a quantitative exercise to assess the empirical relevance of the Scrooge McDuck mechanism. Section 4 concludes.

1 Production Economy

Insatiable preferences for wealth are introduced into a growing production economy to demonstrate that the existence of bubbly equilibria remains robust in the presence of reproducible capital and to connect the model to the stylized facts motivating the analysis. In bubbly equilibria, the wealth-to-output ratio diverges due to a bubble that crowds out capital, while the capital-to-output ratio converges to a level that does not maximize asymptotic aggregate consumption.

1.1 Model Environment

Production. The production function is a Cobb-Douglas with three inputs: labor, reproducible capital, and a rent-generating factor:

$$Y_t = K_t^{\alpha} L_t^{\gamma} (Z_t N_t)^{1-\alpha-\gamma}, \tag{2.1}$$

where K_t denotes the capital stock, L_t the rent-generating factor, N_t the labor supply and Z_t the labor productivity. The rent-generating factor is in fixed supply, normalized to $L_t = 1$ for all t, and each of its units is priced at Q_t after production. The capital stock depreciates at a rate δ. It is reproducible, and its price is normalized to one by assumption, precluding valuations above fundamental value. The rent-generating factor is introduced to study the asset pricing implications of insatiable preferences for wealth. It is interpreted as encompassing all non-reproducible, long-lived assets, for example, land.[3]

The growth rates of productivity and labor supply from period t to $t+1$ are denoted by g_{t+1}^Z and g_{t+1}^N, respectively:

$$g_{t+1}^Z \equiv \frac{Z_{t+1} - Z_t}{Z_t}, \qquad g_{t+1}^N \equiv \frac{N_{t+1} - N_t}{N_t}. \tag{2.2}$$

Asymptotically, the economy grows at a slower rate than the effective labor force. This is due to the presence of the rent-generating factor, which does not scale with output. As the capital-to-output ratio K_t/Y_t converges, the economy tends to grow at the same rate as $(Z_t N_t)^{\frac{1-\alpha-\gamma}{1-\alpha}}$, whose growth rate, denoted by g_t, is given by:

$$g_{t+1} \equiv [(1+g_{t+1}^Z)(1+g_{t+1}^N)]^{\frac{1-\alpha-\gamma}{1-\alpha}} - 1. \tag{2.3}$$

[3] Its real-world definition could be subject to debate, depending on how one defines reproducibility and longevity—whether, for instance, it includes long-lived intangible assets such as brands or market power.

To account for growth, each capital-letter variable X_t has a normalized counterpart x_t, defined as:

$$x_t \equiv \frac{X_t}{(Z_t N_t)^{\frac{1-\alpha-\gamma}{1-\alpha}}}. \tag{2.4}$$

All figures presented below display variables normalized according to Equation 3.5.

Factors of production are paid at their marginal productivity. At time t, one unit of labor is paid a wage W_t. By arbitrage, the rates of return on capital and on the rent-generating factor from period t to $t+1$ are equalized and denoted by R_{t+1}:

$$R_{t+1} = 1 + \alpha k_{t+1}^{\alpha-1} - \delta = (1 + g_{t+1}) \frac{q_{t+1} + \gamma y_t}{q_t}. \tag{2.5}$$

Let $\widetilde{R}_t$ denote the growth-adjusted rate of return:

$$\widetilde{R}_t \equiv \frac{R_t}{1 + g_t}. \tag{2.6}$$

As in the case of the Lucas tree in the endowment economy, the price of the rent-generating factor can be decomposed into a fundamental component, F_t, which corresponds to the present value of future dividends discounted at the rate of return, and a bubbly component, B_t. Normalizing F_t and B_t yields:

$$f_t \equiv \sum_{j=1}^{\infty} \frac{\gamma y_{t+j}}{\prod_{s=1}^{j} \widetilde{R}_{t+s}}, \qquad b_t = \lim_{T\to\infty} \frac{q_T}{\prod_{s=1}^{T} \widetilde{R}_{t+s}} \qquad \text{with} \quad q_t = f_t + b_t. \tag{2.7}$$

The rent-generating factor is assumed to be productive in every period, with $\gamma > 0$. This assumption prevents the rate of return from falling below the growth rate of the economy, following the argument in Rhee (1991). Since the rent-generating factor's share of output does not converge to zero, its return remains above the growth rate asymptotically. Attention is therefore restricted to the empirically relevant case in which rents are present in the economy, at a minimum in the form of land rents, and the rate of return exceeds the growth rate of the economy (Jordà et al., 2019; Reis, 2021).

Households. The population consists of a unit mass of households, each comprising N_t agents. There are I types of infinitely-lived agents, $i \in \{1, 2, ..., I\}$, differing in their labor productivity, and eventually in their initial endowment. Each type represents a share λ^i of the total population. A household of type i holds K_{t+1}^i units of capital and L_{t+1}^i units of the rent-generating factor at the end of period t.

Agents are ranked in decreasing order of labor productivity and initial endowments. Each agent i supplies ζ_t^i units of labor inelastically in period t, where ζ_t^i reflects productivity differences and satisfies,

$$\zeta_t^i > \zeta_t^j \quad \text{for all } i < j. \tag{2.8}$$

For convenience and without loss of generality, productivity levels are chosen such that $\sum_i \zeta_t^i \lambda^i = 1$ for all t, ensuring that aggregate labor supply coincides with population size and equals N_t. Initial endowments of capital, K_0^i, and of the rent-generating factor, L_0^i, are weakly decreasing in agent index, satisfying,

$$K_0^i \geq K_0^j, \quad L_0^i \geq L_0^j \quad \text{for all } i < j. \tag{2.9}$$

The normalized end-of-period wealth of type i households, denoted a_{t+1}^i, is defined as,

$$a_{t+1}^i \equiv k_{t+1}^i + \frac{q_t L_{t+1}^i}{1 + g_{t+1}}. \tag{2.10}$$

Their budget constraint is given by,

$$c_t^i + a_{t+1}^i (1 + g_{t+1}) = R_t a_t^i + \zeta_t^i w_t. \tag{2.11}$$

Preferences are assumed to take a functional form that satisfies Assumption A and allows for insatiable preferences for wealth. Households derive utility from consumption and wealth holdings, and maximize the following intertemporal utility:

$$\sum_{t=0}^{\infty} \beta^t \left[u(c_t) + v(a_{t+1}) \right], \tag{2.12}$$

where the period utility functions are given by,

$$u(c) = \begin{cases} \dfrac{c^{1-\theta}}{1-\theta} & \text{if } \theta \neq 1, \\ \log(c) & \text{if } \theta = 1, \end{cases} \quad \text{and} \quad v(a) = \begin{cases} \psi \cdot \dfrac{(a - \underline{a})^{1-\eta}}{1-\eta} + \kappa a & \text{if } \eta \neq 1, \\ \psi \log(a - \underline{a}) + \kappa a & \text{if } \eta = 1. \end{cases} \tag{2.13}$$

The utility from consumption, $u(c)$, follows a CRRA specification, while the preference for wealth, $v(a)$, consists of two components: a standard concave term and an additional linear component that captures insatiable preferences for wealth. The parameters are chosen such that $\eta < \theta$ and $\underline{a} \geq 0$, ensuring that Assumption A holds and that saving rates increase with both income and wealth. The parameter ψ is a scaling parameter that captures the strength of standard preferences for wealth relative to utility from consumption. The linear term in $v(a)$ imposes a lower bound on the marginal utility of holding wealth at κ, that is,

$$\lim_{a \to \infty} \frac{\partial v(a)}{\partial a} = \kappa. \tag{2.14}$$

Preferences with $\kappa = 0$ are referred to as *standard* preferences for wealth, and to those with $\kappa > 0$ as *insatiable* preferences for wealth. To avoid the counterfactual implication of ever-increasing aggregate saving rates as the economy expands, the arguments of the utility function are expressed in normalized rather than absolute terms.[4]

The solution to the optimization problem for type i households is characterized by the following Euler equation (2.15),

$$c_t^{i\,-\theta} = \beta R_{t+1} \frac{(c_{t+1}^i)^{-\theta}}{1+g_{t+1}} + \psi \frac{(a_{t+1}^i - \underline{a})^{-\eta}}{1+g_{t+1}} + \frac{\kappa}{1+g_{t+1}}, \tag{2.15}$$

and the transversality condition (2.16),

$$\lim_{t\to\infty} \beta^t \left[c_t^{i\,-\theta} - \frac{\psi(a_{t+1}^i - \underline{a})^{-\eta} + \kappa}{1+g_{t+1}} \right] a_{t+1}^i = 0. \tag{2.16}$$

Market Clearing Conditions. The goods market clearing condition writes as,

$$k_{t+1}(1+g_{t+1}) + \sum_i \lambda^i c_t^i = (1-\delta)k_t + y_t. \tag{2.17}$$

The asset market clearing conditions for capital and for the rent-generating factor are given by:

$$k_t = \sum_i \lambda^i k_t^i, \tag{2.18}$$

$$1 = \sum_i \lambda^i L_t^i. \tag{2.19}$$

Equilibrium. Given the initial endowments $\{K_0^i, L_0^i\}$ for each agent $i \in \{1, 2, \ldots, I\}$, an equilibrium $\left\{ \{c_t^i, k_t^i, L_t^i\}_{i\in\{1,2,\ldots,I\}}, q_t \right\}_{t=0}^{\infty}$ is characterized by the solutions to the household optimization problems (2.15, 2.16) and the market clearing conditions for capital (2.18) and the rent-generating factor (2.19). The goods market clearing condition holds by Walras's Law.

Definitions. The definitions of the core elements of the Scrooge McDuck analysis are adapted to account for growth and labor income. Similarly to the endowment economy, insatiable preferences for wealth impose a lower bound on the marginal utility of saving, which in turn determines an upper bound on consumption, $\overline{C}_t$. Its

[4] In the words of Mian et al. (2021), the purpose of preferences for wealth is to break *individual* scale invariance, so that wealthier households exhibit higher saving rates, while preserving *aggregate* scale invariance, thereby preventing the saving rate of the economy as a whole from rising with its size. Aside from the standard preferences-for-wealth case with log utility from consumption, this requires that both consumption and wealth enter the utility function in normalized terms.

normalized counterpart, $\bar{c}_t$, is given by,

$$\bar{c}_t \equiv \left[\frac{1}{1+g_{t+1}}\sum_{s=0}^{\infty}\beta^s\left(\prod_{j=1}^{s}\frac{R_{t+j}}{1+g_{t+j+1}}\right)\kappa\right]^{-\frac{1}{\theta}}, \tag{2.20}$$

where the term in brackets represents the lower bound on the marginal utility of saving. By assumption, the analysis is restricted to parameter values and initial conditions under which not all agents asymptotically consume at the upper bound of optimal consumption. The definition of Scrooge McDuck agents is refined to use normalized rather than absolute consumption and wealth. Agent i is a Scrooge McDuck if it accumulates unbounded *normalized* wealth, a_t^i, over time while keeping its *normalized* consumption, c_t^i, bounded. Finally, the definition of surplus wealth, still referring to wealth held without consumption motives, is modified to account for the fact that part of future consumption is financed by wages. Its normalized version is given by,

$$s_{t+1}^i \equiv a_{t+1}^i - \sum_{j=1}^{\infty}\frac{c_{t+j}^i - \zeta_t^i w_{t+j}}{\prod_{s=1}^{j}\widetilde{R}_{t+s}}. \tag{2.21}$$

From the goods and asset market-clearing conditions, it follows that Proposition 1 holds in the production economy, and that aggregate surplus wealth coincides with the aggregate bubble.

Numerical Solution Method. Solving the model numerically raises two main challenges: characterizing the diverging bubbly transition, given that both the initial and terminal values of the bubble must be determined as part of the equilibrium; and computing the partial-equilibrium household problem with wealth preferences over long transition paths. These issues are addressed through original methods, the details of which are provided in Appendix A.

1.2 Bubbly and Non-bubbly Equilibria

The model is solved numerically, and bubbly equilibria are found to exist even in the presence of reproducible capital. As the insatiable component of wealth preferences strengthens or the degree of labor income inequality rises, the set of parameters supporting a bubbly equilibrium expands, while that supporting a non-bubbly one shrinks.

A Numerical Illustration. An illustrative exercise examines two transition paths corresponding to low and high values of κ, which give rise to a non-bubbly and a bubbly equilibrium, respectively. The emphasis at this stage is on the model's qualitative implications, while a more comprehensive quantitative analysis is presented in Section 3.

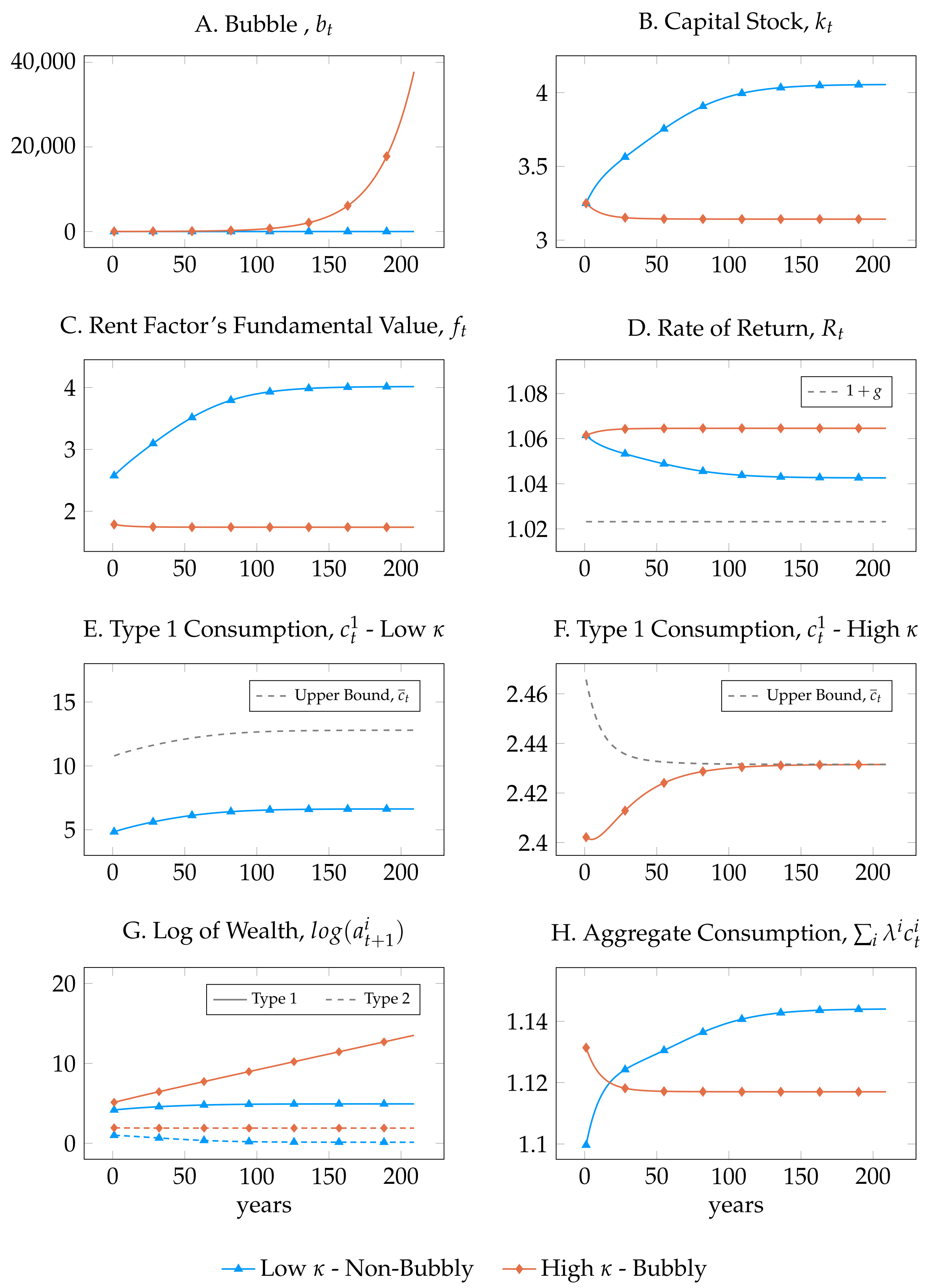
The population is divided into two types. Type 1 represents 5% of the population, $\lambda^1 = 0.05$, earns 25% of labor income in every period, $\zeta_t^1 = 5\ \forall t$, and holds 55% of the initial stock of capital and rent-generating factor endowments. Production parameters are set to $\gamma = 0.05$, $\alpha = 0.3$, $\delta = 0.07$, $g_t^Z = 1.5\%$, and $g_t^N = 1\%$ for all t. The discount factor is $\beta = 0.95$, and the consumption utility and standard preference-for-wealth parameters follow Kumhof et al. (2015): $\theta = 2$, $\eta = 0.92$, $\underline{a} = 0.0$ and $\psi = 0.05$. Two values for the insatiable preference-for-wealth parameter are considered: a low value, $\kappa = 0.0002$, and a high value, $\kappa = 0.002$. Starting from an initial capital stock of $k_0 = 3.25$, the transition dynamics are shown in Figure 2.1.

When κ is high, the upper bound on optimal consumption binds for type 1 agents, which are Scrooge McDuck. From Equation 2.20, the upper bound on optimal consumption, $\bar{c}_t$, decreases with the rate of return and with the strength of the insatiable preference for wealth. As a result, it binds asymptotically for type 1 agents exclusively in the high-κ case, where it is lower than under low κ.[5] Under high κ, type 1 agents' normalized consumption remains bounded, and their income persistently exceeds their consumption. It follows that their wealth diverges over time. They are Scrooge McDuck agents and hold positive surplus wealth, $s_t > 0$, for all t.

When strong enough to generate a bubbly equilibrium, insatiable preferences for wealth produce a diverging wealth-to-output ratio, while the capital-to-output ratio converges. Under high κ, the surplus wealth held by type 1 agents coincides in the aggregate with a rational bubble, $b_t = \lambda^1 s_{t+1}^1$. Since the bubble grows at the rate of return, which exceeds the economy's growth rate, its normalized size diverges. Asymptotically, this divergence in the bubble-to-output ratio raises the wealth-to-output ratio without affecting the capital-to-output ratio. The rational bubble is the key asset-pricing mechanism that disconnects wealth dynamics from those of capital. Bubbly equilibria thus align with the wealth and capital trends observed in many advanced economies in recent decades, which motivates this paper.

In contrast, under low insatiable preferences for wealth—of which standard preferences are a special case—the equilibrium is non-bubbly. In this case, the upper bound on optimal consumption is not binding for type 1 agents, who do not accumulate unbounded wealth. There are no Scrooge McDuck agents, no associated surplus wealth, and thus no rational bubble. Given the initial k_0, the wealth-to-output ratio increases over the transition but does not diverge. This rise is intrinsically linked to the increase in the capital-to-output ratio. The latter raises wealth directly and also increases the

[5] For simplicity, cases in which the upper bound on optimal consumption binds for both types are not considered.



Notes: The transition starts from the same initial capital stock, $k_0 = 3.25$, and asset distribution, with type 1 agents holding 55% of initial wealth, under either low ($\kappa = 0.0002$, blue line) or high ($\kappa = 0.002$, red line) insatiable preferences for wealth.

Figure 2.1: Transition Dynamics in the Production Economy: Bubbly vs. Non-Bubbly Equilibrium

fundamental value of the rent-generating factor through a lower rate of return, R_t, and higher rents (see Equation 2.7).

The presence of a rational bubble crowds out capital. As the capital stock comparison shows, despite a stronger preference for wealth, the high-κ case features a lower capital stock. This occurs because, under high κ, part of agents' savings goes into the bubble, but not under low κ. In turn, lower capital stock levels imply lower output and higher rates of return. It follows that type 2 agents may receive higher asymptotic income in either the low-κ non-bubbly case or the high-κ bubbly case, depending on the relative strength of two channels. On the one hand, they earn lower wages in the high-κ case; on the other hand, their wealth income benefits from the capital gains associated with the rational bubble. The parametrization determines which effect dominates and, consequently, whether type 2 agents consume asymptotically more in the low- or high-κ case.

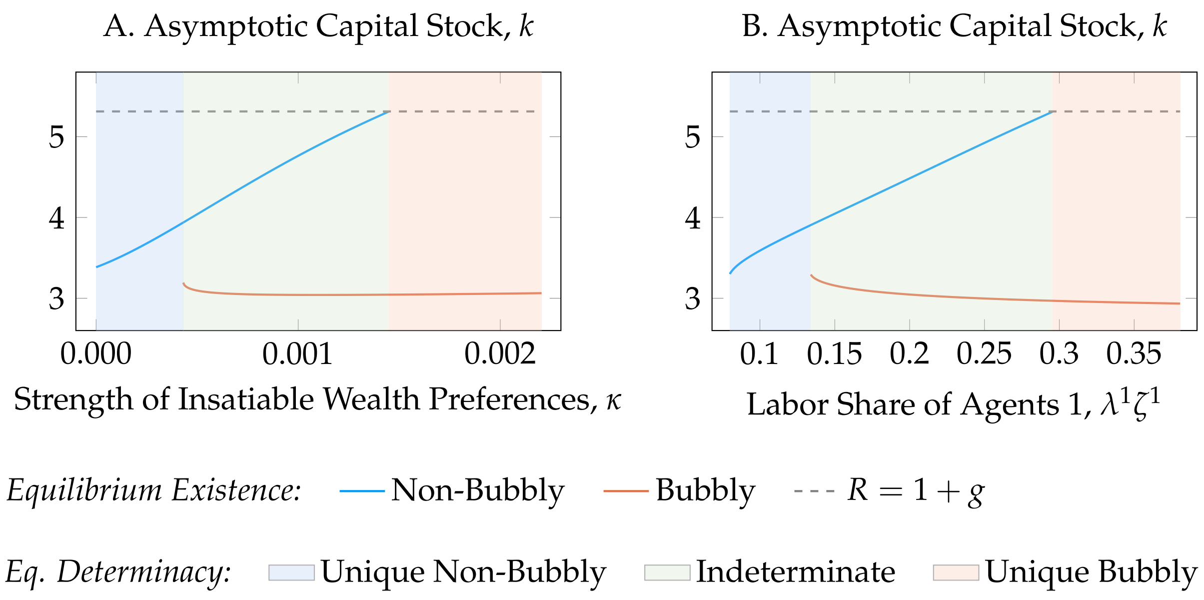
The bubble's crowding-out of capital reduces asymptotic aggregate consumption. A well-established result implies that, whenever $R > 1 + g$, an increase in the capital stock raises asymptotic consumption, whereas it lowers it when $R < 1 + g$.[6] By the assumption of a productive rent-generating factor, the rate of return necessarily exceeds the growth rate, and the bubble's crowding-out reduces asymptotic consumption. This can be seen in Figure 2.1, where aggregate consumption is asymptotically higher in the non-bubbly, low-κ scenario than in the bubbly, high-κ scenario. However, during the transition to this asymptotic state, aggregate consumption is initially higher under high κ, as fewer resources are allocated to capital investment. Notice that neither of the two equilibria considered reaches the capital level that maximizes asymptotic consumption and would imply $R = 1 + g$.

Comparative Statics. Comparative statics are now used to determine the levels of insatiable wealth preference strength and labor income inequality under which non-bubbly and bubbly equilibria exist, as well as the corresponding asymptotic capital stock.

When the insatiable component of wealth preferences κ is sufficiently low, the economy admits a non-bubbly equilibrium; when it is sufficiently high, a bubbly equilibrium exists. The left panel of Figure 2.2 shows the different equilibria and their associated capital levels as a function of κ. At $\kappa = 0$, which corresponds to standard preferences for wealth, the equilibrium is necessarily unique and non-bubbly. More generally, there exist two threshold values of κ: below the lower threshold, the equilibrium is unique and non-bubbly (blue zone), while above the upper threshold, it

[6]This result traces back to Diamond (1965). Asymptotically, a one-unit increase in the capital stock raises output by $\alpha k^{\alpha-1}$ and investment by $\delta + g$ in normalized terms. If $\alpha k^{\alpha-1} > \delta + g$, it raises aggregate consumption, a condition that can be rewritten as $R > 1 + g$.

Figure 2.2: Capital Accumulation by Strength of Insatiable Wealth Preferences and Labor Inequality



Notes: This figure shows the existence and type of equilibria, along with the corresponding asymptotic capital stock k, under alternative parameter values for (Panel A) the strength of insatiable wealth preferences, κ, and (Panel B) the labor share of agents 1, $\lambda^1\zeta^1$. Non-bubbly and bubbly equilibria are represented by the blue and red lines, respectively. The dashed line indicates the capital level at which the rate of return equals the growth rate. Shaded areas indicate equilibrium regimes: regions with a unique non-bubbly equilibrium (blue), a unique bubbly equilibrium (red), or coexistence of both equilibria with indeterminate selection (slate green).

is unique and bubbly (red zone). In the intermediate range, one bubbly and one non-bubbly equilibrium coexist (green zone). As in the endowment economy, equilibrium selection is in that case self-fulfilling: if agents expect high returns, the upper bound on consumption is low and binding, leading to a bubble that sustains high returns; if they expect low returns, the opposite holds.

The asymptotic capital stock rises with stronger insatiable preferences for wealth, as long as the equilibrium does not turn bubbly. A stronger preference for wealth strengthens saving incentives and, consequently, aggregate savings. In the absence of a bubble, this inevitably translates into a larger capital stock.[7] The non-bubbly equilibrium disappears once κ exceeds the threshold at which further capital accumulation would drive the asymptotic return below the growth rate. This is indeed precluded by the presence of a rent-generating factor of production, as discussed in Subsection 1.1. Furthermore, any marginal increase in κ that shifts the economy from a non-bubbly to a bubbly equilibrium necessarily lowers the asymptotic capital stock, reflecting the crowding-out effect of the bubble.

[7] A similar result is obtained when the taste for wealth is increased through the standard wealth-preference component of utility (for instance, a higher value of ψ) rather than through the insatiable one.

The comparative statics with respect to labor income inequality yield results analogous to those obtained for κ. In this simplified two-agent framework, labor income inequality is captured entirely by the labor share of type-1 agents. The right panel of Figure 2.2 shows how the equilibria and their associated capital levels depend on this measure. When labor inequality is low, type-1 agents earn insufficient income to hit the asymptotic upper bound on consumption, which never binds. In this case, the equilibrium is unique and non-bubbly (blue zone). Conversely, when labor inequality exceeds a given threshold, the upper bound always binds and the equilibrium is unique and bubbly (red zone). In the intermediate range, a bubbly and a non-bubbly equilibrium coexist with self-fulfilling equilibrium selection (green zone). Finally, increasing labor inequality raises aggregate savings by redistributing income from low-saving to high-saving agents. Conditional on the equilibrium being non-bubbly, this translates into a higher capital stock, whereas this need not be the case under a bubbly equilibrium.

This comparative static exercise suggests that the widely observed capital-gains-driven increase in the wealth-to-output ratio could stem from the concomitant rise in top income inequality. This is illustrated in Figure 2.2, where rising labor income inequality may translate into the existence, or even the necessity, of a bubbly equilibrium. It is important to note that the mechanism at play is not one in which a one-time increase in income inequality leads to a one-time rise in wealth through valuation effects. Rather, higher inequality shifts the economy from a non-bubbly regime to a bubbly one, generating persistent capital gains. When unrealized capital gains are excluded from the definition of income, income inequality converges in bubbly equilibria while the wealth-to-output ratio diverges. This could rationalize why the wealth-to-output ratio has continued to rise in many economies where income inequality appears to have converged, under a definition of income that excludes unrealized capital gains (Figures 1.1a and 1.1b).

1.3 Discussion

The model generates two unusual outcomes within a neoclassical framework: a non-zero mass of agents with diverging wealth and a diverging rational bubble. In doing so, it provides insights into several strands of the literature.

Return on Wealth. Although the safe rate of return has declined steadily since the 1980s, the evolution of the average return on wealth remains considerably less clear. Several studies examining risk premia or returns on capital argue that the ex-ante expected return on wealth has remained broadly stable, or has declined only slightly (Duarte and Rosa, 2015; Caballero et al., 2017; Reis, 2022). However, under the assumption that assets are priced at their fundamental value, this pattern is difficult to reconcile with evidence of increasing price-to-earning ratios (Kuvshinov and Zimmermann, 2021; Eggertsson et al., 2021; Greenwald et al., 2025). These ratios suggest that the average rate of return has moved in line with the safe rate and declined over time.

The rational bubble developed in this model provides a coherent explanation for the seemingly conflicting evidence regarding the average return on wealth. The bubble is a non-dominated asset because its absolute value rises by R_t between periods t and $t+1$. Hence, it generates new capital gains in every period. On the one hand, these capital gains raise the average return on wealth above the ratio of earnings from wealth to total wealth. This is consistent with evidence that the average return on wealth has remained relatively stable. On the other hand, because the bubble grows at a rate exceeding that of the economy, it increases the ratio of wealth to earnings from wealth over time. This aligns with the observed rise in price-to-earnings ratios.

A Unified Framework for Rising Wealth-to-Output Ratios. Another appealing feature of the framework is that it offers a unified explanation for the widely observed rise in the wealth-to-output ratio. Large capital gains represent a robust stylized fact across advanced economies, yet the types of assets benefiting from them vary substantially across regions. The latter aspect has led some lines of research to develop country-specific explanations for a global phenomenon. For instance, in the U.S., the rise in stock values is often attributed to increasing market power (Farhi and Gourio, 2018; Eggertsson et al., 2021), while the low elasticity of housing supply is frequently cited to explain the rise in real estate prices in Europe (Hilber and Vermeulen, 2016; Muellbauer, 2018). Although these mechanisms account for part of the observed dynamics, the Scrooge McDuck theory links the widespread capital gains to an equally widespread and simultaneous rise in top income inequality. Furthermore, it does so through a rational bubble that can arise in a wide range of real-world assets.

Capital Gains. The substantial capital gains observed since the 1980s have given rise to a growing debate on their welfare implications. This debate has mostly centered on asset price increases driven by a once-and-for-all decline in the discount rate, which raises the fundamental value of assets. Such capital gains mechanically increase absolute wealth inequality without affecting the distribution of income from wealth. These valuation effects therefore cannot be used to sustainably finance consumption. They have been interpreted in various ways: as a pure increase in welfare inequality (Saez et al., 2021), as mere "paper gains" with no welfare effect (Cochrane, 2020; Krugman), or as a mix of both, depending on net asset sales (Fagereng et al., 2024).

In contrast, this paper examines the consumption implications of the recurrent capital gains generated by a rational bubble. As new capital gains emerge each period, non-Scrooge McDuck agents can draw on them to finance a fraction of their consumption.

This result is especially relevant given empirical evidence showing that a large fraction of the population finances consumption through capital gains. In the United States, Mian et al. (2020) show that capital gains realized by the bottom 90 percent of the wealth distribution have been used to a large extent to finance consumption through increased collateralized debt. This study suggests that in a frictionless bubbly world, with the bubble attached partly to the land underlying real-estate, this collateralized debt could continue to rise without bound. This study suggests that, in a frictionless bubbly economy where part of the bubble is attached to the land underlying real estate, consumption could be permanently financed through a process of ever-increasing collateralized borrowing. In practice, this continued rise could ultimately be halted by a binding financial constraint, the analysis of which lies beyond the scope of this paper.

Relation to Piketty's *Capital in the Twenty-First Century*. The present analysis also contributes to the discussion initiated by Piketty (2014) on how the gap between the rate of return, r, and the growth rate, g, influences the dynamics of wealth inequality. A key conjecture of Piketty (2014) posits that "if the rate of return on capital remains significantly above the growth rate for an extended period, the risk of divergence in the distribution of wealth is very high" (p. 34). Addressing this conjecture in a model with standard preferences for wealth, Michau et al. (2025) refutes the possibility of r remaining asymptotically above g when wealth inequality is diverging. As wealth at the top of the distribution diverges, aggregate savings should rise, resulting in greater capital accumulation. The corresponding decline in the rate of return drives it asymptotically toward the growth rate.

In contrast, this paper develops the first microfounded general equilibrium model in which the rate of return remains asymptotically above the growth rate and wealth inequality diverges. The divergence of wealth for a non-zero fraction of agents implies that aggregate wealth diverges. Under fundamental valuation, this would require the discount rate to approach the growth rate. The presence of a rational bubble growing at a rate higher than that of the economy breaks this result and allows the model to align with Piketty (2014). This suggests that the longer an economy experiences rising wealth inequality alongside a rate of return significantly above the growth rate, the more likely it is to be on a bubbly path.[8] This distinction is crucial, since, as shown above, the dynamics of capital accumulation vary sharply between bubbly and non-bubbly equilibria. It also carries significant implications for taxation, as discussed in the next section.

[8] This interpretation is only suggestive: the divergence of aggregate wealth or inequality is inherently untestable, given that the asymptote is never observed.

2 Tax Policy Implications

Whether the economy is in a bubbly or non-bubbly equilibrium has important implications for taxation. The introduction of any wealth tax—and, in some cases, capital income taxes—in a bubbly equilibrium shifts the economy to a non-bubbly one. This can foster capital accumulation by preventing the crowding-out associated with a bubble.

2.1 Capital Tax Instruments: Income vs. Wealth Taxation

Capital income and wealth taxation are introduced into the previous model. The standard equivalence between the two breaks down in the presence of a bubble, which enters the tax base of the wealth tax but not that of the capital income tax.

Capital Income Tax. Consider first the capital income tax. In many sectors, the tax authority typically does not (and cannot) distinguish between rent and capital income, taxing them identically under the so-called capital income tax. Following this terminology, the capital income tax is defined as applying to both capital and rent income. The net-of-depreciation income paid to capital and to the rent-generating factor is taxed at a fixed rate τ^r. The post-capital income tax rate of return from period t to $t+1$ on capital is denoted by $R_{t+1}^{\text{post-CIT}}$:

$$R_{t+1}^{\text{post-CIT}} = 1 + (1 - \tau^r)(\alpha k_{t+1}^{\alpha-1} - \delta). \tag{2.22}$$

By the no-arbitrage condition, the post-capital income tax rate of return on the rent-generating factor is also equal to $R_{t+1}^{\text{post-CIT}}$:

$$R_{t+1}^{\text{post-CIT}} = (1 + g_{t+1})\frac{q_{t+1} + (1 - \tau^r)\gamma y_t}{q_t}. \tag{2.23}$$

The tax revenues from the capital income tax are given by T_t^r:

$$T_t^r = \tau^r \left[(\alpha k_t^\alpha - \delta k_t) + \gamma k_t^\alpha \right]. \tag{2.24}$$

Wealth Tax. A wealth tax is introduced in addition to the capital income tax. Before the production process, a fraction τ^a of each agent's wealth is collected. The rate of return between periods t and $t+1$, net of both the wealth and capital income taxes, is denoted by $R_{t+1}^{\text{after-tax}}$:

$$R_{t+1}^{\text{after-tax}} = (1 - \tau^a) R_{t+1}^{\text{post-CIT}}. \tag{2.25}$$

Let T_t^a denote wealth tax revenues:

$$T_t^a = \tau^a R_t^{\text{post-CIT}} \sum_i \lambda^i a_t^i. \tag{2.26}$$

Revenues from both the capital income tax and the wealth tax are assumed to be redistributed equally across households through lump-sum transfers. The budget constraint of households of type i then becomes:

$$c_t^i + a_{t+1}^i(1+g_{t+1}) = R_t^{\text{after-tax}} a_t^i + \zeta_t^i w_t + T_t^r + T_t^a. \tag{2.27}$$

(Non-)Equivalence Between Wealth and Capital Income Tax. When rates of return are homogeneous, capital income and wealth taxation are generally equivalent. A wealth tax at rate τ^a collects τ^a units of the consumption good for each unit of wealth. By contrast, a capital income tax at rate τ^r on the net-of-depreciation return yields revenues of $\tau^r(R-1)$ per unit of wealth. For any wealth tax, there thus exists an equivalent capital income tax such that $\tau^r = \tau^a/(R-1)$. This equivalence and its breakdown under heterogeneous returns are discussed extensively in Guvenen et al. (2023). With heterogeneous returns, a wealth tax penalizes low-return agents more than a capital income tax, while the reverse holds for high-return agents.

In fact, the equivalence requires a condition more restrictive than mere return homogeneity: at each point in time, the ratio of income paid to price must be uniform across assets. Two assets priced at their fundamental value can have the same value and rate of return, yet differ in the timing of their income payments. In that case, they are taxed the same under a wealth tax but differently under a capital income tax. This is what happens here when comparing capital and rent-generating factor taxation in a non-bubbly equilibrium. The no-arbitrage condition (2.5) ensures homogeneous returns, but does not necessarily imply that the following equality holds:

$$\frac{\gamma y_t}{f_{t-1}/(1+g_{t-1})} = \frac{\alpha y_t - \delta k_t}{k_t}. \tag{2.28}$$

Equation 2.28 illustrates the case in which the ratio of taxable income to taxable wealth is equalized across both assets. In the presence of capital gains on the rent-generating factor ($f_t > f_{t-1}$), this equality no longer holds, and the income-to-value ratio at time t is lower for the rent-generating factor.[9]

[9] One way to restore the equivalence between the two taxes would be to include capital gains, both realized and unrealized ones, as part of capital income. However, since capital gains tend to be taxed only upon realization and often at different rates than capital income, this non-equivalence is likely to be important in practice. Accordingly, the present definition of the capital income tax excludes capital gains from its tax base.

This mechanism can generate persistent differences in the effective taxation of capital and the rent-generating factor. As the economy grows asymptotically at rate g_t, so do rents and the fundamental value of the rent-generating factor. These capital gains on f_t are taxed under a wealth tax but not under a capital income tax. Taxing wealth amounts to taxing the rent-generating factor's fundamental value slightly more heavily than capital, compared with taxing capital income. This asymmetry is asymptotically proportional to the growth rate of the economy: the higher g, the greater the asymmetry. This consideration is, however, of secondary importance for the purposes of this paper.

More importantly for the present analysis, the rational bubble enters the tax base of the wealth tax but not that of the capital income tax. The rational bubble does not enter the production process and pays no dividends. Its taxable income-to-value ratio equals zero, reflecting the fact that it is held solely for its expected appreciation. It is not subject to the capital income tax and is included in the wealth tax base only. This distinction is central to why wealth and capital income taxes have different implications for capital accumulation and the equilibrium type in this model, which will be examined in the next subsection. Before turning to this analysis, it is worth noting that the requirement of homogeneous taxable income-to-value ratios for the equivalence between capital income and wealth taxation extends beyond the specific context of insatiable preferences for wealth. Its generalization and detailed analysis are outside the scope of this paper and are left for future research.

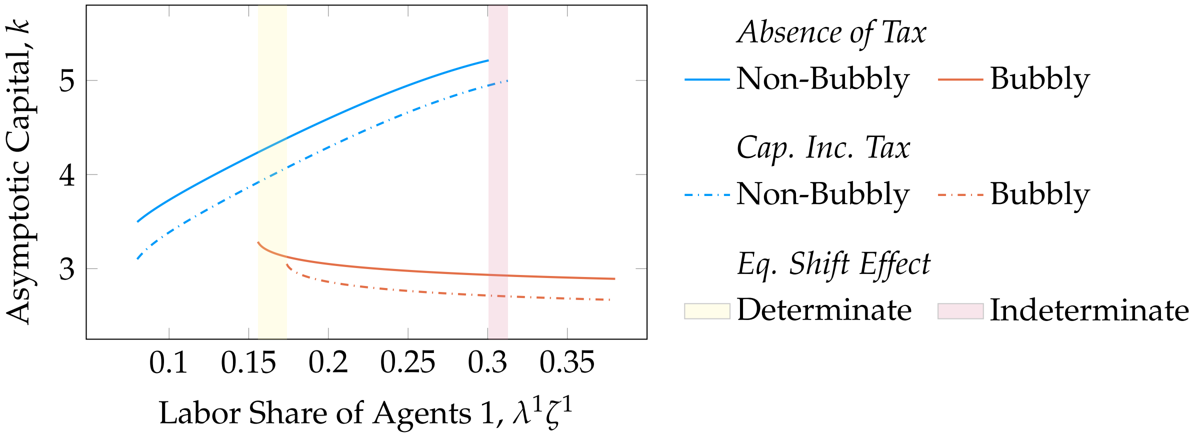
2.2 Capital Accumulation and Tax Policy

The effect of a tax on capital accumulation is ambiguous. Two mechanisms are at play. The standard disincentive effect of taxation reduces capital accumulation, while its *equilibrium-shift effect*—from a bubbly to a non-bubbly equilibrium, when present—operates in the opposite direction.

Asymptotic Capital under a Capital Income Tax. The effect of introducing a capital income tax on capital accumulation is first examined. The focus is on the asymptotic level of capital as a summary measure of the economy's long-run intensity of capital accumulation, abstracting from transition dynamics. To this end, the comparative exercise from Figure 2.2 is replicated with the inclusion of a capital income tax. Figure 2.3 displays the asymptotic capital stock as a function of labor income inequality, contrasting the no-tax scenario (solid lines) with a 25% capital income tax (dash-dotted lines, , $\tau^r = 0.25$). In both cases, the wealth tax rate remains set to zero.

Introducing a capital income tax shifts the curves representing the existence of non-bubbly and bubbly equilibria both downward and to the right. These two movements reflect the channels through which taxation influences the capital stock, with their

Figure 2.3: Equilibria with and without a Capital Income Tax



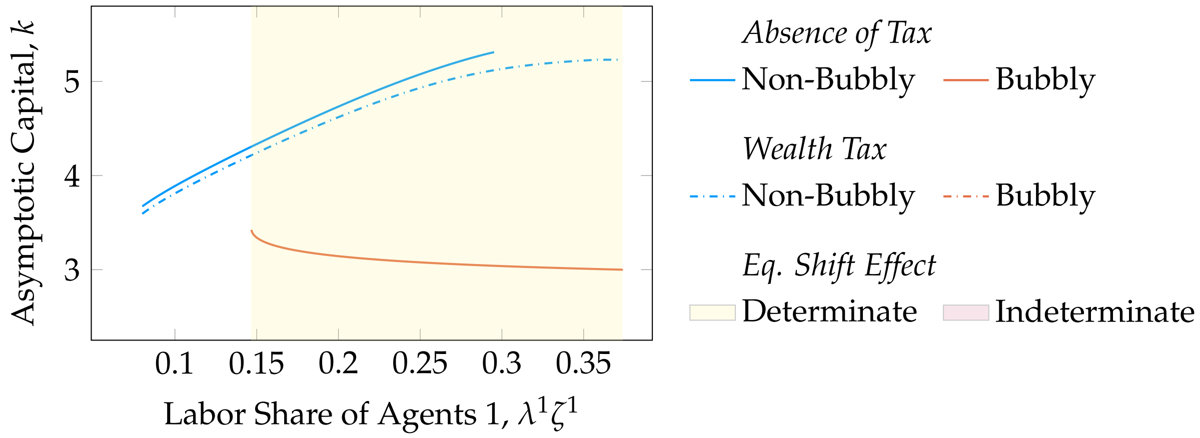
Notes: This figure shows the equilibrium type and corresponding asymptotic capital stock k as the labor share of agents 1, $\lambda^1\zeta^1$, varies, comparing a baseline without taxation to a scenario with a 25% capital income tax. Blue and red lines represent the non-bubbly and bubbly equilibria, respectively; dash-dotted lines correspond to the taxed case. Shaded areas indicate levels of labor income inequality at which introducing a 25% capital income tax in the bubbly no-tax equilibrium would necessarily (yellow) or possibly, depending on equilibrium selection (purple), shift the economy to a non-bubbly equilibrium.

magnitude increasing as the tax rate rises. The first channel is the standard disincentive effect. It is reflected in the downward shift of the dash-dotted lines relative to the solid ones. For a given level of inequality and a given equilibrium type, the capital stock is lower under a capital income tax. Taxation makes saving less attractive by lowering the post-tax rate of return. As a result, agents save less, which translates into a lower asymptotic capital stock.

The second channel is the equilibrium shift effect: a capital income tax can shift the equilibrium from bubbly to non-bubbly. The tax being progressive, it reduces income at the top of the distribution, and relaxes the upper-bound constraint on optimal consumption. It follows that the introduction of a tax increases the range of inequality levels for which a non-bubbly equilibrium exists and reduces the range for which a bubbly equilibrium does not exist. Graphically, this effect is reflected in the rightward shift of the curves. For a given level of taxation and labor inequality, the capital stock is always higher in the absence of a bubble, and the equilibrium shift effect has a positive impact on capital accumulation. In its presence, the impact of taxation on the capital stock is ambiguous and depends on the relative magnitude of the disincentive and equilibrium shift effects.

The equilibrium shift effect of a capital income tax operates only within a limited range of labor inequality levels. The levels of labor inequality, where the introduction of a 25% capital income tax can shift the equilibrium from bubbly to non-bubbly corresponds to the shaded area of Figure 2.3. Two subcases can be distinguished. The yellow-shaded area corresponds to levels of inequality for which two equilibria exist in the absence of taxation, but only one non-bubbly equilibrium remains when a 25%

Figure 2.4: Equilibria with and without a Small Wealth Tax



Notes: This figure shows the equilibrium type and corresponding asymptotic capital stock k as the labor share of agents 1, $\lambda^1\zeta^1$, varies, comparing a baseline without taxation to a scenario with a 0.1% wealth tax. Blue and red lines represent the non-bubbly and bubbly equilibria, respectively, while dashdotted lines correspond to the taxed case. Yellow-shaded areas indicate levels of labor income inequality at which introducing a 0.1% wealth tax in the bubbly no-tax equilibrium would necessarily shift the economy to a non-bubbly equilibrium.

capital income tax is introduced. In this zone, conditional on the economy initially being in a bubbly equilibrium, the equilibrium shift effect is determinate: the economy cannot remain in a bubbly state, as no bubbly equilibrium exists anymore. In contrast, the purple-shaded region identifies inequality levels with a single bubbly equilibrium in the absence of taxation, but two equilibria under a 25% capital income tax. In this zone, conditional on the economy initially being in a bubbly equilibrium, the equilibrium shift effect is indeterminate: the economy may either remain in the bubbly state or transition to the non-bubbly one.

Asymptotic Capital under a Wealth Tax. Attention now turns to the effects of wealth taxation on asymptotic capital. To this end, a comparative statics exercise is performed both with and without taxation. Figure 2.4 displays the asymptotic capital stock as a function of labor income inequality, contrasting the no-tax economy with an economy subject to a 0.1% wealth tax. This relatively low tax rate is chosen for reasons that become clear in the analysis. The same legend styles as in Figure 2.3 are retained to distinguish bubbly and non-bubbly equilibria, as well as the inequality levels for which the equilibrium-shift effect of the tax is determinate or indeterminate.

Figure 2.4 reveals a striking pattern: the wealth tax rules out any bubbly equilibrium. This is evident from the yellow-shaded area, which represents the range where the tax necessarily shifts a bubbly equilibrium to a non-bubbly one. It fully overlaps with the set of values for which a bubbly equilibrium exists in the absence of taxation. This result holds for any positive wealth tax rate. The value of τ^a is set to 0.1% to demonstrate that the result remains valid even for very small taxes. In this framework, a bubbly equilibrium generates a bubble whose size diverges relative to output. Taxing a fixed fraction of it, τ^a, would asymptotically yield diverging tax revenues, redistributed equally across households. Since not all agents are Scrooge McDuck—this is ruled out by assumption in the parametrization—this redistribution would raise asymptotic aggregate consumption above feasible levels, thereby violating the goods market clearing condition.

In the presence of a wealth tax, higher labor inequality always translate into a higher asymptotic capital stock. Conditional on remaining in a non-bubbly equilibrium, increasing labor inequality raises aggregate savings and the capital stock by redistributing income from low-saving to high-saving agents. In the absence of taxation, there are levels of inequality for which the capital stock would need to satisfy $R < 1 + g$ in order to absorb all savings in a non-bubbly equilibrium. This is precluded by the presence of the rent-generating factor. However, when R approaches $1 + g$ in the presence of a wealth tax, tax revenues become arbitrarily large, since so does f_t. This redistributes income from the high-saving type 1 agents t o the type 2 agents, thereby lowering aggregate savings. This mechanism explains why, under a wealth tax, a non-bubbly equilibrium exists for any level of labor income inequality.[10]

In a bubbly equilibrium, one can always find a sufficiently low wealth tax rate that, if implemented, raises the capital stock. It has been established that, regardless of the tax rate, a wealth tax always generates an equilibrium-shift effect that, in itself, increases the capital stock. At the same time, like a capital income tax, it entails a disincentive effect. Graphically, for any given level of inequality, the non-bubbly equilibrium features a higher capital stock in the absence of taxation than under a wealth tax. This disincentive effect is proportional to the tax rate. A small τ^a, such as the one considered in Figure 2.4, generates only a minor disincentive effect, which explains why the curves with and without taxation are close to each other. It follows that, in a bubbly equilibrium, there always exists a range of sufficiently low wealth tax rates for which the disincentive effect is dominated by the rate-invariant equilibrium shift effect.

3 Quantitative Assessment

Thus far, the analysis has focused on the qualitative features of a framework with insatiable preferences for wealth. The quantitative relevance of the Scrooge McDuck

[10]Note that a bequest tax and a wealth tax are equivalent in the context of a model with infinitely-lived agents. This could shed light on a country that appears to have followed a non-diverging path since the early 1990s: Japan. Japan exhibits stable wealth inequality, a growing capital-to-output ratio, and a declining wealth-to-output ratio since the burst of the Japanese asset price bubble. At the same time, it has a significant and highly progressive bequest tax, which accounts for more than 1% of total tax revenues (OECD, 2021) while taxing less than 9% of descendants (National Tax Agency, 2022).

mechanism is now assessed. The production economy is calibrated to the U.S. economy from 1989 onward. Despite being highly stylized, it reproduces the untargeted dynamics of aggregate wealth and wealth inequality. It serves as a benchmark for the counterfactual analysis of a 1.5 percent wealth tax.

3.1 Overview of the Exercise and Calibration

The economy is assumed to start from a non-bubbly steady state. This steady state is used to identify the internally calibrated parameters so as to match economic conditions in 1989.[11] These parameters are chosen to match the wealth-to-output ratio, the capital-to-output ratio, and the distribution of wealth. The economy is then subjected to an exogenous increase in labor income inequality, scaled to match its empirical counterpart. After the shock, the equilibrium can either shift to a bubbly state or remain non-bubbly. The model remains deliberately stylized and features four agent types corresponding to the top 1% $[99, 100]$, the top next 9% $[90, 99)$, the middle 40% $[50, 90)$, and the bottom 50% $[0, 50)$ percentiles of the wealth distribution.

By construction, income and wealth rankings are perfectly aligned across agent types. The levels of effective labor supply are selected to reflect this alignment. The parameter ζ_t^i measures the effective labor of type i relative to the population average at time t. It is calibrated by dividing the population into groups based on their position in the *wealth distribution* and measuring the corresponding labor income of each group. This is done using the Survey of Consumer Finances, which reports both individual pre-tax income and wealth. Because the model abstracts from life-cycle considerations, the calibration is based on data for households aged 35 to 54. The post-tax effective labor income is proxied using the transformation

$$\zeta_t^i = (1 - \tau^w)\zeta_t^{i,pre} + \tau^w, \tag{2.29}$$

where $\zeta_t^{i,pre}$ denotes non-tax-adjusted relative effective labor supply. This corresponds to a linear labor income tax at rate τ^w, rebated lump-sum.

Of the externally calibrated parameters, several warrant particular attention. The first is the discount factor, set to a relatively low value of $\beta = 0.9$. This calibration is consistent with the preference-for-wealth literature, where agents exhibit an additional saving motive beyond intertemporal consumption smoothing. In equilibrium, it implies a rate of return of about 5 percent, which is the typical targeted moment for choosing a discount factor around 0.95 in absence of preferences for wealth. Second, following Guvenen et al. (2023), the capital and labor income tax rates are calibrated using the

[11] 1989 is chosen as the starting point of the analysis because the Survey of Consumer Finances begins in its current form in that year.

Table 2.1: Baseline Calibration

Parameters	Description	Value	Source
Production			
γ	Rent factor share	0.035	Match Initial W/Y
α	Capital factor share	0.325	Labor Share of 0.64
δ	Capital depreciation rate	0.075	Standard
ζ_t^i	Pre-tax labor income share		SCF
Growth	Averages for 1989-2004 and 2004-2023		
g_t^N	Labor Input growth rate	1.3% - 1.2%	Bureau of Labor Statistics (BLS)
g_t^Z	Labor productivity growth rate	1.8% - 1.0%	Calculated from the BLS's TFP estimate
Taxation			
τ^r	Capital income tax rate	0.25	McDaniel (2007)
τ^w	Labor tax rate	0.225	McDaniel (2007)
τ^a	Wealth tax rate	0.0	
Preferences			
β	Discount rate	0.9	Mian et al. (2021)
θ	CRRA coefficient for consumption	2	Standard
ψ	Weight on taste for wealth	0.76	Joint Calibration
η	CRRA coefficient for wealth	1.43	Joint Calibration
$\underline{a}$	Stone-Geary parameter for wealth	-2.46	Joint Calibration
κ	Linear coefficient in the utility from wealth	0.0008	Joint Calibration

Notes: The internally calibrated preference parameters are jointly selected to match the wealth-to-output ratio, the capital-to-output ratio, and the wealth shares observed in 1989, assuming the economy was in a non-bubbly steady state at that time. The rent-generating factor share γ is chosen such that, in a steady state where the capital-to-output ratio equals 2.16 (its 1989 value), total wealth equals 3.16 (its 1989 value). To focus on long-term dynamics rather than business-cycle fluctuations, TFP and labor input growth are smoothed over time. In the initial steady state and from 1989 to 2004, it is set to its 1989-2004 average; from 2004 onward, it is set to its 2004-2023 average. This 2004 break accounts for the growth slowdown observed in the United States over the period considered.

estimates reported by McDaniel (2007). Wealth and bequest taxation are shut down in the baseline. Finally, to account for the growth slowdown observed over the sample period, the growth rates of labor-augmenting productivity, g_t^N, and TFP, g_t^Z, are set to their 1989-2004 averages from 1989 to 2004, and to their 2004-2023 averages thereafter. Although this adjustment represents an additional shock to the initial steady state, it is neither necessary nor sufficient to drive the economy from a non-bubbly to a bubbly equilibrium.

The internal calibration proceeds in two steps. First, the factor shares of rents and capital, γ and α, are chosen so that a non-bubbly steady state matching the 1989 wealth-to-output ratio also reproduces the 1989 capital-to-output ratio given a standard labor share of 64%. The rent share parameter is set to $\gamma = 3.5\%$, a relatively low value given that land rents alone are generally estimated to account for 5-10 percent of output. This modest value likely reflects capital mismeasurement, similar to the mechanisms that have historically driven the macro Tobin's q below one (see Eggertsson et al., 2021 for a discussion). The final calibration step jointly determines the preference parameters ψ, η, $\underline{a}$, and κ so as to match the 1989 wealth-to-output ratio and the corresponding wealth distribution. All the calibration details are summarized in Table 2.1.

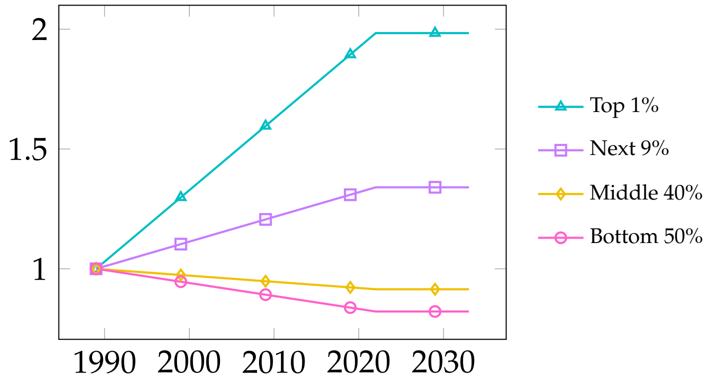
The fit of all targeted moments is summarized in Table 2.2. The model performs well overall, except for the wealth shares of the next 9 percent and the middle 40 percent. Owing to the perfect alignment between income and wealth distributions, this stylized framework cannot simultaneously match both groups' empirical shares: it underestimates the wealth share of the next 9 percent by about 9 percentage points and overestimates that of the middle 40 percent by a similar magnitude.

Table 2.2: Calibration Targets: Initial Steady-State vs. Data (1989)

Moments	**Data**	**Model**
Wealth-to-output	3.16	3.12
Capital-to-output	2.16	2.15
Top 1% Wealth Share	27%	29%
Next 9% Wealth Share	35%	26%
Middle 40% Wealth Share	33 %	42%
Bottom 50% Wealth Share	5 %	3%

This 1989 steady state is subsequently shocked by the rise in effective labor supply inequality illustrated in Figure 2.5. The ζ_t^i values are assumed to transition smoothly from their 1989 levels, computed using Equation 2.29, to their 2022 counterparts. The change in ζ_t^i, computed from the data, amplifies inequality: the higher an agent's position in the wealth distribution, the larger the increase (or the smaller the decrease) in its effective labor supply.

Figure 2.5: Labor Inequality Shock: Dynamics of ζ_t^i (1989 = 1)



Notes: This figure depicts the exogenous shock to ζ_t^i. Effective labor supply is computed from pre-tax labor income data across the wealth distribution, using the Survey of Consumer Finances and adjusted for a 22.5% linear labor income tax as in equation 2.29.

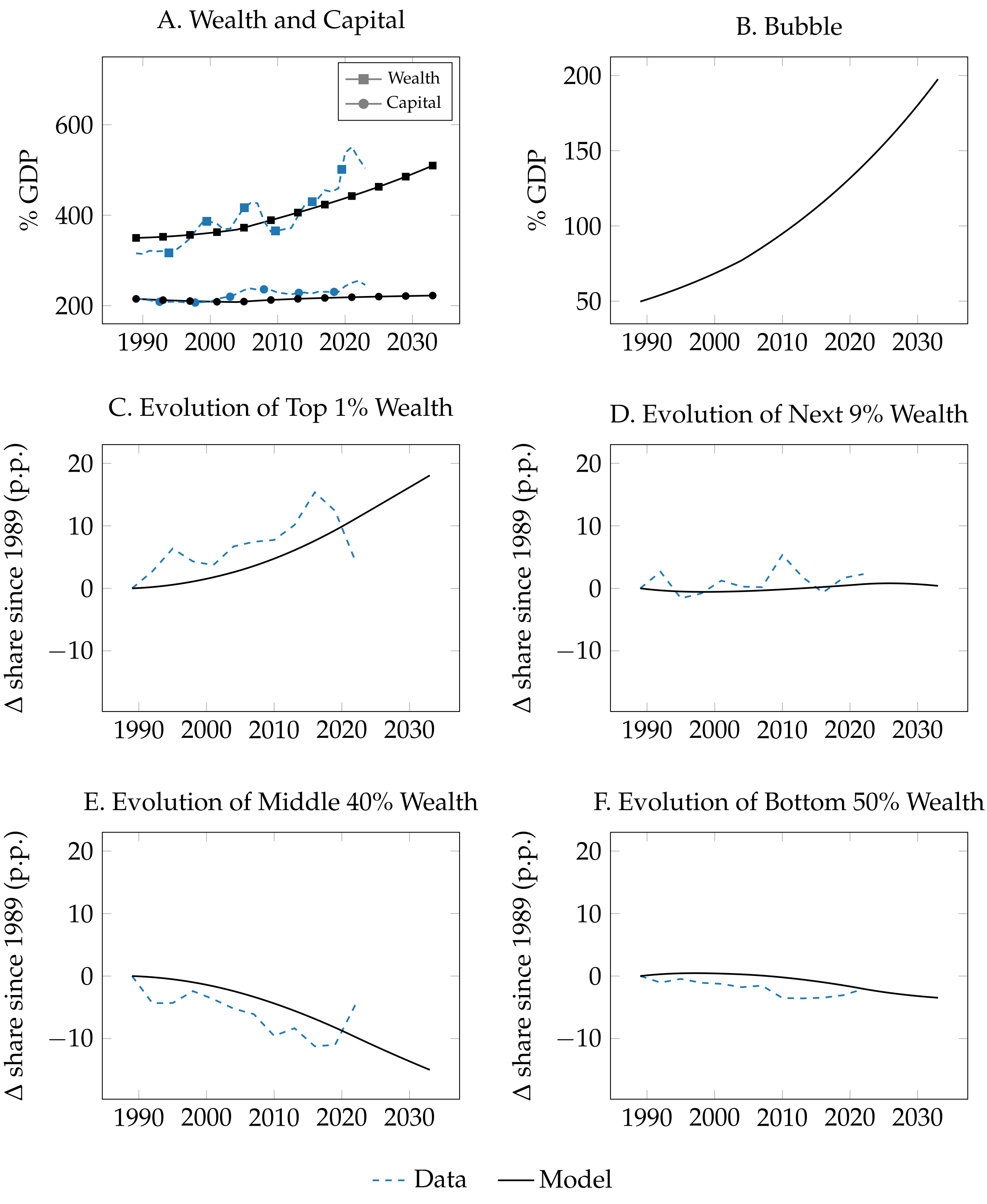
3.2 Results

Figure 2.6 displays the model's predicted adjustment following the labor income shock. Despite being calibrated to match the non-bubbly steady state, the model endogenously switches to a bubbly equilibrium after the shock. The transition path is uniquely determined, and no equilibrium-selection assumption underlies these results.

The model predicts an increase in wealth-to-output ratio driven by a rational bubble. Since agents have perfect foresight, a bubbly equilibrium requires the bubble to be present from the first period onward. The bubble therefore jumps, by assumption, from its steady-state value of zero to a positive value in 1989. This accounts for the model's predicted wealth-to-output ratio starting above its 1989 value. Designed to capture long-run dynamics, the model consistently underestimates wealth during booms and overestimates it during busts. Overall, the model replicates reasonably well both the speed and the magnitude of the increase in the wealth-to-output ratio observed in the data. In the simulated economy, this rise is largely driven by the rational bubble, which reaches about 150 percent of GDP by 2024.

The growth slowdown accelerates slightly the normalized bubble and capital growth. The growth rate of $(Z_t N_t)^{\frac{1-\alpha-\gamma}{1-\alpha}}$, denoted g_t, closely tracks that of the overall economy and equals 3 percent before 2004 and 2.1 percent thereafter. This break in trend makes it possible to visualize the effects of slower growth on the model's dynamics. The normalized bubble then evolves at rate $R_t - (1 + g_t)$, with the return on capital, $R_t - 1$, oscillating around 5.5 percent both before and after 2004. Overall, $R_t - (1 + g_t)$ rises slightly after 2004, accounting for the modest acceleration in the growth of the bubble-to-output ratio. Moreover, the slowdown in growth mechanically induces a modest rebound in the capital-to-output ratio. This rebound may help explain the rise in the capital-to-output ratio documented in the data.

Figure 2.6: Transition Dynamics in the Baseline Scenario



Notes: Model's transition starts from the 1989 non-bubbly steady state. It departs from this steady state and evolves along a bubbly path as the economy experiences an increase in effective labor supply inequality, depicted in Figure 2.5. In addition, the growth rates of labor productivity and labor input exhibit a slowdown after 2004, as reported in Table 2.1. The transition path is uniquely determined. Data on wealth are obtained from the World Inequality Database, capital stock data from the Bureau of Economic Analysis, and wealth share data from the Survey of Consumer Finances (conducted every three years).

Not only the aggregate wealth dynamics is well matched by the model but also its distribution. In both the data and the model, the main change in relative wealth shares occurs between the top 1 percent and the middle 40 percent. By 2019, the wealth share of the top 1 percent had increased by 12.4 percentage points, while that of the middle 40 percent had declined by 11 percentage points in the data. In the model, the corresponding changes are an increase of 9.29 percentage points and a decrease of 8.25 percentage points. The wealth share of the top 1 percent rises as they are Scrooge McDuck, while the rest of the population is not. As a result, their share follows a path that converges asymptotically to one. At the other end of the distribution, some agents must lose in terms of wealth share, and these losses are concentrated among the middle 40 percent. Note that this trend partially reverses in the last SCF wave (2022), likely reflecting temporary factors related to the COVID crisis and its aftermath that are not captured by the model.

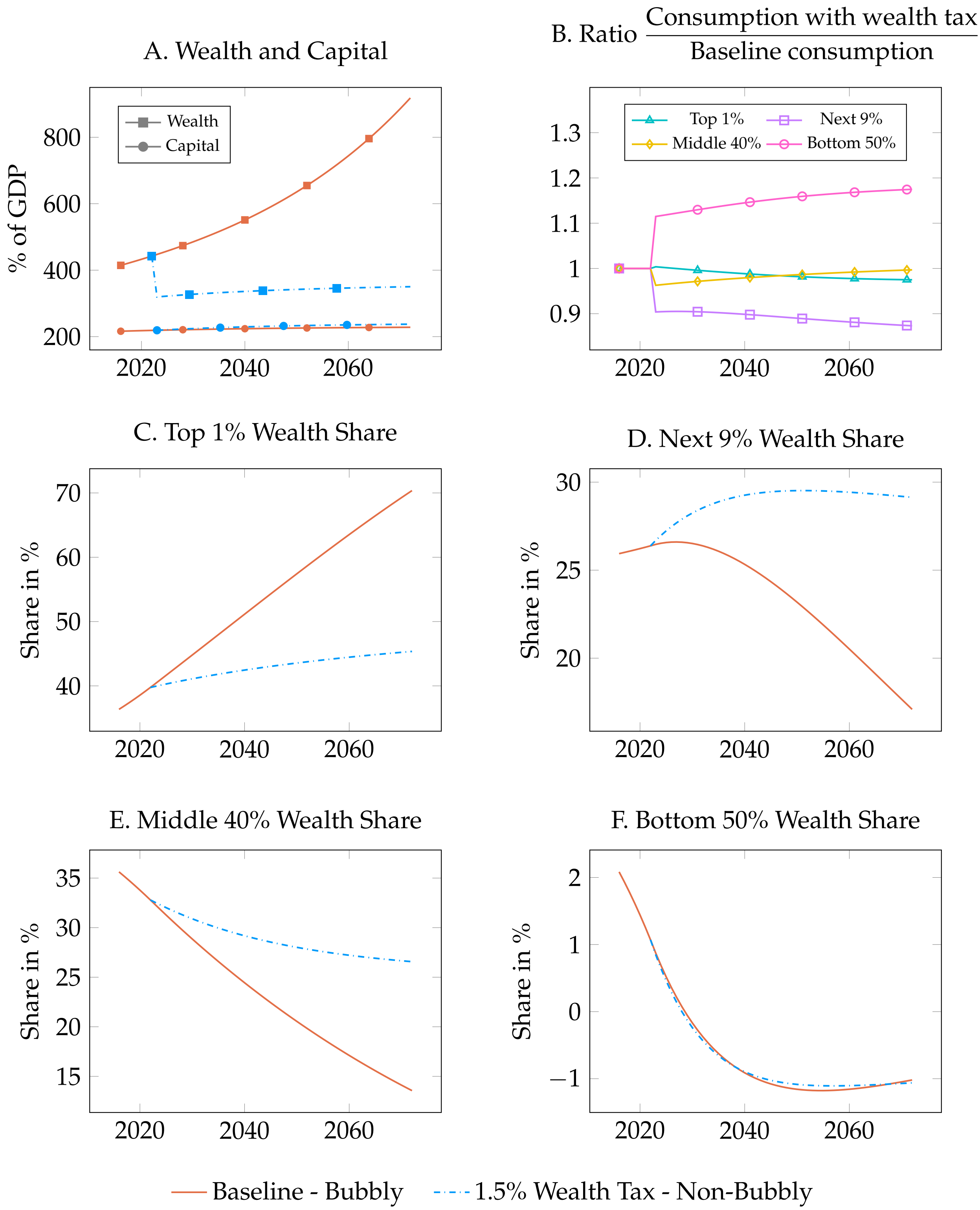
In both the data and the model, the wealth shares of the next 9 percent and the bottom 50 percent remain relatively stable, though for distinct reasons. The wealth share of the bottom 50 percent declines but cannot fall as much as that of the middle 40 percent, given its already low initial level. For the next 9 percent, two opposing forces offset each other over the period considered in the model. On the one hand, as with the middle 40 percent, their wealth share should decline since they are not Scrooge McDuck. On the other hand, they benefit from an increase in relative labor income after 1989 (Figure 2.5), which raises their savings. Yet this second effect is temporary, and their wealth share is projected to decline after 2030.

3.3 Introduction of a 1.5 Percent Wealth Tax

Having established that the model quantitatively replicates the observed wealth dynamics, attention now turns to counterfactual analysis. Specifically, the introduction of a 1.5 percent wealth tax from 2022 onward is considered. The tax is assumed to be unanticipated prior to its implementation. Figure 2.7 illustrates how the economy's trajectory under this policy departs from the baseline between 2016 and 2072.

The primary effect of the wealth tax is to eliminate the bubble. As shown in Section 2, any positive wealth tax rules out bubbly equilibria. Accordingly, the 1.5 percent tax induces an immediate drop in the wealth-to-output ratio and halts its divergence, which had previously been driven by the bubble. The burst of the bubble further affects capital stock, as the crowding-out effect of the bubble disappears. The wealth tax rate is small enough that its disincentive effect on capital accumulation is dominated by this equilibrium shift effect. Relative to the no-tax baseline, the tax raises the capital stock by 2.5 percent in 2032 and 5 percent in 2052, with corresponding increases in the capital-to-output ratio of 1.6 percent and 3.3 percent.

Figure 2.7: Prospective Transition Dynamics: Baseline Scenario vs. 1.5% Wealth Tax



Notes: The figure contrasts the baseline scenario (in red, continuing the model series from Figure 2.6) with a counterfactual scenario featuring a 1.5 percent wealth tax implemented in 2022 (in blue). Panel B shows, for each period and each type, the ratio of consumption c_t^i under the wealth tax to its baseline counterpart.

The wealth tax also modifies the evolution of wealth inequality. The presence of a wealth tax prevents any agent from accumulating unbounded wealth. Because the tax is proportional to wealth, doing so would entail an asymptotically infinite tax burden exceeding aggregate output. Consequently, in the presence of a wealth tax, type 1 agents no longer exhibit Scrooge McDuck consumption-saving behavior. Their wealth share no longer converges to one but stabilizes below 50 percent. The wealth shares lost by the top 1 percent accrue to the next 9 percent and the middle 40 percent. As the wealth of the bottom 50 percent oscillates around zero, their wealth share does as well and remains largely unaffected by the wealth tax.

This counterfactual experiment shows that the standard efficiency-redistribution trade-off vanishes in a bubbly equilibrium for sufficiently low wealth tax rates. In standard model, introducing a tax has a positive effect through redistribution but comes at the cost of lower output due to the disincentive effects of taxation. This mechanism can be further amplified in the presence of non-homothetic saving rates, as discussed in Morrison (2024). In contrast, within the present framework, the capital stock increases following the introduction of the 1.5 percent wealth tax. Taxation no longer entails an output cost, while its redistributive effect remains. This can be seen in Panel B of Figure 2.7: the higher an agent's position in the wealth distribution, the lower their consumption with the tax relative to without it. The only exception to this pattern is the top 1 percent. Their consumption falls by less than that of the next 9 percent. As they cease to be Scrooge McDuck and to hold a surplus wealth, the resulting reduction in saving boosts their consumption.

Yet a new trade-off may appear for the social planner between utility over consumption versus over wealth. Since small taxes do not entail an efficiency-redistribution trade-off, a social planner maximizing utility from consumption only would optimally implement a wealth tax. The 1.5 percent wealth tax considered here, for instance, would result in a consumption-equivalent gain of 7.26 percent. However, a social planner assigning the same weight to preferences for wealth as agents do would face a welfare cost due to the lower wealth-to-output ratio under taxation. In the present case, this cost is outweighed by the utility gains from consumption, with the consumption-equivalent welfare gain from the tax amounting to 4.7 percent. The weight that a social planner should assign to preferences for wealth likely depends on the micro-foundations of these preferences. Some motives, such as status or power seeking, suggest that preferences for wealth are essentially a zero-sum game and should not be valued by the planner. Others, such as the desire to accumulate or to leave bequests, are positive-sum motives, which would lead the planner to place a positive weight on preferences for wealth.

4 Conclusion

This paper proposes a parsimonious explanation for the rise in the wealth-to-output ratio, the stagnation of the capital-to-output ratio, and the increase in wealth inequality observed across advanced economies in recent decades. The core assumption of this framework is that agents derive *insatiable* utility from wealth. As a result, agents at the top of the income distribution accumulate a *surplus wealth* solely for the sake of holding it. They are defined as *Scrooge McDuck* agents, as they ultimately hold unbounded wealth while maintaining bounded consumption relative to output. Their surplus wealth drives asset prices above their fundamental value, leading to a rational bubble. The latter grows at a rate exceeding the economy's growth rate, enabling a disconnect between the wealth-to-output ratio and the capital-to-output ratio.

The Scrooge McDuck theory carries several implications. It suggests that wealth inequality may follow a diverging trajectory; that increases in income or wealth inequality do not necessarily translate into higher investment; and that the rate of return may remain permanently above the dividend-to-price ratio due to capital gains on a diverging rational bubble. By preventing unbounded wealth accumulation, a wealth tax, or some capital income taxation, shift the equilibrium from bubbly to non-bubbly, potentially fostering capital accumulation. In that case the standard efficiency-redistribution trade-off is replaced a trade-off between preferences for consumption and preferences for wealth .

This paper opens a research agenda for studying the wealth-to-output ratio as a diverging rather than a slowly converging process. Natural extensions include, for instance, modeling bubbly boom–bust cycles around a diverging long-run trend to better understand financially driven business cycles.

Bibliography

Maurice Allais. *L'impôt Sur Le Capital et la Réforme Monétaire.* Hermann, 1977.

Ricardo J Caballero, Emmanuel Farhi, and Pierre-Olivier Gourinchas. Rents, technical change, and risk premia accounting for secular trends in interest rates, returns on capital, earning yields, and factor shares. *American Economic Review*, 107(5):614–620, 2017.

Christophe Chamley. Optimal taxation of capital income in general equilibrium with infinite lives. *Econometrica*, pages 607–622, 1986.

John Cochrane. Wealth and taxes, part ii, January 2020. URL https://johnhcochrane.blogspot.com/2020/01/wealth-and-taxes-part-ii.html.

Peter A Diamond. National debt in a neoclassical growth model. *The American Economic Review*, 55(5):1126–1150, 1965.

Fernando Duarte and Carlo Rosa. The equity risk premium: a review of models. *Economic Policy Review*, (2):39–57, 2015.

Gauti B Eggertsson, Jacob A Robbins, and Ella Getz Wold. Kaldor and piketty's facts: The rise of monopoly power in the united states. *Journal of Monetary Economics*, 124: S19–S38, 2021.

Andreas Fagereng, Martin Blomhoff Holm, Benjamin Moll, and Gisle Natvik. Saving behavior across the wealth distribution: The importance of capital gains. Working paper, 2021.

Andreas Fagereng, Matthieu Gomez, Emilien Gouin-Bonenfant, Martin Holm, Benjamin Moll, and Gisle Natvik. Asset-price redistribution. Working paper, 2024.

Emmanuel Farhi and François Gourio. Accounting for macro-finance trends: Market power, intangibles, and risk premia. *Brookings Papers on Economic Activity*, 2018.

Alexandre Gaillard and Philipp Wangner. Wealth, returns, and taxation: A tale of two dependencies. Working paper, SSRN, 2023.

Paul Gomme, B Ravikumar, and Peter Rupert. The return to capital and the business cycle. *Review of Economic Dynamics*, 14(2):262–278, 2011.

Daniel L Greenwald, Martin Lettau, and Sydney C Ludvigson. How the wealth was won: Factor shares as market fundamentals. *Journal of Political Economy*, 133(4): 1083–1132, 2025.

Fatih Guvenen, Gueorgui Kambourov, Burhan Kuruscu, Sergio Ocampo, and Daphne Chen. Use it or lose it: Efficiency and redistributional effects of wealth taxation. *The Quarterly Journal of Economics*, 138(2):835–894, 2023.

Christian AL Hilber and Wouter Vermeulen. The impact of supply constraints on house prices in england. *The Economic Journal*, 126(591):358–405, 2016.

Òscar Jordà, Katharina Knoll, Dmitry Kuvshinov, Moritz Schularick, and Alan M Taylor. The rate of return on everything, 1870–2015. *The Quarterly Journal of Economics*, 134(3): 1225–1298, 2019.

Kenneth L Judd. Redistributive taxation in a simple perfect foresight model. *Journal of Public Economics*, 28(1):59–83, 1985.

Paul Krugman. Pride and prejudice and asset prices. URL https://www.nytimes.com/2021/10/05/opinion/low-interest-rates-monetary-policy.html.

Michael Kumhof, Romain Rancière, and Pablo Winant. Inequality, leverage, and crises. *American Economic Review*, 105(3):1217–1245, 2015.

Dmitry Kuvshinov and Kaspar Zimmermann. The expected return on risky assets: International long-run evidence. Working paper, SSRN, 2021.

Cara McDaniel. Average tax rates on consumption, investment, labor and capital in the oecd 1950-2003. *Manuscript, Arizona State University*, 2007.

Atif Mian, Ludwig Straub, and Amir Sufi. The saving glut of the rich. Working paper, National Bureau of Economic Research, 2020.

Atif Mian, Ludwig Straub, and Amir Sufi. Indebted demand. *The Quarterly Journal of Economics*, 136(4):2243–2307, 2021.

Jean-Baptiste Michau, Yoshiyasu Ono, and Matthias Schlegl. The preference for wealth and inequality: Towards a piketty theory of wealth inequality. Working paper, 2025.

Wendy Morrison. Redistribution and investment. Working paper, 2024.

John Muellbauer. Housing, debt and the economy: A tale of two countries. *National Institute Economic Review*, 245:R20–R33, 2018.

National Tax Agency. The 148th national tax agency annual statistics report. Online report, 2022. Available at: https://www.nta.go.jp/english/publication/statistics/index.htm.

OECD. *Inheritance Taxation in OECD Countries*. Number 28 in OECD Tax Policy Studies. OECD Publishing, Paris, 2021.

Thomas Piketty. *Capital in the twenty-first century*. Harvard University Press, 2014.

Thomas Piketty, Emmanuel Saez, and Gabriel Zucman. Rethinking capital and wealth taxation. *Oxford Review of Economic Policy*, 39(3):575–591, 2023.

Ricardo Reis. The constraint on public debt when r< g but g< m. CEPR Discussion Papers 15950, 2021.

Ricardo Reis. Which r-star, public bonds or private investment? measurement and policy implications. *LSE manuscript*, 2022.

Changyong Rhee. Dynamic inefficiency in an economy with land. *The Review of Economic Studies*, 58(4):791–797, 1991.

Emmanuel Saez, Danny Yagan, and Gabriel Zucman. Capital gains withholding. *University of California Berkeley*, 2021.

Ludwig Straub and Iván Werning. Positive long-run capital taxation: Chamley-judd revisited. *American Economic Review*, 110(1):86–119, 2020.

Chapter 2 – Appendix

A Numerical Algorithm

This appendix describes the algorithm used to compute the converging and diverging transition paths in the production economy. The paper introduces two key innovations to solve the model numerically:

1. It develops a method to compute divergent bubbly transitions arising from insatiable preferences for wealth.
2. It provides a new approach to solving the household partial-equilibrium problem over long horizons in the presence of preferences for wealth.

A.1 Converging Non-Bubbly Equilibrium

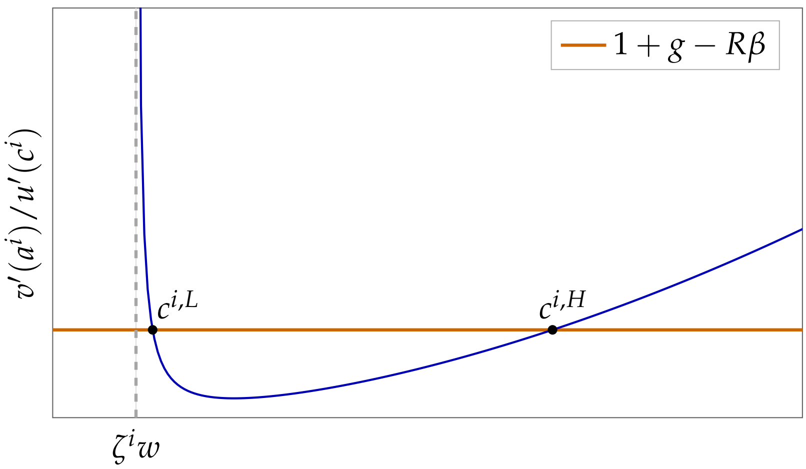
The numerical solution of a converging equilibrium is described first. In this case, all variables converge asymptotically to their balanced growth path (BGP), and they are assumed to have reached their BGP values by the terminal period of the algorithm, $t = T$. The exposition of the numerical procedure proceeds in three steps. It begins with the computation of the asymptotic BGP. The solution of the converging transition path is then explained. Finally, a detailed description is provided of the solution to households' partial-equilibrium optimization problems.

A.1.1 Asymptotic Balanced Growth Path

Consider first the numerical computation of the balanced growth path (BGP) to which the non-bubbly equilibrium converges. By definition, this BGP is characterized by the vector $\left\{\{c^i, a^i\}_{i\in\{1,2,\ldots,I-1\}}, c^I, k, q\right\}$ that satisfies the stationary versions of the households' Euler equation and flow budget constraint (2.15)–(2.11), together with market clearing in the capital (2.18) and rent-generating factor (2.19) markets.

The procedure used to compute the BGP reduces to solving a one-dimensional root-finding problem in aggregate capital. For a candidate value of aggregate capital k, households' BGP problem is solved to obtain individual consumption levels $\{c^i\}_{i\in\{1,\ldots,I\}}$ (see below) and thus aggregate consumption $\sum_i \lambda^i c^i$. Independently, the same candidate k determines output and investment, and delivers an aggregate-consumption level from goods-market clearing condition (2.17). The value of k is chosen such that aggregate aggregate consumption implied by households' optimality coincides with the aggregate

Figure 2.8: Asymptotic Value of $\frac{v'(a^i)}{u'(c^i)}$ as a function of c^i



consumption implied by the resource constraint. The market for the rent-generating factor clears by Walras' law.

The only non-standard step concerns the mapping from aggregate capital k to individual asset holdings a^i. With preferences for wealth, a given steady-state level of capital k—and thus a given pair of prices (R, w)—need not pin down a unique pair (c^i, a^i). This can be seen by writing agent i's steady-state version of their Euler equation 2.15,

$$1 + g - R\beta \;=\; \frac{v'(a^i)}{u'(c^i)}, \tag{2.30}$$

and combining it with the steady-state relationship between wealth and consumption implied by the flow budget constraint (2.11),

$$a^i \;=\; \frac{c^i - \zeta^i w}{R - (1+g)}. \tag{2.31}$$

When marginal utility of consumption declines faster than marginal utility of wealth, as in the calibration with $\theta > \eta$, the right-hand side of (2.30), viewed as a function of c^i after substituting (2.31), is convex. When labor income is positive ($\zeta^i w > 0$), the steady-state Euler condition admits two consumption levels consistent with the same (R, w), denoted $c^{i,L}$ and $c^{i,H}$, as shown in Figure 2.8.

As discussed in detail in Section 3 of Michau et al. (2025), under standard wealth-in-the-utility preferences and for a fixed steady-state level of capital k, the equilibrium associated with $c^{i,H}$ is unstable. In their framework with a continuous distribution of wealth, virtually no agent converges to this asymptotic level of consumption in equilibrium. In the present framework, the implication is slightly different because the wealth distribution features mass points: at most one type i can converge to the high-consumption root $c^{i,H}$. Attention is restricted to configurations in which the wealthiest agents (type 1) may be on the $c^{1,H}$ root. Operationally, the root-finding exercise described above is implemented under both candidate selections for type 1, namely $c^1 = c^{1,L}$ and $c^1 = c^{1,H}$. In the joint calibration reported in Subsection 3.1, the root selection is left open, and the root that provides the best fit to the data is retained. In the baseline calibration of the 1989 BGP, it is the case that $c^1 = c^{1,H}$.

A.1.2 Converging Transitions

Having computed the balanced growth path to which the economy converges by period T, the solution method is now described for the transition dynamics over $t = 0, \ldots, T$.

The main difficulty in solving for the converging transition dynamics is that the initial price of the rent-generating factor is unknown, implying that agents' initial wealth is also endogenous. In the non-bubbly equilibrium, the value of the rent-generating factor coincides with its fundamental price at all dates. While its asymptotic value f_T is pinned down by the BGP, the initial value f_0, and, more generally, the entire path $\{f_t\}_{t=0}^T$, must be determined jointly with the equilibrium transition path of the aggregate capital stock.

To address this challenge, the algorithm combines (i) an *outer loop* that maps a guess for the aggregate capital path into individual wealth dynamics and updates the capital path accordingly, and (ii) an *inner loop* that, conditional on the households' wealth paths, updates the capital sequence so as to satisfy asset-market clearing and the fundamental pricing equation for the rent-generating factor (with terminal condition f_T).

Outer Loop of the Algorithm

The algorithm starts from an initial guess for the aggregate capital path, $\{k_t^{guess}\}_{t=0}^T$. Using the pricing equation of the rent-generating factor (2.7) and the terminal condition f_T, the entire sequence $\{f_t^{guess}\}_{t=0}^T$ is then recovered by backward induction. Given $\{k_t^{guess}\}_{t=0}^T$, factor prices $\{R_t^{guess}, w_t^{guess}\}_{t=0}^T$ are implied by firms' optimality conditions.

Step 1: Truncating the Capital Path. The iterates $\{k_t^{guess}\}_{t=0}^T$ may feature extreme capital levels that should not be used to solve the household problem, as they would generate implausible consumption and saving choices and hinder convergence of the algorithm. To mitigate this issue, a truncated path $\{k_t^{trunc}\}_{t=0}^T$ is computed by imposing lower and upper bounds, $k_{\min}$ and $k_{\max}$. Let t' denote the *first* period such that $k_{t'}^{guess} \notin [k_{\min}, k_{\max}]$. The path $\{k_t^{trunc}\}_{t=0}^T$ is defined as

- if $k_{t'}^{guess} < k_{\min}$, set $k_t^{trunc} = k_{\min}$ for all $t \geq t'$;
- if $k_{t'}^{guess} > k_{\max}$, set $k_t^{trunc} = k_{\max}$ for all $t \geq t'$;

- in either case, set $k_t^{trunc} = k_t^{guess}$ for all $t < t'$.

Step 2: Household Optimization in Partial Equilibrium. Given f_0^{guess} and the price sequences implied by $\{k_t^{trunc}\}_{t=0}^{T}$, the household problem is solved in partial equilibrium following the method described in Subsubsection A.1.3. This yields consumption and wealth paths for each type, $\{c_t^{i,guess}, a_t^{i,guess}\}_{t=0}^{T}$.

Step 3: Inner Loop to Update the Guess of Capital Path In this thrid step, the inner loop described in Subsubsection A.1.2 delivers a capital path $\{k_t^{inner}\}_{t=0}^{T}$ such that:

- the associated sequence $\{f_t^{inner}\}_{t=0}^{T}$ is consistent with the fundamental pricing equation (2.7) and the terminal condition f_T, given $\{k_t^{inner}\}_{t=0}^{T}$; and
- asset markets clear at each date, i.e. the market value of aggregate assets (capital and the rent-generating factor) equals aggregate household wealth implied by $\{a_t^{i,guess}\}_{t=0}^{T}$.

Step 4: Updating the Capital Guess. The capital guess is updated as a period-by-period weighted average between the previous guess and the newly computed capital path:

$$k_t^{guess\prime} = \omega^{outer} k_t^{guess} + (1 - \omega^{outer}) k_t^{inner}, \tag{2.32}$$

with $\omega^{outer} \in [0, 1)$.

The Manhattan distance between $\{k_t^{guess}\}_{t=0}^{T-1}$ and $\{k_t^{new}\}_{t=0}^{T-1}$ is computed. If this distance falls below the tolerance threshold, the algorithm is deemed to have converged, and $\{k_t^{guess}\}_{t=0}^{T}$ is retained as the equilibrium capital path. If the distance exceeds the threshold and the maximal number of iterations has not yet been reached, the iteration continues.

Inner Loop of the Algorithm

Step 3 of the outer loop is now described in detail. It consists in finding a capital path $\{k_t^{inner}\}_{t=0}^{T}$ and a rent-generating factor price path $\{f_t^{inner}\}_{t=0}^{T-1}$ that are jointly consistent with (i) the fundamental pricing equation of the rent-generating factor (2.7) and (ii) asset-market clearing at all dates, taking as given the households' wealth dynamics $\{a_t^{i,guess}\}_{t=0}^{T}$:

$$\sum_i \lambda^i a_{t+1}^i = k_{t+1} + f_t, \tag{2.33}$$

where the left-hand side denotes aggregate wealth carried into period $t + 1$, and the right-hand side is the market value of aggregate assets (capital and the rent-generating factor).

The inner loop starts from a guess $\{k_t^{inner}\}_{t=0}^T$ initialized at the current outer-loop guess, i.e. $k_t^{inner} = k_t^{guess}$, and iterates on the following steps until convergence.

Step 1: Truncating the Capital Path. The iterates $\{k_t^{inner}\}_{t=0}^T$ may feature extreme capital levels, which in turn generate implausible factor prices and can hinder the convergence of the algorithm. To mitigate this issue, a truncated path $\{k_t^{trunc}\}_{t=0}^T$ is constructed using the same method as in the outer loop described above.

Step 2: Updating the Rent-generating Factor Price. Given the truncated capital path $\{k_t^{trunc}\}_{t=0}^T$ and the implied sequences for factor prices and output, the associated fundamental-value path of the rent-generating factor $\{f_t^{inner}\}_{t=0}^{T-1}$ by backward induction from the terminal condition f_T, using the pricing equation (2.7).

Step 3: Capital Path Guess from Asset-Market Clearing. Given the updated price path $\{f_t^{inner}\}$ and the wealth dynamics $\{a_t^{i,guess}\}$, a new implied capital sequence is computed $\{k_t^{inner,bis}\}_{t=0}^T$ by imposing the asset market to clear (2.33) period by period,

$$k_{t+1}^{inner,bis} = \sum_i \lambda^i a_{t+1}^{i,guess} - f_t^{inner}, \qquad t = 0, \dots, T-1. \tag{2.34}$$

Step 4: Updating the Capital Guess. The capital path is updated using relaxation:

$$k_t^{inner\prime} = \omega^{inner} k_t^{inner} + (1 - \omega^{inner}) k_t^{inner,bis}, \qquad \omega^{inner} \in [0, 1). \tag{2.35}$$

Steps 1–4 are iterated until the distance between successive iterates of $\{k_t^{inner}\}_{t=0}^T$ falls below a pre-specified tolerance level (or until a maximal number of iterations is reached). Upon convergence, this procedure delivers a fixed point $\{k_t^{inner}\}_{t=0}^T$ together with the associated price sequence $\{f_t^{inner}\}_{t=0}^{T-1}$.

A.1.3 Household Optimization Problem in Partial Equilibrium

Having described the general structure of the algorithm, a detailed exposition of the solution to households' partial-equilibrium optimization problem is now provided. Since this section focuses exclusively on the converging transition, attention is restricted to the partial equilibrium of non–Scrooge McDuck agents (which, in the present case, comprise the entire set of agents). The partial-equilibrium problem of Scrooge McDuck agents is discussed separately in the diverging-equilibrium section, Subsubsection A.2.3.

Consider the problem of a non-Scrooge McDuck agent i. The presence of preferences for wealth does not fundamentally change the solution method for the optimization problem in partial equilibrium. The objective is to find the path $\{c_t^{i,\star}\}_{t=0}^{T-1}$ such that:

- the budget constraint (2.11) and the Euler equation (2.15) of agent i are satisfied for all $t \in [0 : T-1]$ given the paths of rate of return $\{R_t^{trunc}\}_{t=0}^{T}$, and wages, $\{w_t^{trunc}\}_{t=0}^{T}$,
- $a_T^i = a^i$, with a^i the asymptotic value of the wealth of type i,
- a_0^i is equal to its value determined by the guess of f_0 and the exogenous values of the initial capital stock, k_0, and the wealth shares of each agent type.

To do so, the algorithm is initialized with two arbitrarily chosen values of c_0^i, denoted $c_0^{i,min}$ and $c_0^{i,max}$, which are expected to lie respectively below and above the equilibrium value $c_0^{i,\star}$. For simplicity, the typical choice is $c_0^{i,min} = 0$. The following steps are then iterated:

1. The test value is defined as $c_0^{i,test} = \frac{c_0^{i,min} + c_0^{i,max}}{2}$
2. The paths of consumption, $\{c_t^{i,test}\}_{t=0}^{T-1}$, and wealth, $\{a_t^{i,test}\}_{t=0}^{T}$, associated with $c_t^{i,test}$ are computed forward given the budget constraint (2.11) and the Euler equation (2.15) of agent i
3. If $a_T^{i,test} \geq a_T^i$, the agent has underconsumed compared to the solution of the problem and the new value of $c_0^{i,min}$ is set to $c_t^{i,test}$. Conversely, if $a_T^{i,test} \leq a_T^i$, $c_0^{i,max}$ is set to $c_t^{i,test}$.
4. If the difference $c_0^{i,max} - c_0^{i,min}$ falls below a pre-specified tolerance level, the algorithm is stopped.

Without preferences for wealth, this algorithm converges to a solution coherent with the asymptotic level of wealth, a^i, for reasonable precision levels.

However, when agents derive utility directly from wealth, a snowball effect can occur: two very close initial consumption choices, c_0^i, may lead to sharply different terminal wealth outcomes, a_T^i. The lower the initial consumption level, the more wealth the agent accumulates in period 1, a_1^i, as implied by the budget constraint. This lowers the marginal utility of wealth, $v'(a_1^i)$, thereby reducing the consumption level, c_{t+1}^i consistent with the Euler equation (2.15). The resulting increase in wealth accumulation further reduces the marginal utility of wealth, reinforcing the initial deviation. Repetition of this mechanism over time generates the snowball effect. This is particularly an issue when the two close values of initial consumption, $c_0^{i,min}$ and $c_0^{i,max}$, at which the partial-equilibrium algorithm stops, both generate terminal wealth levels far from the target, i.e. $a_T^i(c_0^{i,min}) \gg a_T^i$ and $a_T^i(c_0^{i,max}) \ll a_T^i$. In this case, the algorithm incorrectly interprets the bracket as having converged, even though both candidate paths are off target.

A new method is proposed to overcome this problem. Since the issue arises from a snowball effect that becomes more severe as the transition horizon increases, the method decomposes the transition into a sequence of shorter sub-transitions. Consider two candidate initial consumption levels, $c_0^{i,\min}$ and $c_0^{i,\max}$, which generate terminal wealth levels above and below the target, respectively. The associated consumption and wealth trajectories bracket the equilibrium paths. Although the two trajectories eventually diverge substantially, they remain nearly indistinguishable over an initial segment of the transition. Over this interval, both trajectories can be regarded as approximate equilibrium paths up to a prescribed tolerance level. Only afterward does the distance between them become sufficiently large for both paths to be considered off-equilibrium.

The proposed algorithm exploits this observation. Let t^{div} denote the first date at which the distance between the two wealth trajectories exceeds a pre-specified tolerance threshold. The transition is split at t^{div}. The portion of the trajectories prior to t^{div} are retained as approximations of the equilibrium path, while the search for the equilibrium consumption path is restarted from t^{div} onward. Because the remaining horizon is shorter, the snowball effect is reduced and the procedure becomes more accurate. Repeating this operation allows long transition paths to be computed despite the presence of strong wealth-preference amplification effects.

Two vectors are constructed to record the equilibrium path: a consumption vector, $\{c_t^{i,*}\}_{t=0}^{T-1}$, and a wealth vector, $\{a_t^{i,*}\}_{t=1}^{T}$. Let t^{conv} denote the first period for which the equilibrium value of consumption has not yet been determined, initialized at $t^{conv} = 0$. Two arbitrary consumption values are chosen to bracket the equilibrium initial consumption level, $c_0^i \in [c_{(1)}^{i,min}, c_{(1)}^{i,max}]$ at iteration 1. The algorithm iterates then over the following steps:

1. The previous partial-equilibrium algorithm is run from period t^{conv} to T with starting wealth $a^i_{t^{conv}}$ and bracket for $c^i_{t^{conv}}$ being $[c_{(n)}^{i,min}, c_{(n)}^{i,max}]$ at iteration n. The divergence period t^{div} is defined (or updated) as the earliest period for which the difference between the two asset paths exceeds a prescribed tolerance threshold.

2. The consumption and wealth values implied by the two candidate paths prior to the divergence period are collected as sufficiently accurate approximations of the equilibrium path. For $t \in [t^{conv}, t^{div}]$, consumption is set equal to the average of the two candidate paths,

$$c_t^{i,*} = \frac{1}{2}\left(c_t^i(c_{(n)}^{i,min}) + c_t^i(c_{(n)}^{i,max})\right), \tag{2.36}$$

and for $t \in [t^{conv}+1, t^{div}+1]$, wealth is set equal to the corresponding average,

$$a_t^{i,*} = \frac{1}{2}\left(a_t^i(c_{(n)}^{i,min}) + a_t^i(c_{(n)}^{i,max})\right). \tag{2.37}$$

3. The equlibrium consumption at t^{div} lies in the interval $[c_{(n+1)}^{i,\min}, c_{(n+1)}^{i,\max}]$, where $c_{(n+1)}^{i,\min}$ and $c_{(n+1)}^{i,\max}$ are defined as:

$$c_{(n+1)}^{i,\min} = c_{t^{div}}^i\left(c_{(n)}^{i,\min}\right), \quad \text{and,} \quad c_{(n+1)}^{i,\max} = c_{t^{div}}^i\left(c_{(n)}^{i,\max}\right) \tag{2.38}$$

   A value $a^{intermediate} \in \left[a_{t^{div}}^i(c_{(n)}^{i,max}), a_{t^{div}}^i(c_{(n)}^{i,min})\right]$ is chosen such that, when the simulation is restarted from $a^{\text{intermediate}}$ at date t^{div}, the consumption paths associated with $c_{n+1}^{i,\min}$ and $c_{n+1}^{i,\max}$ generate terminal wealth levels above and below the target, respectively.

4. The new convergence period is set to $t^{conv} = t^{div}$, and wealth at that date is initialized at $a_{t^{conv}}^i = a^{\text{intermediate}}$.

Iterating this loop until $t^{conv} = T$ allows long transition paths to be computed in the presence of preferences for wealth. The solution of the problem is obtained from the two vectors constructed in the second step: $\{c_t^{i,*}\}_{t=0}^{T-1}$ and $\{a_t^{i,*}\}_{t=0}^{T}$.

A.2 Diverging Bubbly Equilibrium

Having described the numerical solution of converging equilibria, attention now turns to the computation of diverging equilibria.

As in the converging case, all variables that follow a BGP asymptotically are assumed to have reached their BGP values by the terminal period of the algorithm, $t = T$. The exposition follows the same sequence as in the converging case: the computation of the asymptotic BGP variables is described first, followed by the transition dynamics over $t = 0, \dots, T$ and the solution of households' partial-equilibrium problems.

A.2.1 Asymptotic Value of Converging Variables

Consider first the numerical computation of the asymptotic values of the variables that converge to a balanced growth path (BGP). These asymptotic objects are summarized in the vector $\left\{\{c^i, a^i\}_{i\in\{2,\dots,I\}}, c^1, k\right\}$. The computation mirrors the converging case, except for type 1 agents, who are the only Scrooge McDuck agents by assumption. In a bubbly equilibrium, their wealth diverges, while their marginal utility of wealth converges to κ:

$$\lim_{t\to\infty} v'\left(a_t^1\right) = \kappa. \tag{2.39}$$

Their asymptotic consumption is then uniquely pinned down by their steady-state Euler equation,

$$1+g-R\beta \;=\; \frac{\kappa}{u'(c^1)}. \tag{2.40}$$

A.2.2 Diverging Transitions

The solution method for the transition dynamics over $t=0,\ldots,T$ is now described.

The main challenge in solving the diverging transition dynamics is the presence of a rational bubble that itself diverges over time. Both its initial and terminal (at $t=T$) value are endogenous, and must be determined jointly with the equilibrium transition path of the economy. To address this difficulty, the initial value q_0 of the rent-generating factor is iteratively guessed and updated until it converges to its equilibrium level. This subsection describes the solution of the model for a given guess of q_0 (the inner loop of the algorithm), before turning to the procedure used to update this guess (the outer loop).

Inner Loop: Solving the Transition for a Guess of q_0.

The equilibrium transition dynamics are solved conditional on a given initial guess for the rent-generating factor price, q_0. Starting from an initial guess for the paths of consumption $\{c_t^{i,guess}\}_{t=0}^{T-1}$ for all agents i, the following steps are iterated:

Step 1: Updating the Capital Guess. A guess of the transition path of capital is determined from the consumption paths $\{c_t^{i,guess}\}_{t=0}^{T-1}$ of all agents i. The procedure proceeds *backward*. The procedure starts from $k_T^{guess}=k$, with k the asymptotic capital stock, whose value is known. Using the goods market-clearing condition (2.17), one can then recover the value of k_t^{guess} based on k_{t+1}^{guess} and $\{c_t^{i,guess}\}_i$. This value corresponds to the unique positive k_t^{guess} that satisfies the goods market-clearing condition:

$$(1-\delta)k_t^{guess}+(k_t^{guess})^{\alpha}=k_{t+1}^{guess}(1+g_{t+1})+\sum_i \lambda^i c_t^{i,guess}. \tag{2.41}$$

Through iteration, a complete guess for the path of capital $\{k_t^{guess}\}_{t=0}^{T}$ is obtained. By construction, it is consistent with the consumption paths $\{c_t^{i,guess}\}_{t=0}^{T-1}$ and with the equilibrium value of k_t at $t=T$. However, it may be inconsistent with the initial capital stock k_0.

Step 2: Deriving the Truncated Capital Path. This method of step 1 to derive the capital guess can give extreme values of capital. When used directly to compute the solutions to the household optimization problem in partial equilibrium, this approach is likely to trigger a snowball effect. If the level of capital exceeds the value to which it should converge given the guess of q_0, agents receive higher wages and initial wealth. This may translate into too high consumption and thus too high capital, reinforcing the initial deviation.

A modified version of the capital guess must therefore be used to compute the wages and rates of return that enter the household optimization problem in partial equilibrium. Two levels of capital are defined arbitrarily, corresponding to the minimum and maximum values that capital is allowed to take in the partial equilibrium: k_{min} and k_{max}. All values of capital in the modified path lie within the interval $[k_{min}, k_{max}]$, and this adjusted series is referred to as the *truncated* capital path guess. Define t' as the latest period for which k_t^{guess} lies outside the interval $[k_{min}, k_{max}]$. The truncated capital path, $\{k_t^{trunc}\}_{t=0}^{T}$, is defined as follows:

- if $k_{t'}^{guess} < k_{min}$, set $k_t^{trunc} = k_{min}$ for all $t \leq t'$,
- if $k_{t'}^{guess} > k_{max}$, set $k_t^{trunc} = k_{max}$ for all $t \leq t'$,
- in either case, set $k_t^{trunc} = k_t^{guess}$ for all $t > t'$.

Notice that it is important to fix k_t^{trunc} from period 0 to t', otherwise the model could generate jumps from k_{min} to k_{max} that prevent convergence.

Step 3: Household Optimization Problem in Partial Equilibrium. The partial equilibrium of all agents is then solved following the method described in Subsubsection A.2.3. Specifically, the guess of q_0 and the exogenous values of k_0 and of the wealth shares are used to compute the initial wealth a_0^i of each agent, while the truncated capital path, $\{k_t^{trunc}\}_{t=0}^{T}$, is used to compute the rates of return, $\{R_t^{trunc}\}_{t=0}^{T}$ and wages $\{w_t^{trunc}\}_{t=0}^{T}$. The resulting consumption paths are denoted by $\{c_t^{i,new}\}_{t=0}^{T-1}$ for all agent types i.

Step 4: Updating the Consumption Guess. The consumption guess is updated as a period-by-period weighted average between the previous guess and the newly computed consumption path:

$$c_t^{i,guess'} = \omega c_t^{i,guess} + (1-\omega) c_t^{i,new}, \tag{2.42}$$

with $\omega \in [0, 1)$.

The Manhattan distance between $\{c_t^{i,guess}\}_{t=0}^{T-1}$ and $\{c_t^{i,new}\}_{t=0}^{T-1}$ is computed. If this distance falls below the tolerance threshold, the algorithm is deemed to have converged, and $\{c_t^{i,guess}\}_{t=0}^{T-1}$ and $\{k_t^{guess}\}_{t=0}^{T}$ are taken as the vectors associated with the initial guess q_0. If the distance exceeds the threshold and the maximal number of iterations has not yet been reached, the iteration continues. Otherwise, once the maximal number of iterations is reached, the algorithm is restarted from the initial guess with a higher ω. The algorithm allows for a limited number of adjustments to the relaxation parameter

ω. If the algorithm still fails to converge at the highest ω, the algorithm takes the final values, obtained at the highest ω and at the maximal number of iterations, of $\{c_t^{i,guess}\}_{t=0}^{T-1}$ and $\{k_t^{guess}\}_{t=0}^{T}$ as those associated with the initial guess q_0.

Outer Loop: Finding the Value of q_0.

Attention turns to the outer loop of the algorithm, which updates the guess for q_0 until it converges to its equilibrium value. The algorithm is initialized with two arbitrarily chosen values of q_0, denoted q_0^{min} and q_0^{max}, which are expected to lie below and above the equilibrium value of q_0, respectively. For simplicity, a typical choice is $q_0^{min} = 0$. The following steps are iterated:

Step 1: Defining a Test Value for q_0. An initial value of the rent-generating factor to test, denoted q_0^{test}, as the midpoint between q_0^{min} and q_0^{max}:

$$q_0^{test} = \frac{q_0^{min} + q_0^{max}}{2}. \tag{2.43}$$

Step 2: Computing the Associated Capital Path. Following the method described above (A.2.2), the capital path $\{k_t^{guess}\}_{t=0}^{T}$ associated with q_0^{test} is computed.

Step 3: Updating q_0^{min} or q_0^{max} Depending on the value of k_0^{guess}, either q_0^{min} or q_0^{max} is updated to q_0^{test}:

- If $k_0^{guess} > k_0$, consumption is higher than in equilibrium, implying that the economy must start from a level of capital above the exogenously given k_0. It is inferred that the rent-generating factor price guess exceeds its equilibrium value, and q_0^{max} is updated to q_0^{test}.
- Conversely, if $k_0^{guess} < k_0$, q_0^{min} is updated to q_0^{test}.

If, after this update, the bracket width $q_0^{max} - q_0^{min}$ falls below a pre-specified tolerance level, the algorithm is deemed to have converged and $\frac{q_0^{min}+q_0^{max}}{2}$ is taken as the equilibrium initial value of the rent-generating factor. The associated equilibrium transition path is computed using the procedure described in Subsubsection A.2.2. Otherwise, the iteration continues.

A.2.3 Household Optimization Problem in Partial Equilibrium

A distinction is made between the methods used to solve the household optimization problem in partial equilibrium for non–Scrooge McDuck agents and for Scrooge McDuck agents. For non–Scrooge McDuck agents, the partial-equilibrium problem is solved exactly as in the converging case described in Subsubsection A.1.3.

For Scrooge McDuck agents, consumption at time T is known and assumed to be equal to its asymptotic value, $c_T^1 = c^1$, while wealth at time T remains unknown. In contrast to non–Scrooge McDuck agents, for whom the unique consumption path satisfies the sequence of Euler equations at all dates and matches both initial and terminal wealth, the partial-equilibrium problem of Scrooge McDuck agents is solved by selecting the terminal wealth level that is consistent with their terminal consumption, their initial wealth, and the entire sequence of per-period Euler equations.

Given a guess for a_T^1, $a_T^{1,guess}$, one can derive the entire paths of consumption, $\{c_t^{1,predicted}\}_{t=0}^{T-1}$, and wealth, $\{a_t^{1,predicted}\}_{t=0}^{T-1}$, consistent with $(a_T^{1,guess}, c_T^1)$ using their budget constraint (2.11) and their Euler equation (2.15) backward. The predicted wealth at $t = 0$, $a_0^{1,predicted}$ is strictly increasing in $a_T^{1,guess}$. The terminal wealth of type-1 agents is chosen such that $a_0^{1,predicted}$ matches the initial wealth a_0^1 implied by q_0, k_0 and the wealth share of type 1 agents at $t = 0$. The corresponding paths of consumption, $\{c_t^{1,predicted}\}_{t=0}^{T-1}$, and wealth, $\{a_t^{1,predicted}\}_{t=0}^{T-1}$ are then the solutions to the household optimization problem used in other stages of the algorithm.

Chapter 3

Rational Bubbles in the Ecological Transition

This paper studies how an ecological transition, modelled as a permanent decline in productivity growth, affects the pricing of unproductive rational bubbles in an overlapping-generations economy. It identifies two channels: an ambiguous long-run effect, more likely to be negative when the elasticity of intertemporal substitution is high, and a negative transition effect. When the overall effect is negative, a trade-off emerges between the speed of the ecological transition and the size of bubble losses. A new numerical method for solving transition paths with deterministic rational bubbles is developed.

I am particularly grateful to Stéphane Guibaud, Nicolas Cœurdacier and Jeanne Commault for their support throughout this project. I also thank Guillaume Plantin and all seminar attendees at Sciences Po for helpful comments and suggestions. All remaining errors are mine.

Human activity generates large-scale environmental damage, most notably climate change (IPCC, 2023) and biodiversity loss (IPBES, 2019). Addressing these challenges requires a profound transformation of production processes. Because market economies do not fully internalize environmental externalities, this transition must be largely coordinated through public policy. As a result, the announcement by policymakers of a given level of ecological ambition is a critical moment, as it reveals the extent to which environmental regulations will constrain production.

More specifically, a key risk associated with the announcement of ambitious environmental policies is that it may trigger large asset price losses. This concern is particularly salient for productive assets that rely on pollution-intensive technologies. Policies that restrict or tax such activities reduce the expected future dividends of these assets, leading to an immediate decline in their valuation upon announcement. These assets are commonly referred to as *stranded assets* and this mechanism has become central in the macro-financial analysis of climate policy (Van der Ploeg and Rezai, 2020b).

Yet, this leaves open the question of how the transition affects assets whose value does not rest on productive fundamentals. It is well established that, in incomplete-market environments, some assets can trade above their fundamental value, even under perfect foresight (Martin and Ventura, 2018). Under fully rational expectations, the component of the asset price in excess of fundamentals is defined as a rational bubble. These bubbles can exist if their equilibrium rate of return does not exceed the growth rate. Because such assets are not claims on production, they are not directly exposed to climate policy through their cash flows. However, their valuation remains tied to equilibrium growth and returns. A transition that affects these objects may therefore strand not only productive assets, but also rational bubbles.

This question is examined in a standard overlapping-generations (OLG) economy. This framework is canonical for studying rational bubbles (Samuelson, 1958; Diamond, 1965). Agents live for J periods and receive labor income that declines over time as they retire. Retirement generates a motive to save, and agents may be willing to accept a rate of return below the growth rate of the economy. In this case, rational bubbles can arise. They are purchased by young cohorts and sold when old, thereby facilitating intertemporal transfers of resources. As such, bubbles partially alleviate the market incompleteness inherent to OLG economies, which stems from agents' inability to trade with future (unborn) cohorts.

The ecological transition takes the form of a permanent, pre-announced decline in the TFP growth rate. Rather than explicitly distinguishing between brown and green technologies, the analysis focuses on its implications for aggregate productivity. Production constraints associated with the ecological transition are assumed to reduce the level of output relative to the unconstrained benchmark. The sequence of events is as follows. The economy initially lies on a bubbly balanced growth path (hereafter, BGP) with high productivity growth. It is then hit by an unexpected announcement shock: the government announces an ecological transition that will take effect in T periods, leading to a permanent decline in TFP growth from that date onward. The impact of this announcement shock on the rational bubble value can be decomposed into two components.

The first channel operates through a shift in the balanced growth path. On a bubbly BGP, the rate of return equals the growth rate. A transition from a high-growth to a low-growth bubbly path therefore reduces the equilibrium rate of return. The resulting effect on savings is ambiguous and depends crucially on the elasticity of intertemporal substitution. A higher elasticity implies lower aggregate savings relative to the pre-shock BGP. Moreover, the lower rate of return on the low-growth BGP is associated with a higher normalized capital stock. This has two opposing effects on the bubble size. First, it raises labor income and thus savings. Second, for a given level of savings, it reduces the size of the bubble, as aggregate savings must equal the sum of capital and the bubble. Overall, the shift in the balanced growth path has an ambiguous effect on the asymptotic value of the bubble relative to its initial level.

The second channel operates through transition dynamics. Assuming the growth slowdown occurs at the time of its announcement, the economy does not immediately jump to the low-growth balanced growth path. Instead, normalized capital starts below its asymptotic level and gradually converges upward. Along this path, the return on capital remains temporarily above its low-growth BGP value. By no-arbitrage, the bubble also needs to grow faster than the economy and must converge to its asymptotic value from below. This implies that, at the announcement date, the bubble lies below its asymptotic value.

Advance announcements of the ecological transition affect the bubble valuation. When the transition reduces the bubble value, a longer delay between announcement and implementation of the growth slowdown attenuates the initial decline. As in the stranded asset literature for brown assets, the policymaker faces a trade-off between limiting losses in asset value and implementing the ecological transition rapidly.

Methodologically, the paper proposes a novel numerical algorithm to solve transition paths with deterministic rational bubbles. The main challenge is that the initial bubble value is endogenous and must be determined as part of the equilibrium. The algorithm follows an iterative guess procedure. Starting from a candidate initial bubble value, it computes households' initial wealth and solves for the associated transition paths of capital and the bubble. It then compares the implied initial bubble value with its guess and updates the guess until convergence.

Related Literature This paper contributes to the literature on climate transition risk and stranded assets. It is most closely related to structural macroeconomic analyses that quantify the effects of climate policies, such as carbon taxes or green subsidies, on macroeconomic outcomes and asset prices (Van der Ploeg and Rezai, 2020a; Baldwin et al., 2020; Bistline et al., 2023; Capelle et al., 2023; Hambel and Van Der Ploeg, 2025), and, in some cases, incorporate financial frictions that amplify the adverse effects of asset price declines (Carattini et al., 2023; Kaldorf and Rottner, 2024; Berthold et al., 2024).[1] The key distinction is that existing work focuses on productive assets valued at fundamentals and tied to brown or green technologies, whereas this paper studies the revaluation of unproductive assets in the form of rational bubbles.

This focus connects the paper to the literature on rational bubbles. It builds on the classic result that rational bubbles can arise in overlapping-generations economies when saving demand is sufficiently strong (Samuelson, 1958; Diamond, 1965; Tirole, 1985). This paper is the first to study how a productivity shock affects the transition dynamics of a deterministic bubble. To do so, the paper develops a new numerical approach that computes the unique equilibrium transition path of the bubble globally. This contrasts with much of the recent work on rational-bubble dynamics, which typically analyzes the effects of shocks to bubble size in settings with local indeterminacy in bubble value (Martin and Ventura, 2012; Galí, 2014; Miao and Wang, 2018; Galí, 2021).[2]

The remainder of the paper is organized as follows. Section 1 introduces the model. Section 2 compares the bubble value under high- and low-growth balanced growth paths. Section 3 studies the transition dynamics when the growth slowdown occurs immediately upon announcement. Section 4 examines the effects of a delay between announcement and implementation of the growth slowdown. Section 5 concludes.

1 Model

Time is discrete and runs from $t = -\infty$ to $t = \infty$. The economy is a standard overlapping-generations environment with capital.

[1] A complementary empirical literature estimates the impact of past climate policies on brown asset valuations (Faccini et al., 2023; Delis et al., 2024; Kölbel et al., 2024) or quantifies the stock of assets exposed to climate-related revaluation and associated financial risks (Battiston et al., 2017; Allen et al., 2020; Vermeulen et al., 2021; Network for Greening the Financial System, 2022). See Campiglio et al. (2023) for a review.

[2] Such bubble shocks are natural in those settings, where the objective is often to study boom-bust episodes or the effects of monetary policy. This paper instead abstracts from business-cycle dynamics and focuses on how productivity growth shapes bubble size over the medium to long run.

Demographics A cohort born at date s has size N_s and lives for $J+1$ periods, from age 0 to age J. Population grows at rate g^N according to

$$N_{t+1} = (1 + g^N)N_t, \tag{3.1}$$

and agents supply labor inelastically according to an age-efficiency profile $\{e^j\}_{j=0}^{J}$. Aggregate labor input is denoted by $_t$:

$$_t \equiv \sum_{j=0}^{J} e^j N_{t-j}. \tag{3.2}$$

The age-efficiency profile e^j is assumed to decline with j, implying that labor income decreases over the life cycle and generating a motive for retirement saving.

Production A representative firm produces output Y_t with two inputs: labor, $_t$, and reproducible capital, K_t. Production follows a Cobb-Douglas function with labor-augmenting productivity growth and $\alpha \in (0, 1)$,

$$Y_t = K_t^{\alpha}(A_{t\,t})^{1-\alpha}, \tag{3.3}$$

where productivity evolves as

$$A_{t+1} = (1 + g_{t+1}^A)A_t. \tag{3.4}$$

Factors are paid their marginal products and capital depreciates at rate δ. The wage at time t is W_t, and the gross return between t and $t+1$ is R_{t+1}. To account for growth, each capital-letter variable X_t has a normalized counterpart x_t. It is defined as

$$x_t \equiv \frac{X_t}{A_{t\,t}}, \tag{3.5}$$

for any aggregate variable X_t, except wages that are scaled by productivity alone, $w_t \equiv W_t/A_t$. It follows that,

$$w_t = (1 - \alpha)k_t^{\alpha}, \qquad \text{and} \qquad R_{t+1}^K = 1 + \alpha k_{t+1}^{\alpha-1} - \delta. \tag{3.6}$$

Let total growth in efficient labor, g_t, be denoted by

$$1 + g_t \equiv (1 + g_t^A)(1 + g^N). \tag{3.7}$$

Bubble and Asset Holdings In addition to productive capital, the economy features one unit of a bubble, an intrinsically worthless storable asset with price Q_t. Since the bubble pays no dividend, its value comes entirely from its resale price. Its gross return, R^B_{t+1}, is given by

$$R^B_{t+1} = \frac{Q_{t+1}}{Q_t}. \tag{3.8}$$

The cohort of age j holds K^j_{t+1} units of capital and B^j_{t+1} units of the bubble at the end of period t. Household asset holdings must satisfy the following market-clearing conditions:

$$\sum_{j=0}^{J} K^j_t = K_t, \qquad \sum_{j=0}^{J} B^j_t = 1. \tag{3.9}$$

Aggregate savings at the end of period t, denoted by S_t, are defined as

$$S_t \equiv K_{t+1} + Q_t. \tag{3.10}$$

Household Maximization Problem The cohort of age j at date t chooses consumption and end-of-period asset holdings to maximize the following intertemporal utility:

$$\max_{C^j_t, K^j_{t+1}, B^j_{t+1}} \mathbb{E}\Big[\sum_{\tau=0}^{J-j} \beta^\tau u\Big(C^{j+\tau}_{t+\tau}\Big)\Big], \qquad u(c) = \begin{cases} \dfrac{c^{1-\theta}-1}{1-\theta}, & \theta \neq 1, \\ \log(c), & \theta = 1, \end{cases} \tag{3.11}$$

subject to the period budget constraint

$$C^j_t + Q_t B^j_{t+1} + K^j_{t+1} = R^B_t Q_{t-1} B^{j-1}_t + R^K_t K^{j-1}_t + e^j N_{t-j} W_t, \tag{3.12}$$

and to the terminal conditions

$$B^J_{t+1} \geq 0, \qquad K^J_{t+1} \geq 0. \tag{3.13}$$

The solution to the household's problem for the cohort of age j is characterized by the following Euler equations:

$$u'(c^j_t) = \beta_t \left[\frac{R^K_{t+1}}{1+g_{t+1}} u'(c^j_{t+1})\right] = \beta_t \left[\frac{R^B_{t+1}}{1+g_{t+1}} u'(c^j_{t+1})\right], \tag{3.14}$$

together with the terminal conditions (3.13), which bind for the cohort of age J,

$$B^J_{t+1} = 0, \qquad K^J_{t+1} = 0. \tag{3.15}$$

Timing of the Ecological Transition Shock The ecological transition is modeled as an unexpected permanent decline in TFP growth from a high value g^A_H to a low value

g_L^A, with $g_L^A < g_H^A$. This reduced-form shock captures the idea that environmental policy imposes constraints on production and lowers the growth rate relative to an unconstrained case. Appendix B considers an alternative specification in which the transition affects productivity growth only temporarily.

The economy evolves through three stages. For $t < 0$, it is on a high-growth balanced growth path (BGP), with productivity growth equal to g_H^A. Then, the growth slowdown is announced unexpectedly at $t = 0$ and implemented at time T. Formally, this gives,

- For $t < 0$: $g_t^A = g_H^A$ and ${}_t[g_{t'}^A] = g_H^A$ for all $t' > t$ (high-growth BGP).
- For $t \in [0, T)$: $g_t^A = g_H^A$ and ${}_t[g_{t'}^A] = g_H^A$ for all $t' < T$, but ${}_t[g_{t'}^A] = g_L^A$ for all $t' \geq T$ (anticipated slowdown).
- For $t \geq T$: $g_t^A = g_L^A$ and ${}_t[g_{t'}^A] = g_L^A$ for all $t' > t$ (transition to the low-growth BGP).

The economy is thus deterministic for all $t \neq 0$, and agents expect a deterministic path at all dates. It follows that, whenever the bubble has a positive value, the no-arbitrage condition implies $R_{t+1}^K = R_{t+1}^B \equiv R_{t+1}$, for all $t \neq 0$. At time $t = 0$, it must be that $\mathbb{E}_{-1}[R_0^K] = \mathbb{E}_{-1}[R_0^B]$. The portfolio allocation problem is indeterminate, and it is assumed that all agents hold a representative portfolio.

Equilibrium An equilibrium $\left\{ \{c_t^j, k_t^j, B_t^j\}_{j=0}^J, q_t \right\}_{t=-\infty}^{\infty}$, is a sequence of allocations and prices such that households solve their optimization problem (3.14, 3.15), markets clear for capital and the bubble (3.9), and the economy is on a balanced growth path for all $t \leq 0$. The goods market clears by Walras's law.

Bubble Existence and Equilibrium Selection A bubbly equilibrium is defined as an equilibrium in which the bubble is valued strictly positively along the path, $q_t > 0 \ \forall t$.

In the absence of shocks, a bubbly equilibrium must converge to a balanced growth path such that $R^B \leq 1 + g$. Otherwise, the value of the bubble, which grows asymptotically at rate R^B, would diverge relative to aggregate output. Such paths cannot be sustained in equilibrium: young cohorts would not receive sufficiently high wages to purchase the bubble from the old, violating the latter's terminal condition (3.15). On a balanced growth path, no-arbitrage implies $R^B = R^K$. It follows that a rational bubble can arise only if, absent the bubble, aggregate savings are high enough to drive the capital stock to a level where the return on capital falls below the growth rate of the economy.

Moreover, if $R^B < 1 + g$, the balanced growth path is non-bubbly. In this case, the bubble shrinks asymptotically relative to the size of the economy, as it grows at a slower rate. A bubbly balanced growth path therefore requires $R^B = 1 + g$. In line with the paper's motivation, the remainder of the analysis focuses on parameter values for which bubbly equilibria exist and assumes the equilibrium to be selected such that the economy starts and converges to bubbly balanced growth paths.[3]

2 Balanced Growth Path Comparaison

The first step in understanding how the transition from a high-growth to a low-growth regime affects bubble pricing is to compare bubbly balanced growth paths under high and low productivity growth. For notational convenience, x_m, with $m \in \{H, L\}$, denotes the value of variable x on the balanced growth path associated with high or low growth.

Aggregate Savings and Capital By definition, the value of the bubble corresponds to the difference between aggregate savings and aggregate capital:

$$q_m = s_m - k_m(1 + g_m), \qquad m \in \{H, L\}. \tag{3.16}$$

The evolution of the bubble across balanced growth paths can be decomposed through the comparative statics of aggregate savings and capital.

A decline in growth has an unambiguously positive effect on the normalized capital stock. In any bubbly balanced growth path, the return on assets equals the growth rate, so that

$$R_m = 1 + g_m, \qquad m \in \{H, L\}, \tag{3.17}$$

which pins down the equilibrium capital stock:

$$k_m = \left(\frac{R_m - 1 + \delta}{\alpha} \right)^{\frac{1}{\alpha - 1}}, \qquad m \in \{H, L\}. \tag{3.18}$$

A decline in growth lowers the rate of return, thereby increasing the capital stock: $k_L > k_H$.

Nonetheless, the normalized amount of goods invested in capital at the end of a period does not necessarily increase as g_m declines. To obtain a normalized capital stock of k_m in one period, it is necessary to invest $k_m(1 + g_m)$ in the previous period—an expression that appears in Equation 3.16. This reflects the fact that productivity growth between periods increases the scaling factor, A_{tt}, mechanically reducing the normalized capital stock. This effect is weaker in the low-growth case, and could theoretically offset the need to invest more to achieve a higher level of normalized capital. However, for

[3] As is standard in the OLG literature, when a bubbly equilibrium exists, there is a unique bubbly BGP and a unique initial bubble size B_0^* at $t = 0$ such that the economy converges to it. For any $B_0 \in (0, B_0^*)$, the economy instead converges to the non-bubbly balanced growth path, yielding a continuum of bubbly equilibria.

sufficiently short periods and plausible calibrations, normalized investment in capital, $k_m(1+g_m)$ decreases with the growth rate.[4]

Attention now turns to the effect of lower growth on normalized aggregate savings along the balanced growth path. As with capital accumulation, the effect is ambiguous. The analysis begins with a tractable two-period version of the model before extending to a more complex setting.

Two-period Case The two-period version of the model makes the comparative statics of aggregate savings as a function of growth transparent. Suppose agents live for two periods, working when young, $e^0 = 1$, and retiring when old, $e^1 = 0$. In this case, aggregate savings equal the savings of the young, and the Euler equation (3.14) implies

$$s_m = \varsigma_m \, w_m, \qquad \text{where} \quad \varsigma_m \equiv \frac{(\beta R)^{1/\theta}}{(\beta R)^{1/\theta} + R}, \tag{3.19}$$

and ς_m denotes the saving rate of young agents out of wage income.

A change in trend growth affects normalized aggregate savings through a wage effect and a return effect. As shown in Equation 3.18, a lower growth rate increases normalized capital, thereby raising wages, w, and lowering the rate of return, R. These price changes influence households' saving decisions, as captured by the sensitivity of aggregate savings to the growth rate.

$$\frac{ds}{d(1+g)} = \underbrace{\frac{\partial w}{\partial k} \frac{dk}{d(1+g)} \varsigma}_{\text{wage effect (+)}} + \underbrace{\frac{\partial \varsigma}{\partial R} \frac{dR}{d(1+g)} w}_{\text{return effect (+/-)}} \tag{3.20}$$

Higher wages increase the savings of the young, who save a fraction ς of this additional income (a positive wage effect). By contrast, the effect of a lower return R on aggregate savings is ambiguous. The elasticity of intertemporal substitution governs the sign of this effect. When $\theta < 1$, substitution effects dominate, and a lower return discourages saving. When $\theta > 1$, income effects dominate, so a lower return increases saving, as households accumulate more to smooth retirement consumption.

Overall, a slowdown in growth has an ambiguous effect on both aggregate savings and on the savings allocated to capital along the balanced growth path in the two-period model. As a result, the bubble value may be higher or lower in the low-growth BGP than in the high-growth one. A higher elasticity of intertemporal substitution, i.e. a

[4] A sufficient condition for $k_m(1+g_m)$ to be higher under low growth than under high growth is $\frac{\alpha}{g+\delta} \geq 1$, a condition satisfied under standard annual calibrations. More generally, this effect is not the focus of the analysis, as it is merely an artifact of discrete timing and would not arise under continuous time.

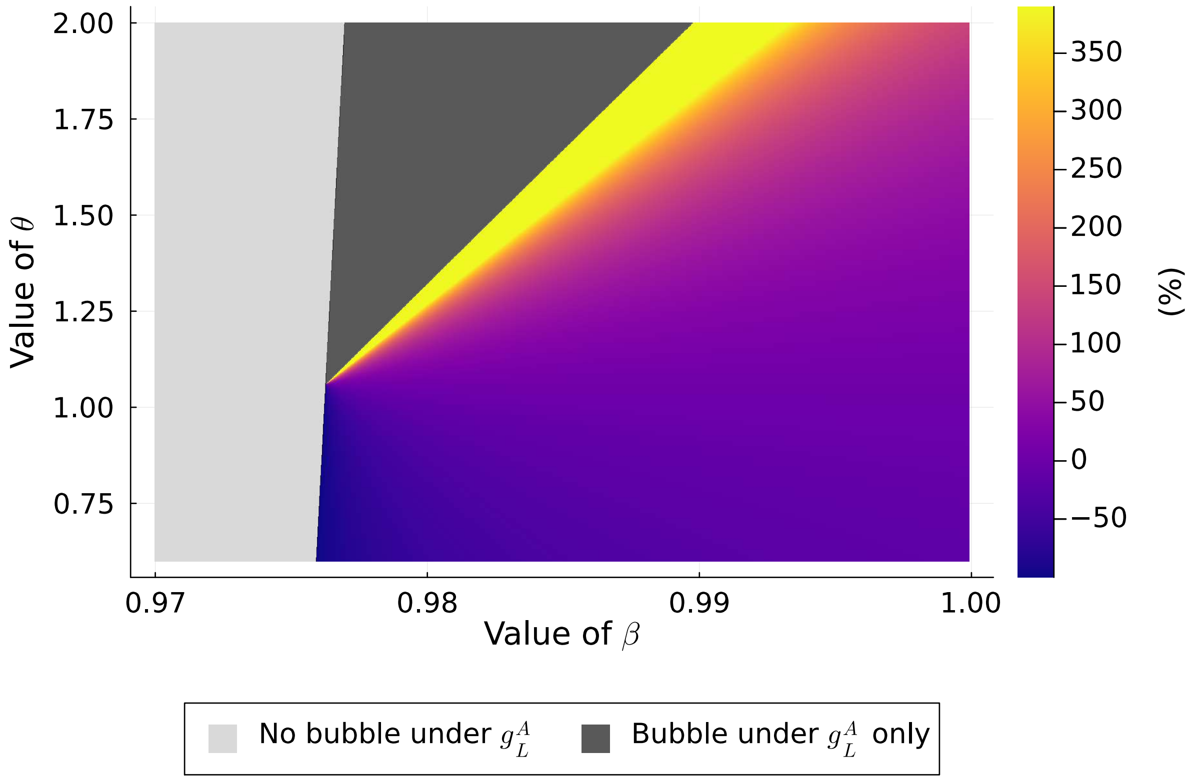


Figure 3.1: Percentage change in the steady-state bubble, $(q_L - q_H)/q_H$, in the 55-period economy.

Notes: This heatmap reports the percentage difference in the BGP value of the bubble, $(q_L - q_H)/q_H$, between a low-growth BGP with $g_L^A = 1.5\%$ (bubble q_L), and a high-growth BGP with $g_H^A = 3\%$ (bubble q_H), as a function of preference parameters β and θ. Other parameters are set to the following values: $g^N = 1\%$, $\alpha = 1/3$, $\delta = 0.15$, $J = 54$, $e^j = 1$ for $j \in [0, 39]$ and $e^j = 0$ for $j \in [40, 54]$. Parameter values for which no bubbly BGP exists under g_L^A are shaded in light gray. Parameter values for which no bubbly BGP exists under g_H^A, and for which the percent change in the bubble value is undefined, are shaded in dark gray.

lower θ, increases the likelihood that the normalized bubble size is higher under the high-growth BGP, $q_L < q_H$.

Multi-period economy The analysis now turns to a richer version of the model to assess the robustness of the two-period results and to consider shorter time periods. The balanced growth path of the model is computed numerically, focusing on the following illustrative parameter values. Agents live for 55 periods ($J = 54$), with periods interpreted as years. They work from age 0 to 39 ($e^j = 1$ for $j \in [0, 39]$) and retire from age 40 to 54 ($e^j = 0$ for $j \in [40, 54]$). Production parameters are set to $\alpha = 1/3$ and $\delta = 0.15$. The ecological transition shock is modeled as a decline in TFP growth from $g_H^A = 3\%$ to $g_L^A = 1.5\%$.

As in the two-period economy, the growth slowdown has an ambiguous effect on the BGP bubble value. Figure 3.1 illustrates this ambiguity by reporting the percentage change in the BGP bubble, $(q_L - q_H)/q_H$, over a grid of preference parameters (β, θ).

As shown above, the growth slowdown raises the normalized capital stock. Under the chosen parametrization, it also increases the amount of savings absorbed by capital, $(1+g_m)k_m$. This crowding-out force puts downward pressure on the bubble. However, as in the two-period economy, it can be offset by a sufficiently large increase in aggregate savings when the elasticity of intertemporal substitution is low enough, that is, when θ is high. Finally, a higher discount factor β raises aggregate savings and increases the bubble value under both high- and low-growth BGPs. This explains why the percentage change in the bubble is smaller for high values of β. Notice that, when β is too low, aggregate savings are insufficient to sustain a bubble, corresponding to the gray areas in the figure.

3 Transition Dynamics under Immediate Implementation

The discussion now turns from the comparison of balanced growth paths to the effect of the transition toward the low-growth bubbly balanced growth path on bubble pricing. To abstract from delayed-implementation effects, this section focuses on the case of immediate implementation of the ecological transition ($T = 0$). The analysis focuses on the 55-period version of the model.

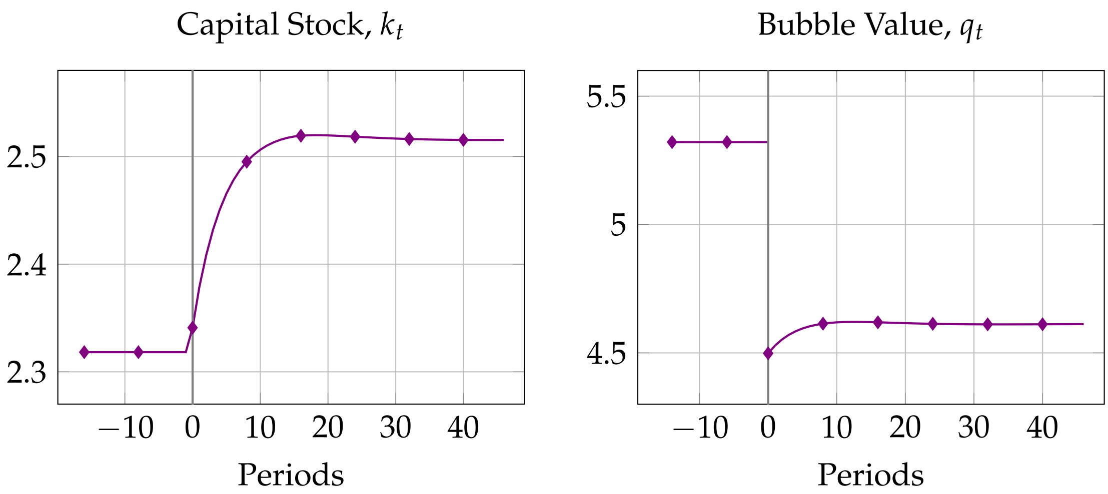
The transition is solved numerically using a novel approach to compute rational bubble dynamics away from the neighborhood of a steady state. The main difficulty lies in the fact that the initial bubble value, q_0, is unknown and must be determined through an iterative guess method. Starting from a candidate value q_0^{guess}, the algorithm computes agents' initial wealth at $t = 0$ and solves for the joint dynamics of capital and the bubble that are consistent with household optimization, asset-market clearing, and the terminal bubble value. Comparing the implied initial bubble value—obtained from the transition—with the candidate q_0^{guess} indicates whether the guess lies above or below the equilibrium value. The algorithm iterates on q_0^{guess} until it converges to the unique equilibrium value q_0. Appendix C provides a detailed description of the procedure.

The analysis distinguishes between a low-θ case and a high-θ case. As shown in Section 2, the elasticity of intertemporal substitution plays an important role in shaping how savings, and thus the bubble value, responds to a shift from a high-growth to a low-growth balanced growth path. For this reason, two illustrative transition paths are computed for low and high values of θ. Figures 3.2a and 3.2b report the corresponding dynamics for the capital stock and bubble value. In the low-θ case ($\theta = 0.75$), the elasticity of intertemporal substitution is high, and the low-growth balanced growth path features a smaller normalized bubble than the high-growth one. In the high-θ case

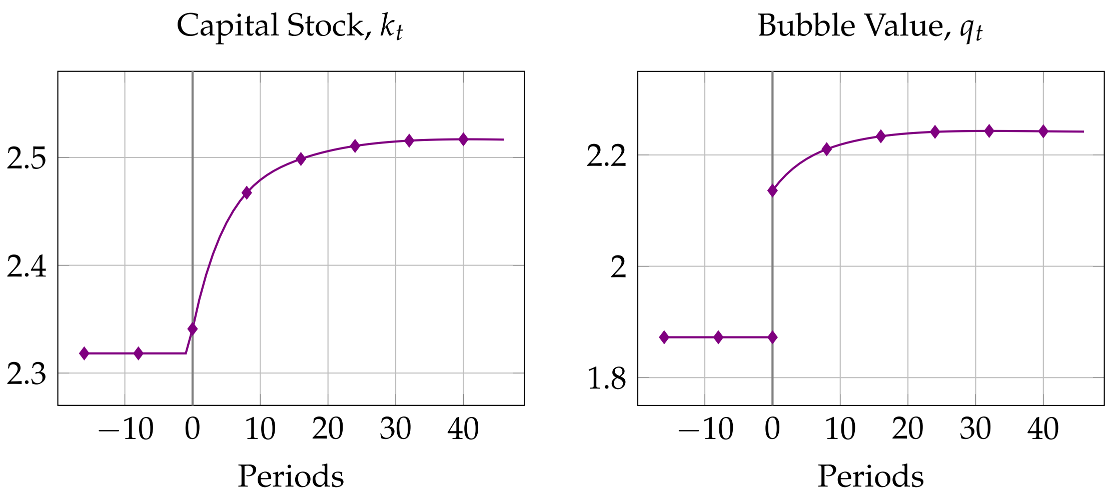
($\theta = 1.25$), the elasticity of intertemporal substitution is low, and the low-growth BGP features a larger bubble than the high-growth one.

Figure 3.2: Transition dynamics following an immediate productivity growth shock

(a) Low-θ case.



(b) High-θ case.



Notes: Illustrative graphs produced under the following parameter values: $\alpha = 1/3$, $\delta = 0.15$, $J = 54$, $e^j = 1$ for $j \in [0, 39]$, $e^j = 0$ for $j \in [40, 54]$, and $g^N = 1\%$. The low-θ and high-θ values are set to 0.75 and 1.25, respectively. The ecological transition shock is modeled as a decline in TFP growth from $g^A_H = 3\%$ to $g^A_L = 1.5\%$.

In both cases, the bubble jumps on impact at $t = 0$. In a deterministic environment, Equation 3.8 implies that, for any $t > 0$, the bubble value is equal to the discounted value of its asymptotic level:

$$q_t = \lim_{S \to \infty} \left[\prod_{s=1}^{S} \frac{1 + g_{t+s}}{R^B_{t+s}} \right] q_{t+S}, \qquad \forall t > 0. \tag{3.21}$$

The bubble value at $t = 0$ is determined backward and does not depend on its pre-announcement value at $t = -1$. Upon the announcement of the ecological transition, agents immediately revise their expectations about future returns, growth, and the asymptotic value of the bubble. As a result, the bubble revalues on impact. Let Δ_0 denote this unexpected normalized jump in value,

$$\Delta_0 \equiv \frac{Q_0 - \mathbb{E}_{-1}[Q_0]}{A_0 N_0}. \tag{3.22}$$

Equation 3.21 implies that Δ_0 can be decomposed as

$$\Delta_0 = \left[q_L - \frac{1+g_H}{1+g_L} q_H\right] + \lim_{S\to\infty}\left[\prod_{s=1}^{S} \frac{R_s^B}{1+g_s} - 1\right] q_L. \tag{3.23}$$

The first term reflects the change in the asymptotic balanced growth path. The second reflects the effect of transition dynamics.

The transition effect lowers the bubble valuation in both cases. For plausible calibrations, the capital stock at $t = 0$ is below its asymptotic level, $k_0 < k_L$.[5] Along the transition path, the capital stock and the return on capital tend, respectively, to lie below and above their asymptotic levels.[6] As a result, the bubble grows faster than effective labor during the transition. The bubble converges to its balanced growth path value from below, implying that its equilibrium value at $t = 0$ must lie below its asymptotic level. This transition effect exerts downward pressure on the bubble value at the time of the announcement of the ecological transition.

The BGP effect, however, may go in either direction depending on the value of θ. As shown in Section 2, the term $q_L - \frac{1+g_H}{1+g_L} q_H$ is negative when the elasticity of intertemporal substitution is sufficiently high, and positive when it is sufficiently low (that is, respectively for low and high θ). In the low-θ case, the BGP effect and the transition effect reinforce each other, so the bubble is revalued downward at $t = 0$, $\Delta_0 > 0$. The unexpected loss in bubble value at the announcement date, $\mathbb{E}_{-1}[Q_0] - Q_0 > 0$, is denoted as *stranded bubble*. In the high-θ case, by contrast, the two effects operate in opposite directions. Under our calibration, the BGP effect dominates, so the bubble is revalued upward on impact, $\Delta_0 < 0$.

[5] At $t = 0$, the normalized capital stock is equal to $k_H \frac{1+g_H}{1+g_L}$, as agents saved at $t = -1$ to maintain the high-growth balanced growth level of capital. While the case $k_0 = k_H \frac{1+g_H}{1+g_L} > k_L$ is theoretically possible, it is not economically relevant and largely reflects a discrete-time artifact. As the length of the period shrinks, the ratio $\frac{1+g_H}{1+g_L}$ converges to one; since $k_H < k_L$, this implies $k_0 < k_L$.

[6] Capital is described as tending to lie below its asymptotic value, as this need not hold in all periods. In particular, a slight overshooting arises in the low-θ case, which correspondingly leads the bubble to temporarily exceed its asymptotic value.

4 Policy Timing and Stranded Bubbles

An unexpected decline in the bubble at the time of the announcement of an ecological transition may be a source of financial instability. In the model, this drop reflects a pure valuation effect, but in practice it could have broader consequences that a government may want to avoid. For instance, if the bubble were attached to public debt, the loss could translate into an explicit or implicit partial default. If instead it is tied to private financial claims, it could weaken the balance sheets of firms or banks. While analyzing these amplification mechanisms lies beyond the scope of the paper, they provide a rationale for why policymakers may seek to limit such asset price declines.

The question then arises whether policy timing can reduce the size of the stranded bubble. In particular, the stranded-asset literature suggests that early announcements of an ecological transition may reduce the initial drop in asset prices (Van der Ploeg and Rezai, 2020b). In economies with both brown and green capital, tighter regulation or the phase-out of brown technologies reduces the value of brown capital. However, a longer delay between the announcement and the implementation of such policies allows agents to gradually let existing brown capital depreciate and reallocate investment toward green capital. This reduces the magnitude of the surprise valuation loss at the announcement date.

To assess whether a similar mechanism operates in this setting, attention is restricted to the low-θ economy, in which the bubble declines at the time of the announcement of the ecological transition. The analysis compares three anticipated transitions in which the growth slowdown is announced at date 0 but implemented only after $T = 1$, $T = 10$, or $T = 30$ periods. The long transitions ($T = 10$, or $T = 30$) are compared to the quasi-immediate transition ($T = 1$) rather than to the immediate transition ($T = 0$), so that the bubble jump is measured under a common normalization at $t = 0$ in all three scenarios (3.5). Figure 3.3 reports the corresponding transition paths. It shows that a longer delay between announcement and implementation of the growth slowdown reduces the initial, unexpected decline in the bubble, Δ_0.

As the bubble value declines at $t = 0$, part of the savings that would have been absorbed by it is reallocated toward productive capital. Between dates 0 and T, the economy accumulates more capital than along the high-growth balanced growth path. This depresses the return on capital below its high-growth balanced-growth-path value and implies that the bubble declines during the announcement window. The longer this pre-implementation period, from $t = 0$ to T, the larger the decline in the bubble over this interval. In turn, the greater this decline, the higher the initial bubble value at $t = 0$ must be to ensure convergence to the bubbly balanced growth path.

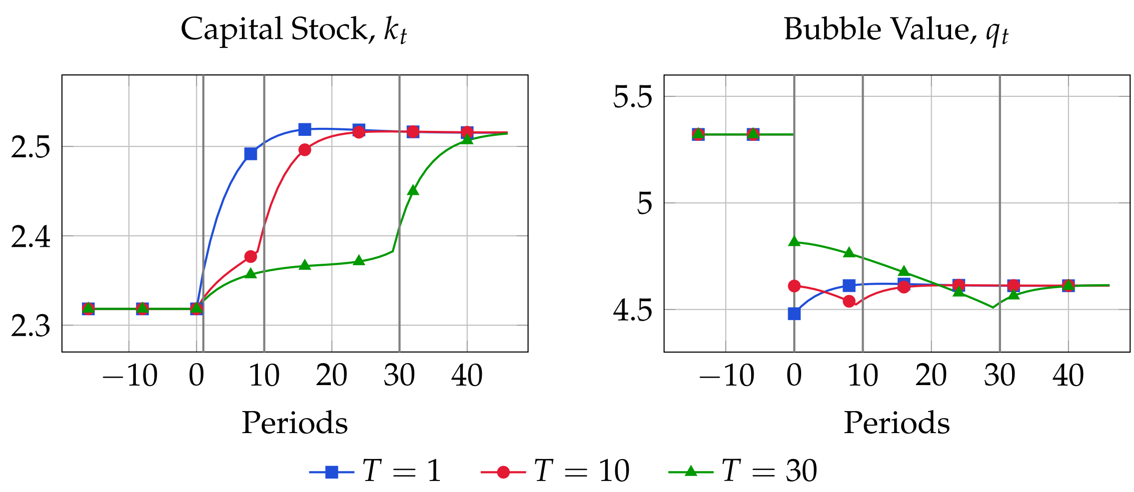


Figure 3.3: Delayed implementation and stranded bubbles in the low-θ case.

Notes: Illustrative graph produced under the following parameter values: $\theta = 0.75$, $\alpha = 1/3$, $\delta = 0.15$, $J = 54$, $e^j = 1$ for $j \in [0, 39]$, $e^j = 0$ for $j \in [40, 54]$, and $g^N = 1\%$. The ecological transition shock is modeled as a decline in TFP growth from $g_H^A = 3\%$ to $g_L^A = 1.5\%$.

As a result, when the bubble declines following the announcement of a growth slowdown, the bubble value at date 0 increases with the length of the announcement horizon. In practice, policymakers face a trade-off between containing the initial decline in bubble valuations—and thereby mitigating the risk of financial disruption—and implementing the ecological transition without delay. Conversely, when the elasticity of intertemporal substitution is sufficiently low for the bubble to increase after a growth slowdown, a longer delay reduces the magnitude of this initial upward jump (see Appendix A).

5 Conclusion

The ecological transition has general equilibrium effects that extend beyond the revaluation of brown assets. Modeled here in reduced form as a decline in growth, it also affects the valuation of rational bubbles through its impact on growth and equilibrium returns. Two policy implications follow. First, the announcement of a growth slowdown has an ambiguous effect on rational bubble valuations on impact, depending in particular on the elasticity of intertemporal substitution. When the elasticity is high, the bubble tends to decline; when it is low, the bubble may instead increase. Second, when the transition lowers bubble valuations, policymakers can mitigate the resulting stranded bubble at the cost of delaying implementation.

The analysis developed in this paper points toward two research directions. The first would build on its methodological contribution to study how rational bubbles respond to productivity shocks in alternative macroeconomic environments. The second would consist in bringing the model closer to the paper's main narrative by providing stronger foundations for the decline in TFP growth induced by the ecological transition. This would require explicitly modeling the transition between brown and green technologies and could be complemented by a quantitative analysis.

Bibliography

Daron Acemoglu, Philippe Aghion, Leonardo Bursztyn, and David Hemous. The environment and directed technical change. *American economic review*, 102(1):131–166, 2012.

Thomas Allen, Stéphane Dees, Carlos Mateo Caicedo Graciano, Valérie Chouard, Laurent Clerc, Annabelle de Gaye, Antoine Devulder, Sebastien Diot, Noëmie Lisack, Fulvio Pegoraro, et al. Climate-related scenarios for financial stability assessment: An application to france. Banque de france working paper, 2020.

Elizabeth Baldwin, Yongyang Cai, and Karlygash Kuralbayeva. To build or not to build? capital stocks and climate policy. *Journal of Environmental Economics and Management*, 100, 2020.

Stefano Battiston, Antoine Mandel, Irene Monasterolo, Franziska Schütze, and Gabriele Visentin. A climate stress-test of the financial system. *Nature Climate Change*, 7(4): 283–288, 2017.

Brendan Berthold, Ambrogio Cesa-Bianchi, Federico Di Pace, and Alex Haberis. The heterogeneous effects of carbon pricing: macro and micro evidence. Working paper, Bank of England, 2024.

John Bistline, Neil Mehrotra, and Catherine Wolfram. Economic implications of the climate provisions of the inflation reduction act. *Brookings Papers on Economic Activity*, 2023(1):77–182, 2023.

Emanuele Campiglio, Louis Daumas, Pierre Monnin, and Adrian von Jagow. Climate-related risks in financial assets. *Journal of Economic Surveys*, 37(3):950–992, 2023.

Mr Damien Capelle, Mr Divya Kirti, Mr Nicola Pierri, and Mr German Villegas Bauer. Mitigating climate change at the firm level: Mind the laggards. Imf working paper, 2023.

Stefano Carattini, Garth Heutel, and Givi Melkadze. Climate policy, financial frictions, and transition risk. *Review of Economic Dynamics*, 51:778–794, 2023.

Manthos D Delis, Kathrin de Greiff, Maria Iosifidi, and Steven Ongena. Being stranded with fossil fuel reserves? climate policy risk and the pricing of bank loans. *Financial markets, institutions & instruments*, 33(3):239–265, 2024.

Peter A Diamond. National debt in a neoclassical growth model. *The American Economic Review*, 55(5):1126–1150, 1965.

Renato Faccini, Rastin Matin, and George Skiadopoulos. Dissecting climate risks: Are they reflected in stock prices? *Journal of Banking & Finance*, 155, 2023.

Jordi Galí. Monetary policy and rational asset price bubbles. *American Economic Review*, 104(3):721–752, 2014.

Jordi Galí. Monetary policy and bubbles in a new keynesian model with overlapping generations. *American Economic Journal: Macroeconomics*, 13(2):121–167, 2021.

Christoph Hambel and Frederick Van Der Ploeg. Policy transition risk, carbon premiums, and asset prices. *Journal of Monetary Economics*, 152, 2025.

Jean-Charles Hourcade, Antonin Pottier, and Etienne Espagne. The environment and directed technical change: comment. Cired working papers series, 2011.

IPBES. Global assessment report on biodiversity and ecosystem services of the intergovernmental science-policy platform on biodiversity and ecosystem services. Technical report, 2019.

IPCC. Climate change 2023: Synthesis report. contribution of working groups i, ii and iii to the sixth assessment report of the intergovernmental panel on climate change. Technical report, 2023.

Matthias Kaldorf and Matthias Rottner. Climate minsky moments and endogenous financial crises. Deutsche bundesbank discussion paper, 2024.

Julian F Kölbel, Markus Leippold, Jordy Rillaerts, and Qian Wang. Ask bert: How regulatory disclosure of transition and physical climate risks affects the cds term structure. *Journal of Financial Econometrics*, 22(1):30–69, 2024.

Alberto Martin and Jaume Ventura. Economic growth with bubbles. *American Economic Review*, 102(6):3033–3058, 2012.

Alberto Martin and Jaume Ventura. The macroeconomics of rational bubbles: a user's guide. *Annual Review of Economics*, 10(1):505–539, 2018.

Jianjun Miao and Pengfei Wang. Asset bubbles and credit constraints. *American Economic Review*, 108(9):2590–2628, 2018.

Network for Greening the Financial System. Ngfs climate scenarios for central banks and supervisors, September 2022. URL https://www.ngfs.net/en/publications-and-statistics/publications/ngfs-climate-scenarios-central-banks-and-supervisors.

Timothée Parrique, Jonathan Barth, François Briens, Christian Kerschner, Alejo Kraus-Polk, Anna Kuokkanen, and Joachim H Spangenberg. Decoupling debunked. Technical report, European Environment Bureau, 2019.

Paul A Samuelson. An exact consumption-loan model of interest with or without the social contrivance of money. *Journal of political economy*, 66(6):467–482, 1958.

Jean Tirole. Asset bubbles and overlapping generations. *Econometrica: Journal of the Econometric Society*, pages 1499–1528, 1985.

Frederick Van der Ploeg and Armon Rezai. The risk of policy tipping and stranded carbon assets. *Journal of Environmental Economics and Management*, 100, 2020a.

Frederick Van der Ploeg and Armon Rezai. Stranded assets in the transition to a carbon-free economy. *Annual review of resource economics*, 12(1):281–298, 2020b.

Robert Vermeulen, Edo Schets, Melanie Lohuis, Barbara Kölbl, David-Jan Jansen, and Willem Heeringa. The heat is on: A framework for measuring financial stress under disruptive energy transition scenarios. *Ecological Economics*, 190, 2021.

Chapter 3 – Appendix

A Policy Timing under Low Elasticity of Intertemporal Substitution

Section 4 studies how policy timing affects bubble valuation in the low-θ case, since this is the calibration under which the ecological transition generates stranded assets that a government may wish to mitigate. This appendix presents the same policy-timing comparison in the high-θ case, in which the bubble jumps upward discretely at $t = 0$. Figure 3.4 compares the capital and bubble paths under transitions announced 1, 10, or 30 periods in advance.

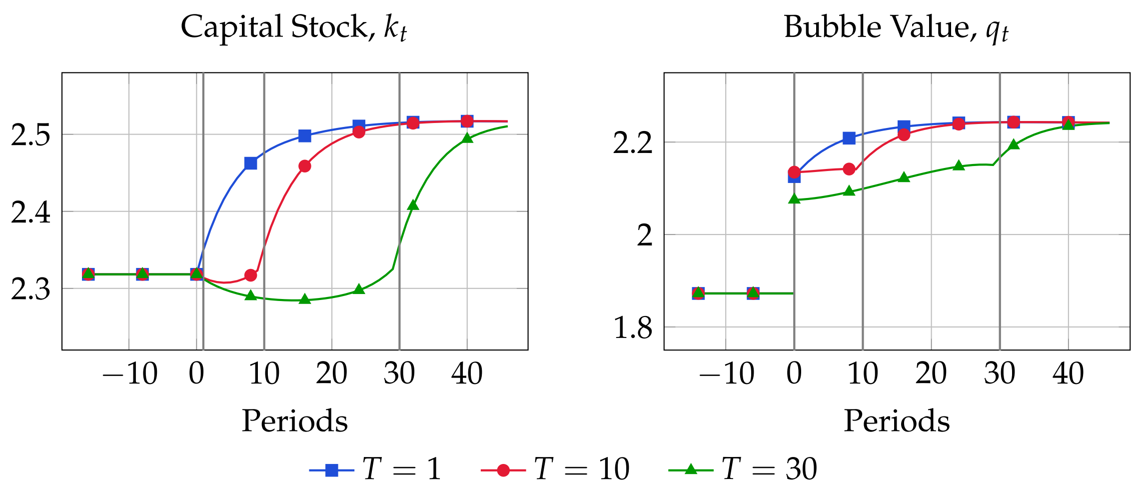


Figure 3.4: Delayed implementation and windfall bubbles in the high-θ case.

Notes: Illustrative graph produced under the following parameter values: $\theta = 1.25$, $\alpha = 1/3$, $\delta = 0.15$, $J = 54$, $e^j = 1$ for $j \in [0, 39]$, $e^j = 0$ for $j \in [40, 54]$, and $g^N = 1\%$. The ecological transition shock is modeled as a decline in TFP growth from $g^A_H = 3\%$ to $g^A_L = 1.5\%$.

When $T > 0$, the upward jump in the bubble at $t = 0$ crowds out productive capital over the pre-implementation period $[0, T]$. Capital falls below its high-growth balanced growth path level. As the implementation date T approaches, however, capital starts to recover, as households accumulate wealth while the rate of return remains temporarily high before converging to its lower level along the transition to the low-growth balanced growth path.

Over the interval $[0, T]$, the rate of return remains above the growth rate, implying that the normalized bubble increases during the pre-implementation phase. The longer this phase lasts, the more the bubble expands between $t = 0$ and T. It follows that a lower initial bubble value at $t = 0$ is required for the economy to converge to the low-growth balanced growth path. Accordingly, the bubble jump at $t = 0$ is larger for

$T = 10$ than for $T = 30$. From a policy perspective, a government seeking to maximize the windfall gain in bubble value at $t = 0$ while minimizing the associated crowding-out of productive capital would prefer the transition to be implemented relatively quickly.

However, the increase in the bubble is slightly smaller under $T = 1$ than under $T = 10$. Once the decline in TFP growth materializes, the capital stock grows more slowly in the former case than in the latter. As a result, the rate of return also converges more gradually, implying a stronger post-T growth rate of the bubble under $T = 1$. Equilibrium therefore requires the bubble to start from a lower initial value. The faster convergence toward the asymptotic capital stock under $T = 10$ likely reflects life-cycle effects associated with the different timing structures, although the precise mechanism warrants further investigation.

B Transition under Transitory Shock

The idea that internalizing the negative externalities of production may reduce output is fairly intuitive, although whether the resulting slowdown in productivity growth is permanent or temporary remains an open question. Acemoglu et al. (2012), for instance, find in their baseline calibration that environmental objectives can be achieved "without sacrificing (much or any) long-run growth." However, such conclusions have been challenged in subsequent discussions of the calibration and modeling assumptions (see, for example, Hourcade et al. (2011), written as a contemporaneous comment on Acemoglu et al. (2012) or Parrique et al., 2019). This issue lies beyond the scope of this paper. This section presents the simulations obtained under the alternative assumption that the decline in TFP growth is temporary.

The exercise follows the same timing structure as in the main analysis. For $t < T$, the productivity growth rate remains at its baseline level $g_t^A = g_{base}^A$. At time T it is hit by a negative shock of magnitude $\epsilon_T^A \geq 0$ with persistence ρ, such that

$$g_{t+1}^A = g_{base}^A - \rho^{t+1-T}\,\epsilon_T^A \qquad \text{for all } t \geq T. \tag{3.24}$$

The ecological transition shock is completely unanticipated before $t = 0$ and perfectly anticipated thereafter:

$$\mathbb{E}_t\left[g_{t'}^A\right] = \begin{cases} g_{\text{base}}^A, & \forall\, t' > t \quad \text{if } t < 0, \\ g_{t'}^A, & \forall\, t' > t \quad \text{if } t \geq 0. \end{cases} \tag{3.25}$$

The economy is assumed to initially lie on, and converge to, a bubbly balanced growth path. New parameter values are set to $g^A_{\text{base}} = 3\%$, $\epsilon^A_T = 1.5\%$, and $\rho = 0.8$ in the following illustrations, while all other parameter values remain unchanged.

Figures 3.5a and 3.5b display the transition dynamics of capital and the bubble following an immediate ($T = 0$) temporary ecological transition under, respectively, a high elasticity of intertemporal substitution (low θ) and a low elasticity of intertemporal substitution (high θ). As under the permanent shock, a growth slowdown initially raises the *normalized* capital stock. This, in turn, affects normalized aggregate savings through two channels: positively through higher wages and ambiguously through the decline in the rate of return. When the elasticity of intertemporal substitution is sufficiently low, a lower rate of return may increase aggregate savings, whereas the opposite occurs when the elasticity is high.

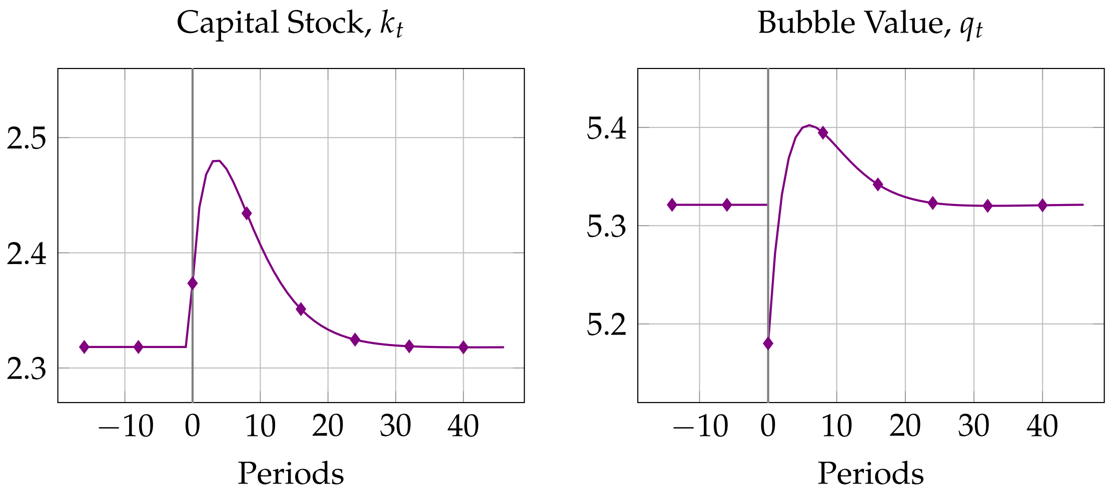
Given the ambiguous effect on aggregate savings and the positive effect on the capital stock, the impact of the shock on the rational bubble valuation is itself ambiguous. The two examples illustrate this ambiguity. Under a high elasticity of intertemporal substitution, aggregate savings decline following the shock, leading the rational bubble to contract as well at $t = 0$. By contrast, in the low-elasticity case, aggregate savings increase sufficiently to allow the bubble to expand on impact despite the higher capital stock. This mirrors the mechanism emphasized in the main text for the case of a permanent productivity growth slowdown. Notice that, although the temporary shock is chosen to generate movements in capital of a magnitude comparable to those obtained under the permanent shock, its effect on the bubble is of much smaller magnitude. The reason is that the bubble valuation is forward-looking and the productivity shock is only short-lived in the present case, unlike in the baseline specification.

Finally, the policy-timing implications of the model remain unchanged relative to the permanent growth slowdown specification. Figure 3.6 presents, in the low-θ case, the transition dynamics of capital and the bubble for alternative implementation horizons, $T \in \{1, 10, 30\}$. Given the high elasticity of intertemporal substitution, the bubble falls on impact. However, the longer the transition horizon, the smaller the initial decline in the bubble. As in the baseline exercise, the capital stock remains above its initial level over the interval $[0, T]$, implying a declining normalized bubble during this pre-implementation phase. It follows that the longer this period lasts, the higher the equilibrium bubble value at $t = 0$ must be in order for the economy to converge to the bubbly BGP.[7]

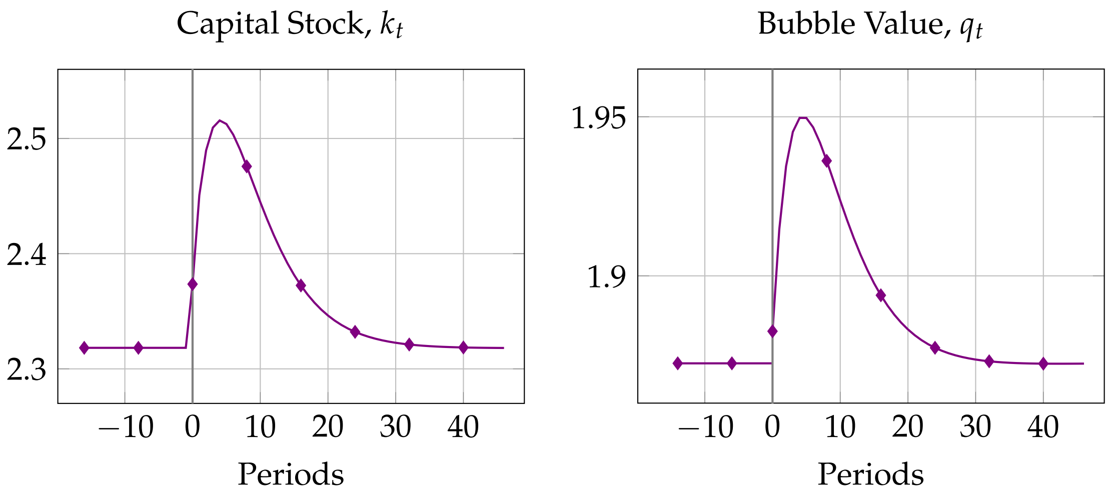
[7] A lower initial bubble value would instead lead to convergence toward the non-bubbly BGP.

Figure 3.5: Transition dynamics following an immediate temporary productivity growth shock.

(a) Low-θ case.



(b) High-θ case.

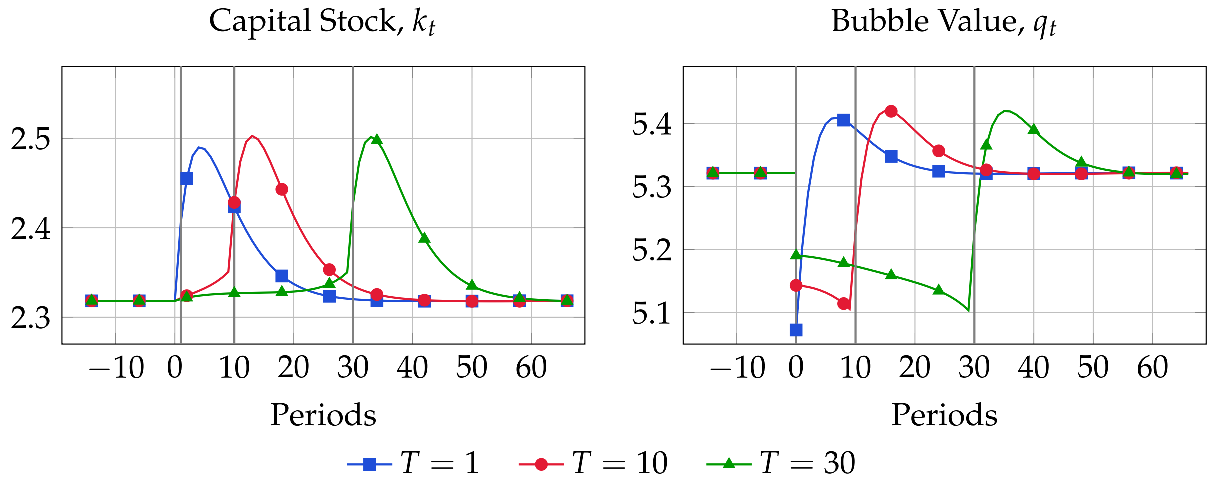


Notes: Illustrative graphs produced under the following parameter values: $\alpha = 1/3$, $\delta = 0.15$, $J = 54$, $e^j = 1$ for $j \in [0, 39]$, $e^j = 0$ for $j \in [40, 54]$, and $g^N = 1\%$. The low-θ and high-θ values are set to 0.75 and 1.25, respectively. The ecological transition is modeled as a temporary shock to TFP growth, reducing it from 3% to 1.5% at time T, with persistence parameter $\rho = 0.8$.

C Numerical Algorithm

This appendix describes the numerical procedure used to compute the deterministic transition path after the announcement of the ecological transition. The method applies to any deterministic sequence of growth rates $\{g_t^A\}_{t \geq 0}$ induced by the policy timing described in Section 1. Because the bubbly balanced growth paths are obtained using standard methods, this appendix focuses on the transition dynamics, which constitute the non-trivial part of the solution.

Figure 3.6: Delayed implementation and stranded bubbles in the low-θ case after a temporary shock.



Notes: Illustrative graph produced under the following parameter values: $\theta = 0.75$, $\alpha = 1/3$, $\delta = 0.15$, $J = 54$, $e^j = 1$ for $j \in [0, 39]$, $e^j = 0$ for $j \in [40, 54]$, and $g^N = 1\%$. The ecological transition is modeled as a temporary shock to TFP growth, reducing it from 3% to 1.5% at time T, with persistence parameter $\rho = 0.8$.

The main difficulty is that the initial normalized bubble value, q_0, is endogenous. As noted in the main text, unlike the capital stock at $t = 0$, which is predetermined by past saving decisions, the bubble value is a forward-looking asset price (see Equation 3.21), and jumps on impact at $t = 0$. To determine its equilibrium value, the algorithm relies on an iterative guess procedure. Intuitively, if the candidate bubble were too large, it would absorb an excessive share of aggregate saving, leading to strong crowding-out of productive capital and driving the implied capital path toward implausibly low, and ultimately negative, values. If instead the candidate bubble were too small, too much saving would be allocated to productive capital and the economy would converge toward the non-bubbly balanced growth path. The equilibrium value of q_0 is the one that exactly separates these two cases.

The algorithm proceeds through two nested loops. Given a candidate value for the initial bubble, q_0^{guess}, agents' initial wealth at $t = 0$ is determined. Conditional on this initial wealth, an inner loop solves for the transition path of capital and the bubble, alternating between household optimization and asset-market clearing, while imposing the terminal bubble value and the no-arbitrage condition. Solving this fixed-point problem yields a transition path for $\{k_t, q_t\}$ associated with the candidate value q_0^{guess}. The implied bubble value at $t = 0$ along this transition is then compared with the initial guess. This comparison indicates whether the candidate bubble is too large or too small. It governs the outer loop of the algorithm, which updates the guess for q_0 until the equilibrium initial bubble value is found.

The algorithm compute transition dynamics over a horizon $\mathcal{H}$. It is chosen large enough for the economy to be sufficiently close to the low-growth bubbly balanced growth path by date $\mathcal{H}$. Accordingly, beyond $\mathcal{H}$ all prices and allocations are set equal to their low-growth bubbly balanced growth values. Notice that the algorithm is described for the baseline case of a permanent growth slowdown, in which the economy transitions from a TFP growth rate g_H to g_L. However, it can easily be adapted to temporary productivity growth shocks by setting $g_H = g_L$. The procedure could even accommodate transition paths in which the initial growth rate is lower than the asymptotic one.

C.1 Inner loop conditional on a candidate q_0

Fix a candidate initial bubble value, denoted q_0^{guess}. This candidate is used only to determine agents' wealth at the beginning of period $t = 0$, which is held fixed throughout the inner loop. Conditional on these initial wealth levels, the algorithm computes the transition path of capital and the bubble through an iterative procedure that alternates between imposing households' optimality conditions and asset-market clearing.

Initialization. The capital stock at date 0 is predetermined by saving decisions taken at date -1 on the pre-announcement high-growth BGP. In normalized terms,

$$k_0 = \frac{1 + g_H}{1 + g_0} \, k_H. \tag{3.26}$$

For the remaining dates, the capital stock must be determined as part of the transition. The algorithm starts from an initial guess that sets the capital stock equal to its terminal bubbly value, $k_t^{(0)} = k_L$ for all $t \geq 1$.

Agents' asset holdings at the beginning of period $t = 0$ are obtained from the high-growth balanced growth path as

$$k_0^j = \frac{1 + g_H}{1 + g_0} \, k_H^{j-1}, \quad \text{and} \quad B_0^j = B_H^{j-1}. \tag{3.27}$$

Following the equilibrium selection described in the text, agents are indifferent between holding the bubble and capital along the high-growth balanced growth path, and all cohorts are assumed to hold the same representative portfolio. The total wealth of cohort j at the beginning of period $t = 0$, after returns have been realized, is denoted $a_{j,0}(q_0^{\text{guess}})$ and defined as

$$a_{j,0}(q_0^{\text{guess}}) \equiv R_0^K \, k_0^{j-1} \, + \, B_0^{j-1} \, q_0^{\text{guess}}. \tag{3.28}$$

Starting from this initialization, Steps 1–4 below are iterated until a fixed point is reached. Let $\{k_t^{(n)}\}_{t=1}^{\mathcal{H}}$ and $\{q_t^{(n)}\}_{t=0}^{\mathcal{H}}$ denote, respectively, the candidate transition paths for capital and the bubble at iteration n of the loop.

Step 1: Household optimization. Given the candidate capital path $\{k_t^{(n)}\}_{t=1}^{\mathcal{H}}$ and k_0, the algorithm recovers the associated sequences of returns and wages, $\{R_t^{(n)}, w_t^{(n)}\}_{t=0}^{\mathcal{H}}$. Conditional on these price paths and on the initial asset wealth of each cohort, $a_{j,0}(q_0^{\text{guess}})$, it computes cohort-by-cohort consumption and saving paths consistent with the budget constraints (3.12), the Euler equations (3.14), and the terminal condition (3.15).

To avoid keeping track separately of capital and bubble holdings, whose composition changes across transition candidate $\{q_t^{(n)}\}_{t=0}^{\mathcal{H}}$, it is convenient to rewrite the budget constraint (3.12) in terms of end-of-period saving. End-of-period saving of cohort of age j at time t is defined as,

$$S_t^j = K_{t+1}^j + Q_t B_{t+1}^j. \tag{3.29}$$

For any period $t \geq 1$, end-of-period saving of cohort j satisfies

$$s_t^j = \frac{R_t^{(n)}}{1+g_t} s_{t-1}^{j-1} + \omega^j w_t^{(n)} - c_t^j, \tag{3.30}$$

with $s_{-1,t} = 0$ for all t by definition, and where ω^j denotes the share of total labor supplied by cohort j,

$$\omega^j = \frac{e^j}{\sum_{s=0}^{J} e^s (1+g^n)^{j-s}} \qquad \left(= \frac{e^j N_{t-j}}{t} \text{ for all } t \right). \tag{3.31}$$

At $t = 0$, the budget constraint becomes

$$s_{j,0} = a_{j,0}(q_0^{\text{guess}}) + \omega^j w_0^{(n)} - c_0^j. \tag{3.32}$$

Terminal optimality further requires agents to die with zero wealth, as stated in equation (3.15),

$$s_t^J = 0 \qquad \text{for all } t. \tag{3.33}$$

Step 2: Computing the bubble path backward. The next step is to recover the updated bubble path, $\{q_t^{(n+1)}\}_{t=0}^{\mathcal{H}}$, implied by the candidate return path $\{R_t^{(n)}\}_{t=0}^{\mathcal{H}}$ and by the terminal bubbly equilibrium. The terminal condition is given by $q_{\mathcal{H}}^{(n+1)} = q_L$. Starting from this value, the full bubble path is then constructed backward using the no-arbitrage condition between the bubble and capital: along the transition, the bubble must yield the same gross return as capital in order not to be dominated. In normalized units, this implies

$$q_t^{(n+1)} = \frac{1+g_{t+1}}{R_{t+1}^{(n)}} q_{t+1}^{(n+1)}. \tag{3.34}$$

Step 3: Capital implied by asset-market clearing. Given the cohort saving decisions obtained in Step 1, $\{s_t^{j\,(n)}\}_{j=0,\dots,J;\,t=0,\dots,\mathcal{H}}$, and the bubble path obtained in Step 2, $\{q_t^{(n+1)}\}_{t=0}^{\mathcal{H}}$, the algorithm next recovers the capital stock implied by asset-market clearing. Let $\{k_{t+1}^{\text{mc}}\}_{t=0}^{\mathcal{H}-1}$ denote this market-clearing capital path. It is given by

$$k_{t+1}^{\text{mc}} = \frac{\sum_{j=0}^{J} s_t^{j\,(n)} - q_t^{(n+1)}}{1+g_{t+1}}, \qquad t = 0,\dots,\mathcal{H}-1. \tag{3.35}$$

Equation (3.35) follows directly from combining the asset market clearing condition for capital and bubble (3.9), with the defintion of aggregate savings (3.10) and cohort saving (3.29).

Step 4: Updating the capital path. The next step is to update the candidate capital path using the market-clearing capital sequence obtained in Step 3. The updated capital path is constructed through a relaxation step that combines the new market-clearing sequence with the previous iterate,

$$k_t^{(n+1)} = \frac{1}{1+\lambda_n} k_t^{\text{mc}} + \frac{\lambda_n}{1+\lambda_n} k_t^{(n)}, \qquad t = 1,\dots,\mathcal{H}, \tag{3.36}$$

where $\lambda_n \geq 0$ is a relaxation parameter. This relaxation step stabilizes the fixed-point iteration by preventing excessively large updates when the current iterate remains far from convergence.

Steps 1–4 are repeated until the distance between two successive capital paths falls below a pre-specified tolerance.

Convergence of the inner loop. A natural question is why the algorithm computes capital and bubble paths that are consistent with the *terminal* value of the bubble, rather than with the initial guess $q_{0,\text{test}}^{(m)}$. In principle, one could proceed in the opposite direction and use equation (3.34) forward, starting from the candidate initial bubble and recovering the implied bubble path period by period. Combined with asset-market clearing, this would deliver an implied capital path. The economic interpretation would be straightforward: if capital were to lie below its low-growth bubbly balanced growth path at the horizon, the candidate bubble would be too large and would crowd out productive capital too strongly; if instead capital were above its low-growth bubbly level at the horizon, the candidate bubble would be too small and would crowd out productive capital too weakly.

In practice, however, this alternative approach substantially complicates the numerical solution. Solving the model forward from the initial bubble guess would no longer turn the inner loop into a convergent fixed-point problem. For guesses of q_0 far from equilibrium—especially when the candidate bubble is too large—the implied capital path would quickly take extreme (potentially negative) values. Handling such off-equilibrium paths numerically, notably through a bounded capital path used to compute wages and returns, makes convergence slower and less robust. By contrast, the procedure adopted here avoids generating implausibly low capital values during the iterations and solves a well-behaved fixed-point problem, yielding more reliable convergence.

C.2 Outer loop on the initial bubble value

The previous subsection takes q_0^{guess} as given. The aim of the outer loop is to determine this initial bubble value. Let $\Phi(q_0^{\text{guess}})$ denote the date-0 bubble value implied by the transition obtained after running the inner loop for a candidate q_0^{guess}. An equilibrium initial bubble value is therefore a fixed point of the mapping

$$\Phi(q_0) = q_0. \tag{3.37}$$

The algorithm starts from two values, $q_{0,min}^{(1)}$ and $q_{0,max}^{(1)}$, chosen so that the equilibrium q_0 lies in the interval $[q_{0,min}^{(1)}, q_{0,max}^{(1)}]$. The code iterates the following steps.

Step 1: Test value and inner loop. At iteration m, the midpoint of the current bracket, $[q_{0,\text{min}}^{(m)}, q_{0,\text{max}}^{(m)}]$, is taken as the test value,

$$q_{0,\text{test}}^{(m)} = \frac{q_{0,\text{min}}^{(m)} + q_{0,\text{max}}^{(m)}}{2}. \tag{3.38}$$

Conditional on this test value, the inner loop is run to compute the associated transition paths of capital and the bubble that satisfy household optimization, asset-market clearing, and convergence to the bubbly balanced growth path (see the previous subsection).

Step 2: Updating the bracket. The sign of the error $\Phi(q_{0,\text{test}}^{(m)}) - q_{0,\text{test}}^{(m)}$ indicates whether the test value is too low or too high.

If

$$\Phi\left(q_{0,\text{test}}^{(m)}\right) > q_{0,\text{test}}^{(m)}, \tag{3.39}$$

the test bubble is too small. Intuitively, if agents were to start with a bubble equal to $q_{0,\text{test}}^{(m)}$ and the capital stock were then determined period by period from the asset-market-clearing condition, too little aggregate wealth would be absorbed by the bubble, so too much would be allocated to productive capital. Capital accumulation would become excessive and the economy would drift toward the non-bubbly balanced growth path. The lower bound of the bracket is updated accordingly,

$$q_{0,\min}^{(m+1)} = q_{0,\text{test}}^{(m)} \quad \text{and} \quad q_{0,\max}^{(m+1)} = q_{0,\max}^{(m)}. \tag{3.40}$$

Conversely, if

$$\Phi\left(q_{0,\text{test}}^{(m)}\right) \leq q_{0,\text{test}}^{(m)}, \tag{3.41}$$

the test bubble is too large. If agents were to start with a bubble equal to $q_{0,\text{test}}^{(m)}$, too much aggregate wealth would be absorbed by the bubble. The associated capital path would ultimately turn negative and lie clearly outside equilibrium. The upper bound of the bracket is updated as follows:

$$q_{0,\max}^{(m+1)} = q_{0,\text{test}}^{(m)}, \quad \text{and} \quad q_{0,\min}^{(m+1)} = q_{0,\min}^{(m)}. \tag{3.42}$$

Step 3: Convergence. The outer loop is iterated until the bracket width $q_{0,\max}^{(m+1)} - q_{0,\min}^{(m+1)}$ falls below a pre-specified tolerance level. The resulting midpoint is taken as the equilibrium initial bubble value,

$$q_0 = \frac{q_{0,\min}^{(m+1)} + q_{0,\max}^{(m+1)}}{2}. \tag{3.43}$$

The equilibrium transition path is then obtained by one final call to the inner loop.

Institut d'études politiques de Paris

ÉCOLE DOCTORALE DE SCIENCES PO

Programme doctoral d'économie

Département d'économie

Doctorat en Sciences économiques

Essais sur les Bulles Rationnelles

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soutenue le 8 juin 2026

Jury

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Résumé en français

Cette thèse est composée de trois chapitres, qui traitent tous de bulles rationnelles. Une bulle désigne une déviation du prix d'un actif par rapport à sa valeur fondamentale, c'est-à-dire à la valeur actualisée des flux de revenus futurs qui lui sont associés. Elle est dite rationnelle lorsque cette déviation est comprise comme telle par les agents : ceux-ci acceptent de détenir l'actif à un prix supérieur à sa valeur fondamentale parce qu'ils anticipent que la bulle sera valorisée dans le futur. Ce résumé en français présente successivement les trois chapitres. Les deux premiers étant étroitement liés, ils sont résumés conjointement.

Chapitre 1 et 2: Une Théorie à la Picsou des Dynamiques de Richesse

Les chapitres 1 et 2 proposent une théorie de la dynamique contemporaine de la richesse fondée sur l'interaction entre inégalités de revenu et préférences insatiable pour la richesse.

Le point de départ est un ensemble de faits stylisés observés dans les économies avancées depuis les années 1980. D'une part, le ratio richesse agrégée sur production a fortement augmenté. Dans les pays du G7, il est passé d'environ 300 % du PIB à près de 500 %. D'autre part, cette hausse ne s'est pas accompagnée d'une augmentation comparable du ratio capital productif sur production, qui est resté relativement stable. La hausse de la richesse agrégée provient donc essentiellement de la hausse du prix des actifs plutôt que d'une accumulation de capital reproductible. Enfin, cette période a également été marquée par une forte augmentation des inégalités de revenu au sommet de la distribution, en particulier de la part du revenu perçue par le top 1 %.

La question centrale est de savoir si la hausse des inégalités de revenu peut expliquer une hausse du ratio richesse/production tirée par les prix d'actifs plutôt que par l'accumulation de capital. Les modèles dans lesquels les ménages riches épargnent une part plus élevée de leur revenu (Hubmer et al., 2021; Elina and Huleux, 2023) permettent d'expliquer pourquoi une hausse des inégalités de revenu augmente l'épargne agrégée : elle redistribue des ressources d'agents à faible propension à épargner vers des agents à forte propension à épargner. Toutefois, dans ces modèles, l'épargne supplémentaire se traduit par une accumulation de capital. Cette prédiction est difficile à réconcilier avec les données. Les chapitres 1 et 2 rompent le lien mécanique entre l'augmentation de l'épargne et de la richesse agrégée, d'une part, et l'accumulation de capital, d'autre part. Ce découplage repose sur l'existence d'une bulle rationnelle dont le taux de croissance excède celui de l'économie.

Le mécanisme proposé repose sur l'introduction de préférences insatiables pour la richesse. La littérature sur les préférences pour la richesse (Kumhof et al., 2015; Mian et al., 2021) suppose déjà que les agents tirent une utilité directe de la détention de patrimoine. Cette hypothèse permet de reproduire le fait empirique selon lequel le taux d'épargne augmente avec le revenu permanent et avec la richesse. Dans sa spécification standard, cependant, l'utilité marginale de la richesse tend vers zéro lorsque la richesse diverge. Les chapitres 1 et 2 introduisent une déviation parcimonieuse par rapport à cette littérature: l'utilité marginale de la richesse est bornée inférieurement par une valeur strictement positive. Ainsi, même lorsque la richesse devient arbitrairement grande, les agents continuent de valoriser une unité supplémentaire de patrimoine.

Les préférences insatiables pour la richesse induisent une borne inférieure sur l'utilité marginale de l'épargne, qui définit en retour un seuil maximal de consommation optimale. Intuitivement, si un agent consommait au-delà de ce seuil, l'utilité marginale de la consommation deviendrait inférieure à la borne inférieure de l'utilité marginale de l'épargne. Il lui serait alors optimal de réduire sa consommation et d'épargner davantage. Ce niveau maximal de consommation optimale n'existe pas sous des préférences standards pour la richesse, puisque l'utilité marginale de l'épargne n'est alors pas bornée inférieurement par une valeur strictement positive : elle tend vers zéro lorsque la richesse diverge.

Asymptotiquement, lorsque les agents situés en haut de la distribution reçoivent un revenu supérieur à cette borne de consommation, ils ne consomment pas tout leur revenu. Ils accumulent alors une richesse qui diverge relativement à la taille de l'économie, tout en maintenant un niveau de consommation borné. Les chapitres 1 et 2 désignent ces agents comme des agents *à la Picsou* (Scrooge McDuck). Ils ne détiennent pas seulement de la richesse afin de financer leur consommation future ; ils détiennent une partie de leur richesse uniquement parce que sa détention leur procure directement de l'utilité. Cette partie du patrimoine est définie comme une *richesse excédentaire* (surplus wealth). Elle correspond à la différence entre la richesse totale d'un agent et la valeur actualisée de la part de sa consommation future qui n'est pas financée par ses revenus du travail futurs. Tous les dividendes associés à cette richesse excédentaire sont intégralement réinvestis. Cette demande persistante pour les actifs pousse leur prix d'équilibre au-dessus de leur valeur fondamentale et conduit à l'existence d'une bulle rationnelle.

L'économie de dotation. Le chapitre 1 développe ce mécanisme dans une économie de dotation à la Lucas (1978) à deux types d'agents. L'économie contient un arbre de Lucas, en quantité fixe, qui verse un dividende fixe chaque période. Les agents diffèrent seulement par leur dotation initiale en actif : les agents de type 1 sont initialement plus riches que les agents de type 2. Les préférences sont identiques entre types, mais elles sont non homothétiques : l'utilité de la richesse a une courbure plus faible que celle de la consommation. Par conséquent, les agents plus riches ont une propension à épargner plus élevée. Ils achètent progressivement les parts de l'actif détenues par les agents moins riches. Asymptotiquement, les agents de type 1 détiennent la totalité de l'arbre de Lucas.

Dans cette économie d'échange, deux types d'équilibres se distinguent selon la présence ou non d'une bulle dans le prix de l'arbre de Lucas. Dans un équilibre sans bulle, la borne supérieure de consommation est suffisamment élevée pour que les agents riches consomment l'intégralité de la dotation agrégée asymptotiquement. Ils ne sont pas des agents à la Picsou : leur richesse reste bornée et il n'existe pas de richesse excédentaire. Le prix de l'actif est alors égal à sa valeur fondamentale. Asymptotiquement, les agents de type 2 finissent avec une richesse et une consommation nulles. Dans un équilibre avec bulle, au contraire, la borne supérieure de consommation est inférieure au revenu asymptotique des agents de type 1. Ces agents ne veulent pas consommer tout le dividende qu'ils reçoivent. Ils accumulent une richesse excédentaire qui diverge au cours du temps et se traduit par une bulle divergente dans le prix de l'arbre de Lucas. La bulle croît au taux de rendement de l'actif, qui est lui-même supérieur au taux de croissance de l'économie. Les agents de type 2 bénéficient à chaque période de plus-values sur leurs unités d'arbre de Lucas. Leur richesse et leur consommation convergent vers des valeurs strictement positives.

Un résultat important est que l'existence d'un équilibre avec bulle dépend de la valeur de la borne inférieure de l'utilité marginale de la richesse. Si ce terme est nul, c'est-à-dire sous préférences standards pour la richesse, l'équilibre est unique et sans bulle. Si il est suffisamment élevé, l'équilibre est unique et avec bulle. Pour des valeurs intermédiaires, un équilibre avec bulle et un équilibre sans bulle peuvent coexister. La sélection de l'équilibre est alors auto-réalisatrice. Si les agents anticipent un rendement élevé, la borne de consommation est basse, les agents riches deviennent des Picsous, la bulle existe, et le rendement élevé est validé. Si les agents anticipent un rendement plus faible, la borne de consommation est plus élevée, les agents riches ne sont pas contraints par cette borne, aucune bulle n'apparaît, et le rendement plus faible est validé. Cette multiplicité n'est toutefois pas centrale: elle repose sur une forte élasticité de la consommation au taux de rendement chez les agents situés au sommet de la distribution. Cette propriété théorique des préférences insatiables pour la richesse reste à tester empiriquement.

L'économie de production. Le chapitre 2 introduit les préférences insatiables pour la richesse dans une économie de production. Cette extension est indispensable pour étudier la distinction entre richesse et capital. L'économie produit à l'aide de trois facteurs : du travail, du capital reproductible et un facteur en quantité fixe. Ce dernier peut représenter des actifs tels que la terre ou, plus généralement, des actifs durables difficilement reproductibles, comme certains droits de propriété, marques ou actifs intangibles. Le prix de ce facteur, déterminé de manière endogène, peut être décomposé entre une valeur fondamentale, correspondant à la valeur actualisée de ses dividendes futures, et une bulle. Les agents diffèrent par leur offre effective de travail et, éventuellement, par leur dotation initiale.

Les inégalités de revenu jouent alors un rôle central dans la détermination du régime d'équilibre, avec ou sans bulle. Lorsque les inégalités sont modérées, les revenus des agents les plus riches restent inférieurs à leur borne supérieure de consommation optimale. Ces agents épargnent davantage que les autres, mais l'économie demeure dans un régime sans bulle. En revanche, lorsque les inégalités sont suffisamment élevées, les revenus des agents du haut de la distribution dépassent asymptotiquement le seuil maximal de consommation optimale. Ces agents deviennent alors des agents à la Picsou : leur consommation est bornée, tandis que leur richesse diverge. Leur richesse excédentaire soutient une demande permanente pour le facteur fixe, dont le prix contient une bulle rationnelle divergente.

Cette économie de production permet ainsi de montrer que la bulle rompt le lien entre richesse et capital. Dans un équilibre sans bulle, une hausse de l'épargne agrégée accroît le capital reproductible. Le ratio richesse/production ne peut alors augmenter sans accumulation de capital. Dans un équilibre avec bulle, en revanche, une partie de l'épargne alimente la bulle plutôt que l'accumulation de capital productif. Cet effet d'éviction sur le capital réduit la consommation agrégée de long terme.

Le modèle peut ainsi reproduire une hausse du ratio richesse/production accompagnée d'une stabilité du ratio capital/production. Cette propriété permet d'interpréter les faits stylisés contemporains. La hausse des inégalités de revenu au sommet peut avoir déclenché une transition vers un régime avec bulle. Une fois ce régime atteint, la richesse peut continuer à croître relativement à la production, même lorsque les inégalités de revenu, mesurées hors effets de valorisation, se stabilisent. En effet, la bulle croît de manière auto-entretenue au taux de rendement de l'actif. Cette interprétation permet de comprendre pourquoi le ratio richesse/production a pu continuer à augmenter dans des économies où la part du revenu des plus riches n'a pas nécessairement continué à croître au même rythme.

Implications pour la taxation du capital et de la richesse. Une autre contribution du chapitre 2 consiste à analyser l'impact de la fiscalité du capital et de la richesse dans le modèle. Deux instruments sont considérés : une taxe sur le revenu du capital et une taxe sur la richesse. Cette distinction est importante, car des actifs peuvent avoir le même rendement total tout en différant dans la composition de ce rendement entre dividendes et plus-values. La taxe sur le revenu du capital s'applique aux flux de revenus, dont les plus-values sont ici exclues. La taxe sur la richesse, en revanche, s'applique à la valeur de l'actif et taxe donc implicitement les plus-values, y compris lorsqu'elles ne sont pas réalisées. La bulle ne verse pas de dividendes : son rendement provient uniquement de l'appréciation de son prix. Elle entre donc dans l'assiette fiscale d'une taxe sur la richesse, mais pas dans celle d'une taxe sur le revenu du capital. La présence d'une bulle brise ainsi l'équivalence usuelle entre taxation du revenu du capital et taxation de la richesse en présence de rendements homogènes Guvenen et al., 2023.[8]

L'effet d'une taxe sur l'accumulation de capital est ambigu, car deux forces opposées sont à l'œuvre. La première est l'effet désincitatif standard : en réduisant le rendement après impôt de l'épargne, la taxe réduit les incitations à accumuler du capital. La seconde est un effet de changement d'équilibre. En réduisant les revenus nets des agents du haut de la distribution, une taxe peut empêcher les agents riches d'accumuler une richesse excédentaire et faire disparaître la bulle. Dans ce cas, l'épargne qui aurait été absorbée par la bulle est redirigée vers le capital productif. Cet effet augmente le stock de capital. L'effet net dépend donc de l'importance relative de l'effet désincitatif et de l'effet de changement d'équilibre.

Le résultat est particulièrement fort pour la taxe sur la richesse. Dans le modèle, toute taxe positive sur la richesse exclut l'existence d'un équilibre avec bulle divergente. En effet, si la bulle diverge relativement à la production, une taxe proportionnelle sur la richesse générerait des recettes fiscales elles-mêmes divergentes. Comme ces recettes sont redistribuées aux ménages, elles impliqueraient une consommation agrégée incompatible avec la contrainte de ressources de l'économie. Une bulle divergente ne peut donc pas survivre à une taxe positive sur la richesse. Cela implique qu'une taxe sur la richesse, même faible, peut provoquer un changement d'équilibre de l'économie avec bulle vers une économie sans.

Ce résultat remet en cause l'arbitrage traditionnel entre redistribution et efficacité associé à la taxation. Dans un modèle standard, la redistribution fiscale réduit les inégalités, mais au prix d'une moindre accumulation de capital et donc d'un niveau de production plus faible. A l'inverse, en présence d'agents à la Picsou, certaines taxes peuvent à la fois réduire les inégalités et augmenter le capital productif.

[8] Cet argument, développé ici pour la première fois, pourrait être généralisé à tout modèle dans lequel les rendements des actifs incluent des plus-values.

Discussions. Le modèle contribue au débat ouvert par Piketty (2014) sur le rôle de l'écart entre le taux de rendement du capital et le taux de croissance dans la dynamique des inégalités de richesse. Selon l'observation empirique de Piketty, un rendement durablement supérieur au taux de croissance favorise la divergence des patrimoines. Michau et al. (2025) montrent toutefois que la conjonction de ces deux faits stylisés est difficile à obtenir dans un modèle en équilibre général. En effet, si les agents riches accumulent une richesse divergente et ont une forte propension à épargner, l'épargne agrégée augmente, le capital s'accumule et le rendement du capital diminue progressivement jusqu'à se rapprocher du taux de croissance. Le mécanisme à la Picsou rompt cette logique. La richesse des agents situés au sommet de la distribution peut diverger sans entraîner une accumulation correspondante de capital, car une partie de l'épargne est investie dans une bulle rationnelle. Le rendement peut donc rester durablement supérieur au taux de croissance, tandis que les inégalités patrimoniales divergent, conformément à la conjecture de Piketty. Ce papier est le premier à générer conjointement ces deux prédictions asymptotiques dans un modèle d'équilibre général microfondé.

Le papier offre également un éclairage sur le débat autour des effets en matière de bien-être des plus-values, réalisées ou latentes. Dans ce débats, une hausse du prix des actifs causée par une baisse du taux d'actualisation est parfois décrite comme un gain purement comptable: elle accroît la valeur du patrimoine, mais ne crée pas de flux de dividendes supplémentaires permettant de financer durablement une consommation plus élevée. Ici, les plus-values sont de nature différente, car elles sont récurrentes: la bulle s'apprécie à chaque période. Les agents qui ne sont pas des Picsous peuvent alors utiliser ces plus-values pour financer leur consommation, par exemple en vendant une partie de leurs actifs ou en empruntant contre leur valeur. Cela permet de relier le modèle à des phénomènes empiriques tels que l'usage de la hausse des prix immobiliers comme collatéral pour soutenir la consommation aux États-Unis.

Exercice Quantitatif. La dernière partie du chapitre 2 évalue quantitativement la pertinence du mécanisme pour les États-Unis. Le modèle est calibré à partir de 1989. Cette date est choisie parce que le Survey of Consumer Finances commence alors dans sa forme moderne. L'économie est supposée initialement dans un état stationnaire sans bulle. Les paramètres sont choisis pour reproduire le ratio richesse/production, le ratio capital/production et la distribution initiale de la richesse. La population est divisée en quatre groupes : le top 1 %, les 9 % suivants, les 40 % du milieu et les 50 % du bas de la distribution de richesse. Le modèle est ensuite soumis à une hausse exogène des inégalités de revenu du travail, mesurée dans les données.

Malgré sa simplicité, le modèle reproduit plusieurs dimensions importantes de l'évolution américaine. Après le choc d'inégalité, l'économie bascule de manière endogène vers un équilibre avec bulle. Le ratio richesse/production augmente, alors que le ratio capital/production reste relativement stable. La hausse de la richesse est largement tirée par la bulle, qui atteint environ 150 % du PIB en 2024. Le modèle reproduit également l'évolution de la distribution de la richesse : la part du top 1 % augmente fortement, tandis que celle des 40 % du milieu diminue, et que les parts des 9 % du top et des 50 % du bas restent relativement stables.

Le mécanisme est le suivant. Les agents du top 1 % deviennent des agents à la Picsou et accumulent une richesse qui diverge, ce qui se traduit par une hausse marquée de leur part dans la richesse totale au détriment principalement des 40 % du milieu. La stabilité relative de la part de richesse des 9 % du top s'explique par la compensation de deux effets opposés. D'un côté, ces agents ne sont pas des agents à la Picsou et voient leur part de richesse diminuer relativement à celle du top 1 %. De l'autre, ils bénéficient d'une hausse de leur revenu du travail relatif, qui soutient leur épargne et limite cette baisse. Enfin, la part de richesse des 50 % du bas reste relativement peu affectée, principalement parce que leur niveau initial de richesse est trop faible pour que leur part puisse diminuer fortement.

Le papier utilise ensuite ce cadre pour étudier une politique fiscale contrefactuelle : l'introduction d'une taxe sur la richesse de 1,5 % à partir de 2022. Cette taxe est supposée non anticipée. Dans le modèle, elle élimine immédiatement la bulle et empêche la poursuite de la divergence du ratio richesse/production. Elle réduit fortement la part de richesse détenue par le top 1 % et stabilise la distribution patrimoniale. Surtout, elle augmente le stock de capital relativement au scénario de référence sans taxe. L'effet de changement d'équilibre domine l'effet désincitatif : la suppression de la bulle libère de l'épargne pour l'accumulation productive. Le stock de capital est environ 5 % plus élevé trente ans après l'introduction de la taxe.

Du point de vue du bien-être, la taxe sur la richesse produit des gains via deux canaux. D'abord, elle redistribue la consommation vers les agents moins riches, dont l'utilité marginale de la consommation est plus élevée. Ensuite, elle augmente le capital productif, ce qui soutient la consommation agrégée. Ces gains doivent être comparés à la perte d'utilité liée à la baisse de la richesse détenue, puisque les agents valorisent directement la richesse. Le papier montre que, dans son exercice quantitatif, les gains de consommation et de redistribution dominent cette perte. Le gain pourrait même être plus fort selon le poids que le planificateur social attribue aux préférences pour la richesse. Si ces préférences reflètent un désir de statut ou de pouvoir, elles peuvent être considérées comme socialement peu pertinentes, voire comme un jeu à somme nulle. Au contraire, si elles reflètent un motif de legs ou une préférence intrinsèque pour l'accumulation, elles peuvent recevoir un poids similaire dans les préférences du planificateur social que dans celles des agents.

Conclusion. En résumé, les chapitres 1 et 2 proposent une théorie dans laquelle une hausse exogène des inégalités de revenu permet d'expliquer conjointement trois faits stylisés : la hausse du ratio richesse/production, la stabilité relative du capital productif et la divergence des inégalités patrimoniales. Le mécanisme central est que des préférences insatiables pour la richesse peuvent conduire les agents du haut de la distribution à accumuler une richesse excédentaire, non destinée à financer leur consommation future. Cette richesse excédentaire soutient une bulle rationnelle qui fait augmenter les prix d'actifs au-delà de leur valeur fondamentale. La bulle croît plus vite que l'économie, ce qui fait diverger la richesse relativement à la production. En même temps, elle a un effet d'éviction sur le capital, ce qui permet de comprendre pourquoi la hausse de la richesse ne s'accompagne pas d'une hausse du capital productif.

Chapitre 3: Bulles Rationnelles et Transition Écologique

Ce chapitre étudie les effets d'une transition écologique sur la valorisation des bulles rationnelles. Les politiques climatiques et environnementales peuvent réduire la valeur de certains actifs existants en imposant de nouvelles contraintes à la production ou en renchérissant l'usage de technologies polluantes. Une centrale à charbon, une réserve de pétrole ou une entreprise dépendante d'une technologie brune peuvent ainsi perdre de la valeur si leur utilisation devient moins rentable. La littérature désigne ces actifs sous le terme de *stranded assets*, ou actifs échoués (Van der Ploeg and Rezai, 2020; Vermeulen et al., 2021; Delis et al., 2024; Kölbel et al., 2024). Cette littérature se concentre toutefois principalement sur des actifs productifs, dont la valeur repose sur des profits futurs. Ce chapitre pose une question complémentaire: comment la transition écologique affecte-t-elle la valeur des bulles rationnelles, qui sont des actifs improductifs ?

Cette question est étudiée dans un modèle à générations imbriquées avec capital productif. Les agents vivent plusieurs périodes : ils travaillent durant la première partie de leur vie, puis se retirent du marché du travail. Cette structure de cycle de vie crée un motif d'épargne, les agents accumulant des actifs lorsqu'ils sont jeunes afin de financer leur consommation future. Une bulle peut exister lorsque cette demande d'épargne est suffisamment forte pour que le rendement d'équilibre des actifs soit inférieur ou égal au taux de croissance de l'économie. Ce mécanisme est au cœur de la littérature classique sur les bulles rationnelles (Samuelson, 1958; Diamond, 1965; Tirole, 1985). Dans le modèle, l'économie contient deux types d'actifs : du capital productif et reproductible, et un actif improductif en quantité fixe, qui ne verse aucun dividende. La valeur de ce second actif dépend uniquement de la possibilité de le revendre aux générations futures. Lorsqu'il a un prix positif en équilibre, il constitue une bulle rationnelle.

La modélisation retenue représente la transition écologique comme une baisse du taux de croissance de la productivité totale des facteurs. Cette hypothèse ne cherche pas à décrire explicitement le passage d'une technologie brune à une technologie verte. Elle capture plutôt l'idée qu'une transition environnementale ambitieuse impose des contraintes sur la production et réduit la trajectoire de croissance par rapport à un scénario sans contrainte environnementale. L'économie est initialement sur un chemin de croissance équilibrée caractérisé par une forte croissance et une bulle positive. Au moment de l'annonce de la transition, les agents apprennent que la croissance de la productivité sera durablement plus faible, soit immédiatement, soit après un certain délai.

Le premier résultat du chapitre est que l'effet de cette baisse de croissance sur la valeur de long terme de la bulle — normalisée par la taille de l'économie — est ambigu. Sur un chemin de croissance équilibrée avec bulle, le rendement du capital, le rendement de la bulle et le taux de croissance de l'économie sont égaux. Une baisse de la croissance réduit donc le rendement d'équilibre. Comme le rendement du capital diminue avec le stock de capital, le nouveau chemin de croissance est associé à un niveau plus élevé de capital normalisé.

Cette hausse du capital affecte d'abord l'épargne agrégée à travers deux canaux. D'une part, elle augmente les salaires, ce qui accroît l'épargne des ménages. D'autre part, elle réduit le rendement des actifs, ce qui a un effet ambigu sur l'épargne et dépend des préférences intertemporelles. Lorsque l'élasticité intertemporelle de substitution est élevée, la baisse du rendement rend l'épargne moins attractive et tend à réduire l'épargne agrégée. À l'inverse, lorsque cette élasticité est faible, l'effet revenu domine : les ménages épargnent davantage pour maintenir leur consommation future malgré un rendement plus faible.

L'effet final sur la bulle dépend de l'évolution relative de l'épargne agrégée et de celle absorbée par le capital productif. Comme la valeur de la bulle correspond à l'épargne non investie en capital, la hausse du capital normalisé réduit la taille de la bulle. Elle peut toutefois être compensée si l'épargne agrégée augmente suffisamment, ce qui est plus probable lorsque l'élasticité intertemporelle de substitution est faible. Les simulations numériques du modèle à 55 périodes confirment cette ambiguïté : selon les valeurs du facteur d'actualisation et de l'élasticité intertemporelle de substitution, le passage à une économie de plus faible croissance peut soit réduire, soit augmenter la valeur de long terme de la bulle.

Le deuxième résultat concerne la dynamique de transition. Même si la baisse de croissance est mise en œuvre immédiatement, l'économie ne rejoint pas instantanément son nouveau chemin de croissance équilibrée. Le capital normalisé augmente progressivement vers son nouveau niveau de long terme. Pendant cette phase de transition, le rendement du capital reste temporairement supérieur à sa valeur de long terme. Par condition de non-arbitrage, la bulle doit offrir le même rendement que le capital. Elle doit donc croître plus vite que l'économie pendant la transition pour atteindre sa valeur asymptotique. Cela implique que, au moment de l'annonce, la bulle doit se situer en dessous de sa valeur de long terme. La transition exerce donc un effet négatif sur la valeur initiale de la bulle.

L'effet total de la transition écologique sur la bulle combine deux forces : un effet de long terme, dont le signe dépend des préférences des ménages, et un effet de transition, qui déprime la valeur initiale de la bulle relativement à sa valeur asymptotique. Lorsque l'élasticité intertemporelle de substitution est élevée, ces deux forces vont dans le même sens : la baisse de croissance réduit la valeur de long terme de la bulle et la dynamique de transition amplifie la baisse initiale. L'annonce de la transition écologique entraîne alors une perte immédiate de valeur, que le chapitre définit comme une *stranded bubble*, par analogie avec les actifs échoués traditionnels dont la valeur diminue à la suite d'une annonce de politique environnementale. À l'inverse, lorsque l'élasticité intertemporelle de substitution est faible, la baisse du rendement peut accroître suffisamment l'épargne agrégée pour augmenter la valeur de long terme de la bulle, et éventuellement compenser l'effet négatif de transition. L'effet de la transition écologique sur les bulles rationnelles n'est donc pas mécaniquement négatif.

Le chapitre étudie ensuite le rôle du calendrier de la politique écologique. Lorsque la transition réduit la valeur de la bulle, le gouvernement peut-il limiter la perte initiale en annonçant la politique à l'avance ? Cette question est importante car une chute brutale des prix d'actifs peut avoir des effets financiers plus larges. Même si le modèle ne formalise pas les frictions financières, une baisse de valorisation pourrait, en pratique, fragiliser les bilans des banques, des entreprises ou de l'État si la bulle est incorporée dans des actifs financiers ou dans la dette publique.

Les résultats montrent qu'un délai plus long entre l'annonce et la mise en œuvre de la baisse de croissance réduit la chute initiale de la bulle. Intuitivement, après l'annonce, une partie de l'épargne qui aurait été absorbée par la bulle est réallouée vers le capital productif. Pendant la période précédant la mise en œuvre de la transition, le capital augmente, ce qui réduit le rendement du capital et conduit la bulle à diminuer progressivement. Plus cette période d'ajustement est longue, plus la baisse cumulée de la bulle peut être importante. Par conséquent, la perte de valeur au moment précis de l'annonce est plus faible lorsque la transition est annoncée longtemps à l'avance. Ce résultat met en évidence un arbitrage de politique économique. Une mise en œuvre rapide de la transition écologique permet de réduire plus tôt les dommages environnementaux, mais elle peut provoquer une baisse plus forte de la valeur des bulles rationnelles. À l'inverse, une annonce anticipée et une mise en œuvre plus graduelle peuvent atténuer cette perte de valeur, mais au prix d'un retard dans la transition. Cet arbitrage est proche de celui mis en avant dans la littérature sur les actifs échoués.

Enfin, le chapitre apporte une contribution méthodologique en résolvant globalement la trajectoire de transition vers le chemin de croissance équilibrée avec bulle. Calculer la transition d'une bulle rationnelle déterministe est difficile car la valeur initiale de la bulle est déterminée de manière endogène. Contrairement au capital, qui résulte des décisions d'épargne passées, le prix de la bulle est une variable prospective qui saute au moment de l'annonce de la transition. Le chapitre propose un algorithme numérique original. L'idée est de partir d'une valeur initiale candidate pour la bulle, de calculer les trajectoires de consommation, d'épargne, de capital et de bulle compatibles avec les conditions d'équilibre, puis de comparer la valeur initiale implicite de la bulle avec la valeur initiale candidate. L'algorithme ajuste ensuite cette valeur jusqu'à convergence.

Bibliographie

Manthos D Delis, Kathrin de Greiff, Maria Iosifidi, and Steven Ongena. Being stranded with fossil fuel reserves? climate policy risk and the pricing of bank loans. *Financial markets, institutions & instruments*, 33(3):239–265, 2024.

Peter A Diamond. National debt in a neoclassical growth model. *The American Economic Review*, 55(5):1126–1150, 1965.

Eustache Elina and Raphaël Huleux. From labor income to wealth inequality in the u.s. Working paper, 2023.

Fatih Guvenen, Gueorgui Kambourov, Burhan Kuruscu, Sergio Ocampo, and Daphne Chen. Use it or lose it: Efficiency and redistributional effects of wealth taxation. *The Quarterly Journal of Economics*, 138(2):835–894, 2023.

Joachim Hubmer, Per Krusell, and Anthony A Smith Jr. Sources of us wealth inequality: Past, present, and future. *NBER Macroeconomics Annual*, 35(1):391–455, 2021.

Julian F Kölbel, Markus Leippold, Jordy Rillaerts, and Qian Wang. Ask bert: How regulatory disclosure of transition and physical climate risks affects the cds term structure. *Journal of Financial Econometrics*, 22(1):30–69, 2024.

Michael Kumhof, Romain Rancière, and Pablo Winant. Inequality, leverage, and crises. *American Economic Review*, 105(3):1217–1245, 2015.

Atif Mian, Ludwig Straub, and Amir Sufi. Indebted demand. *The Quarterly Journal of Economics*, 136(4):2243–2307, 2021.

Jean-Baptiste Michau, Yoshiyasu Ono, and Matthias Schlegl. The preference for wealth and inequality: Towards a piketty theory of wealth inequality. Working paper, 2025.

Thomas Piketty. *Capital in the twenty-first century*. Harvard University Press, 2014.

Paul A Samuelson. An exact consumption-loan model of interest with or without the social contrivance of money. *Journal of Political Economy*, 66(6):467–482, 1958.

Jean Tirole. Asset bubbles and overlapping generations. *Econometrica*, pages 1499–1528, 1985.

Frederick Van der Ploeg and Armon Rezai. Stranded assets in the transition to a carbon-free economy. *Annual review of resource economics*, 12(1):281–298, 2020.

Robert Vermeulen, Edo Schets, Melanie Lohuis, Barbara Kölbl, David-Jan Jansen, and Willem Heeringa. The heat is on: A framework for measuring financial stress under disruptive energy transition scenarios. *Ecological Economics*, 190, 2021.